



**ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA**

Department of Statistical Sciences “Paolo Fortunati”

**Two Year Master in:
Greening Energy Market and Finance**

Continuous-Time Modelling of Wind Speed for Derivatives Pricing and Risk Management

**Supervisor:
Prof.ssa
Silvia Romagnoli**

**Candidate:
Enrico Butali**

**October 2026 session
Academic Year 2025/2026**

Contents

Chapter 1 — Introduction, Literature Review and Data	3
1.1 Motivation and Background	6
1.2 Research Questions and Contribution	8
1.3 Literature Review	9
1.3.1 Statistical Properties of Wind Speed	10
1.3.2 Time-Series Models and Long Memory	11
1.3.3 Conditional Volatility	12
1.3.4 Continuous-Time and Lévy-Driven Processes	13
1.3.5 Stochastic Volatility	14
1.3.6 Weather and Energy Derivatives, and the Market Price of Risk	14
1.3.7 Gap Statement	16
1.4 Data	18
1.4.1 Dataset A: Bologna Proxy (10 m, 2015–2018)	18
1.4.2 Dataset B: Nordfriesland Reference (100 m, 2015–2024)	19
1.4.3 SMARD Generation and EEX Day-Ahead Prices	21
1.4.4 Cross-Dataset Comparison	22
1.5 Thesis Outline	23
Chapter 2 — Methodology and Empirical Wind-Speed Modelling	25
2.1 Deseasonalisation and the Box–Cox Transformation	25
2.1.1 Marginal Distribution	25
2.1.2 Box–Cox Transformation	27
2.1.3 Seasonal Mean and Seasonal Variance	27
2.1.4 The Standardised Series $\tilde{W}(t)$	28
2.2 Conditional Mean: AR(4)	29
2.2.1 Order Selection	29
2.2.2 AR(4) Coefficients	30
2.2.3 Residual Diagnostics	31
2.2.4 Forecast Accuracy versus Persistence	31
2.2.5 Long Memory	33

2.3 Conditional Volatility: GARCH(1,1)	34
2.3.1 GARCH(1,1) Estimates	34
2.3.2 Two-Component Variance Decomposition	35
2.3.3 Standardised Residual Diagnostics	36
2.3.4 Cross-Dataset Comparison	38
2.4 Continuous-Time Embedding: CAR(4)	38
2.4.1 Euler Discretisation and the Companion Matrix	38
2.4.2 Eigenvalues and Stationarity	39
2.4.3 Conditional Forecasts	40
2.4.4 Normal-Inverse-Gaussian Tails	41
2.4.5 Seasonal Variance Carried Forward	44
2.4.6 The Samuelson Effect	44
2.5 Stochastic-Volatility Framework	44
2.5.1 Motivation	44
2.5.2 Two-SDE System Under P	45
2.5.3 Feller Condition	45
2.5.4 Remark on the Change of Measure	47
2.5.5 Characteristic Function and Round-Trip Consistency	47
2.5.6 Variance Term Structure	49
2.5.7 Why Maximum-Likelihood Estimation Is Deferred	51
2.6 Summary and Handover to Chapter 3	51
Chapter 3 — Derivatives Pricing, Risk Management and Conclusions	53
3.1 The Synthetic Wind Variance Swap	53
3.1.1 Instrument Definition and the Two Tracks	53
3.1.2 Market Data: Generation and Day-Ahead Prices	54
3.1.3 Track A: The Empirical Fair Strike	55
3.1.4 Track B: The Model-Implied Fair Strike	57
3.1.5 Commensurability of the Two Tracks	58
3.1.6 Power-Curve Linkage	61
3.1.7 The Merit-Order Effect	64
3.1.8 Payoff Simulation	65
3.2 Cross-Dataset Forecast Validation	67
3.2.1 The Diebold–Mariano Test and Its Conventions	67
3.2.2 Results	67
3.2.3 GARCH Volatility Diagnostics	69
3.2.4 Forecast-Error Distribution	70

3.3 Value of Information	72
3.3.1 The Height Correction	72
3.3.2 Raw and Height-Corrected Mispricing	73
3.3.3 The Empirical-Distance Metric	75
3.3.4 Delta Sensitivity and the Stress Scenario	76
3.3.5 Out-of-Time Validation	77
3.4 Risk Management: The CAWS Instrument	77
3.4.1 Definition and Fair Strike	77
3.4.2 Distribution Metrics on Non-Overlapping Annual Observations	79
3.4.3 Reading the Tail	80
3.5 Conclusions	82
3.5.1 Summary of Findings	82
3.5.2 Limitations	84
3.5.3 Future Work	85
3.5.4 Implications for Producers and Risk Managers	85

Chapter 1 — Introduction, Literature Review and Data

1.1 Motivation and Background

Wind is not a tradable asset. It cannot be stored, its delivery cannot be guaranteed, and no counterparty can be compelled to supply it on demand. Yet an entire class of energy-market participants, from onshore wind producers and transmission system operators to portfolio managers hedging renewable-heavy generation mixes, carries direct financial exposure to its fluctuations. This is the defining tension of weather-linked risk management: the underlying quantity that determines cash flow is physical and non-tradable, while the instruments used to hedge it must nonetheless be priced, marked to market, and settled in monetary terms. Wind speed is, in this sense, a demanding test case for derivative pricing methodology, because it resists the standard toolkit built for tradable underlyings.

For a German onshore wind producer, this exposure has two distinct legs. The first is volumetric: low-wind periods reduce generation directly, and because wind speed is positively autocorrelated at short lags and seasonally heteroskedastic rather than i.i.d., a run of calm days is not diversified away by the following week. The second is a price risk correlated with, rather than independent of, the first: German day-ahead electricity prices exhibit a merit-order effect, in which episodes of high aggregate wind generation push low-marginal-cost supply onto the grid and depress the clearing price when producers are generating the most. A storm system moving across the North Sea can lift aggregate German wind output to a large share of instantaneous demand within hours, driving the day-ahead price toward zero or, on occasion, below it, while onshore producers are running near full capacity.

Because revenue is the product of quantity and price, this negative correlation is partially self-offsetting at the revenue level: it dampens revenue variance relative to the independent case rather than compounding it. That natural hedge is, however, both weak and asymmetric. It is weak because the empirical correlation between German onshore wind generation and the day-ahead price is only $\rho = -0.16$ over 2015–2024 (Chapter 3), reflecting the many other drivers of the clearing price. It is asymmetric because it erodes the producer’s upside, capture prices falling when output is highest, without placing a floor under the downside: a sustained low-wind episode is not reliably compensated by a price move of matching magnitude. What remains after the natural hedge is an uncompensated volumetric exposure, and it is that residual leg which a derivative written on wind speed itself, rather than on the electricity price alone, is designed to address.

The economic stakes are not abstract. To illustrate the order of magnitude involved, consider a representative 100 MW onshore wind farm operating at the capacity factor computed for the Nordfriesland site in Section 1.4.2 (25.70%, from the Weibull-based annual energy production baseline; this is a gross resource figure, and a real project would forgo a further 8–15% to wake losses and availability). Annual generation is

then $100 \text{ MW} \times 8,760 \text{ h} \times 0.257 \approx 225,100 \text{ MWh}$. The mean German day-ahead price computed directly from the SMARD dataset used in this thesis (Section 1.4.3) differs sharply across three windows: €34.60/MWh for the 2015–2020 subsample, €126.47/MWh for the 2021–2024 subsample (inflated by the European energy-price crisis), and €71.38/MWh over the full 2015–2024 sample. Applying these three prices to the illustrative farm gives an annual revenue of roughly €7.8 million, €28.5 million, and €16.1 million respectively. These figures are offered strictly as an order-of-magnitude illustration, built transparently from a stated capacity factor and observed price data, not as a precise valuation of any actual asset. Against a revenue stream of this size, a partial mispricing of a hedging instrument written against it is economically material long before one reaches portfolio scale: as reported in Section 1.2, calibrating the same pricing model on a misaligned proxy dataset misprices the variance-swap fair strike by 88.0% even after correcting for measurement height.

Two features of the German market make it a natural setting for this study. First, Nordfriesland, on Germany’s North Sea coast in Schleswig-Holstein, is the country’s highest-density onshore wind corridor, and its geographic centroid is a reasonable proxy for the aggregate wind generation fleet reported by the German transmission system operators’ joint platform, SMARD (Bundesnetzagentur, smard.de). That alignment is not merely asserted: regressing daily SMARD onshore generation on the cube of Nordfriesland hub-height wind speed yields $R^2 = 0.59$, against $R^2 = 0.0001$ for the identical regression run on the Bologna proxy (Chapter 3). Second, the European Energy Exchange (EEX) lists the most liquid German power options market in Europe, providing at least the theoretical possibility of extracting a market-implied price of risk for wind-linked instruments. That possibility is more remote than data access alone would suggest: EEX options are written on the power price, not on wind speed or wind generation, so recovering a wind risk premium from them would require a joint model of price and generation capable of separating the wind premium from the power-price premium. As discussed in Section 1.3.6 and revisited under the $\theta = 0$ benchmark assumption in Chapter 2, no such option-implied calibration is attempted in this thesis. Together, these two facts motivate treating Nordfriesland hub-height wind as the reference underlying for a derivatives-pricing exercise grounded in German electricity market structure, rather than as an arbitrary meteorological time series.

The modelling programme that this thesis reports on was not designed around Nordfriesland data from the outset. It began, for reasons of data availability, on a deliberately low-cost proxy: four years of 10 m wind speed observations for Bologna, Italy, obtained via the OpenWeatherMap API (Section 1.4.1). Every methodological step, from the Box–Cox transformation and the AR(4)/CAR(4) mean dynamics to the GARCH(1,1) volatility layer and the stochastic-volatility embedding, was built, tested, and validated on this proxy dataset before being applied, unchanged in its functional form, to the Nordfriesland reference data. This two-dataset structure is deliberate rather than incidental: it allows the thesis to separate two questions that are usually conflated in applied derivatives work, namely whether the *methodology* is correct and whether

the *data* it is fed are fit for purpose. Chapter 3 turns this separation into a quantitative argument, the Value of Information, that measures the pricing error attributable to using a geographically and physically misaligned proxy instead of an aligned reference dataset.

1.2 Research Questions and Contribution

The thesis is organised around three research questions.

- RQ1.** To what extent does a continuous-time autoregressive model of order four, CAR(4), combined with a GARCH(1,1) conditional-volatility layer and calibrated on wind speed aligned in both geography and measurement height with the generation aggregate being hedged, yield a tractable closed-form fair strike for a synthetic wind variance swap?
- RQ2.** What is the Value of Information, defined here as the relative mispricing in the model-implied variance-swap fair strike attributable to proxy misalignment, from upgrading a geographically and physically misaligned proxy dataset (Bologna, 10 m) to a reference dataset aligned with the German onshore wind fleet (Nordfriesland, 100 m hub height), measured both before and after a wind-profile height correction?
- RQ3.** How does a Cumulative Accumulated Wind Speed (CAWS) instrument behave as a left-tail complement to the variance swap, in the sign and dispersion of its payoff, when evaluated on serially uncorrelated annual observations?

RQ1 is addressed methodologically in Chapter 2, which develops the full modelling pipeline from raw wind speed to a continuous-time stochastic-volatility representation, and applied in Chapter 3, which uses the resulting parameters to price a synthetic variance swap under a piecewise-constant-volatility convention built on the GARCH-in-wind-speed specification of Tol [1997]. RQ2 and RQ3 are addressed in Chapter 3, where the cross-dataset validation exercise (Diebold–Mariano forecast tests, the Value of Information comparison, and the CAWS hedging simulation) is carried out on both datasets side by side.

The thesis makes three contributions, corresponding one-to-one with the three gaps identified in Section 1.3.7. First, it calibrates the CAR(4)+GARCH pricing framework on wind speed that is simultaneously at turbine hub height and geographically aligned with the traded generation aggregate being hedged, a combination absent from the wind-derivatives literature reviewed below. Second, it produces an explicit, quantified Value of Information estimate for that alignment decision, expressed both as a raw pricing gap and, after applying a wind-profile height correction, as the gap that survives height adjustment. That residual gap is itself the informative quantity: it shows that geographic climate mismatch, not measurement height alone, is the dominant source of proxy-data pricing error. Third, it constructs and stress-tests a CAWS-based instrument addressing the left tail of wind-linked revenue risk, the sustained low-wind episodes

to which a variance swap is by construction indifferent, evaluated on non-overlapping annual observations to avoid the serial-correlation bias that overlapping rolling windows would otherwise introduce into the risk metrics. That evaluation choice trades statistical precision for freedom from serial-correlation bias: it leaves four annual observations for Dataset A and ten for Dataset B, so the resulting tail statistics are reported as descriptive rather than inferential, and the cross-dataset comparison rests on the sign and dispersion of the payoff rather than on a 5% quantile estimated from four points.

The principal findings can be stated at the outset. The model-implied fair strike for the synthetic wind variance swap is $K_{\text{var}}^B = 0.873$ on the Nordfriesland reference dataset against 0.054 on the Bologna proxy, a raw relative mispricing of 93.8% that falls only to 88.0% once the $1.931\times$ wind-profile height correction is applied. The $16.1\times$ ratio between the two strikes factorises into $1.82\times$ attributable to the GARCH conditional-variance state $h(t_0)$ and $8.85\times$ attributable to the seasonal variance level C_0 , which locates the mispricing in the seasonal-variance level of the site rather than in its volatility state, and is the quantitative basis for the claim that geography dominates height. Both components require qualification, supplied in Chapter 3 and signalled here so that the headline is not read for more than it carries: the $1.82\times$ term compares end-of-sample volatility states at two pricing dates six years apart rather than two fixed site characteristics, and Section 3.1.5 shows that a substantial part of the $8.85\times$ term is a Box–Cox measurement-scale artefact and a difference in the absolute level of wind speed, the underlying difference in *relative* variability between the two sites being far smaller. The claim that geography dominates height survives both qualifications intact, and Section 3.3.1 strengthens it by showing that the residual stays above 80% across every admissible wind-profile exponent. On forecast accuracy, all four fitted specifications reject the persistence benchmark decisively (Diebold–Mariano statistics of 8.26 on Dataset A and 16.76 on Dataset B, computed on the full standardised series; Chapter 2 reports the corresponding held-out-split statistics, which carry the opposite sign under a reversed loss-differential convention) while being observationally equivalent to one another in one-step point-forecast terms, a result developed in Section 1.5 and Chapter 3.

1.3 Literature Review

The methodology developed in this thesis draws on six distinct strands of literature, corresponding to the five successive layers of the modelling pipeline developed in Chapter 2 and the pricing framework applied in Chapter 3: the marginal statistical properties of wind speed, discrete-time forecasting models, conditional-volatility modelling, continuous-time and Lévy-driven processes, stochastic-volatility option-pricing theory, and, finally, weather and energy derivatives pricing under an explicit market price of risk. This section reviews each strand in turn and closes with a statement of the specific gap this thesis addresses.

1.3.1 Statistical Properties of Wind Speed

The starting point for any statistical treatment of wind speed is the observation that it is a positive, right-skewed, seasonally heteroskedastic quantity, and therefore not amenable to Gaussian time-series methods without a preliminary transformation. Tuller and Brett [1984] provide the classical physical justification for the marginal distribution most commonly used to describe it. Their argument proceeds from first principles: if the two horizontal components of the wind velocity vector are independently and identically normally distributed with zero mean and equal variance (a reasonable approximation for wind in the free atmosphere, away from surface friction effects), then the magnitude of the vector, which is what an anemometer records, follows a Weibull distribution with shape parameter $k = 2$, the Rayleigh special case. More generally, departures from this idealised isotropy produce Weibull-distributed speeds with k in the neighbourhood of two but not necessarily equal to it. This is a physical derivation, not merely a curve-fitting convention, and it explains why the Weibull distribution has become the default parametric choice in wind-resource assessment.

Garcia et al. [1998] test this claim empirically across twenty meteorological stations in Navarre, Spain, comparing Weibull and lognormal fits to hourly-mean wind speed via a log-log linearisation of the Weibull cumulative distribution function (the “Weibull probability paper” method) and nonlinear regression for the lognormal alternative. They find the Weibull distribution provides a materially better fit at essentially all stations, with R^2 values in the high-0.90s versus the high-0.80s for the lognormal, and they use the fitted distributions to estimate wind-farm energy production, establishing the now-standard link between the marginal distribution and the integral of a turbine power curve against its density. Their result is a useful benchmark against which to read the distributional analysis conducted on the two datasets used in this thesis (Section 1.4): the Weibull distribution is confirmed as the better fit for the hub-height Nordfriesland reference dataset, consistent with Garcia et al. [1998] and Tuller and Brett [1984], whereas the 10 m Bologna proxy dataset is better described by a lognormal distribution, a departure from the Weibull consensus that Section 1.4.1 attributes to the proxy’s low measurement height, where surface-layer friction and diurnal boundary-layer effects violate the isotropy assumption underlying the Rayleigh–Weibull derivation.

Because neither marginal family is Gaussian, a variance-stabilising transformation is required before the Gaussian time-series machinery of the following sections can be applied. The Box–Cox power transformation serves this role throughout the thesis, with the transformation parameter λ estimated by profile likelihood on each dataset and then rounded to a value admitting the closed-form pricing formulas of Section 1.3.6; the specific values adopted, and the cost of the rounding, are reported in Sections 1.4.1 and 1.4.2.

A separate physical literature governs the comparison of wind speeds measured at different heights, which the Value of Information exercise of Chapter 3 requires. Peterson and Hennessey [1978] assess the power-law profile $u(z) \propto z^\alpha$ against measured profiles and conclude that an exponent of $\alpha = 1/7$ is adequate for realistic but conservative

estimates of available wind power, except at extremely rough sites. This is the exponent adopted for the single height correction applied in this thesis, giving a variance scaling of $(100/10)^{2/7} \approx 1.931$ between 10 m and 100 m. It is important to be clear that α is an empirical parameter rather than a physical constant: it varies with surface roughness and atmospheric stability, with values nearer 1/10 typical over open water and larger values over rough terrain, so a coastal German site sits near the middle of the plausible range. The sensitivity of the correction to this choice is treated explicitly in Chapter 3, where the correction is confined to the cross-dataset comparison and applied nowhere else.

1.3.2 Time-Series Models and Long Memory

Once the marginal distribution and the deterministic seasonal component have been dealt with, the residual dynamics of wind speed remain strongly autocorrelated, and a substantial literature has developed autoregressive and long-memory models to capture this. Torres et al. [2005] fit month-specific ARMA models to hourly wind speed at five Navarre stations after a Weibull-shape-based power transformation and per-hour standardisation, and show that ARMA forecasts significantly outperform the naive persistence benchmark, particularly beyond the first few forecasting horizons, although the improvement narrows as the horizon lengthens. This paper establishes both the transform-then-model pipeline adopted throughout the present thesis and a natural point of comparison for the incremental value of the AR(4) model over persistence reported in Chapter 2.

A parallel strand of the literature asks whether wind speed exhibits long memory (autocorrelation that decays hyperbolically rather than exponentially), which would argue for a fractionally integrated ARFIMA specification rather than a finite-order AR or ARMA model. Haslett and Raftery [1989], in one of the earliest and most influential papers in this tradition, model daily wind speed across twelve Irish synoptic stations by combining spatial kriging with a fractionally differenced temporal component on the square-root-transformed series, and show that naive short-memory estimators materially understate the standard errors of long-run wind-power assessments once long-range dependence is properly accounted for. Kavasseri and Seetharaman [2009] apply fractional ARIMA models directly to day-ahead wind speed forecasting at four North Dakota sites, reporting an average 42% reduction in forecast mean squared error relative to a persistence benchmark and consistent gains over standard (short-memory) ARIMA. Ailliot et al. [2006] take a different route to capturing structure beyond a fixed-coefficient AR model, proposing a Markov-switching autoregression with time-varying spatial coefficients to track the translation of synoptic weather systems across a wind field, calibrated on ECMWF North Atlantic hindcast data. Their application is spatial rather than the single-location setting used in this thesis, but the underlying idea, that AR coefficients should be allowed to vary with the state of the system rather than held fixed, foreshadows the seasonally modulated and stochastic-volatility extensions developed in Chapter 2.

Taken together, this cluster motivates a specific modelling choice made in Chapter 2: an AR(4)/ARFIMA order-selection exercise is carried out on both datasets used in this thesis, following the diagnostic logic of Haslett and Raftery [1989] and Kavasseri and Seetharaman [2009], but the finite-order AR(4) specification is ultimately retained as the operating model because it embeds directly into the continuous-time CAR(4) framework of Section 1.3.4 without requiring a separate fractional-integration apparatus, and because the long-memory evidence on the datasets in this thesis, reported in Chapter 2, does not decisively favour the more heavily parameterised alternative.

1.3.3 Conditional Volatility

The residuals of an AR model fit to deseasonalised wind speed are typically not homoskedastic: calm periods and volatile periods cluster in time, in the same way that volatility clusters in financial return series. The tools for modelling this clustering originate in econometrics. Engle [1982] introduces the autoregressive conditional heteroskedasticity (ARCH) process, in which the conditional variance of a serially uncorrelated, zero-mean process is itself an autoregressive function of past squared innovations; Engle motivates the model with UK inflation data and develops both a maximum-likelihood estimator and a Lagrange-multiplier test for the presence of ARCH effects that remains the standard pre-estimation diagnostic to this day. Bollerslev [1986] generalises ARCH to GARCH by allowing the conditional variance to depend on its own lagged values as well as on lagged squared innovations (the volatility analogue of extending an AR model to an ARMA model), and shows that a low-order GARCH(1, 1) specification typically achieves the same fit as a much higher-order ARCH model with far fewer parameters. GARCH(1, 1) is the specification adopted for the conditional-volatility layer $h(t)$ throughout this thesis.

Tol [1997] is the paper that transplants this machinery specifically into wind speed modelling. Working with daily wind data from Shearwater, Canada, Tol fits Gamma-AR(2)-GARCH(1,1) and Normal-AR(2)-GARCH(1,1) specifications and demonstrates that the heteroskedastic model outperforms a homoskedastic AR-only benchmark, with the ARCH effect noticeably stronger in the underlying wind velocity components than in the scalar wind speed series itself. This is the direct precedent for the two-component volatility decomposition used throughout this thesis, $\sigma_{\text{total}}^2(t) = \sigma_{\text{seasonal}}^2(t) \times h(t)$, in which a deterministic Fourier-based seasonal variance is multiplied by a GARCH(1, 1) conditional-variance factor. It is also the source of the two-component volatility specification on which the pricing convention adopted in Chapter 3 rests. Tol's contribution is econometric rather than one of derivatives pricing: the paper contains no variance swap and no fair-strike formula, and it is cited here for establishing $h(t)$ as the conditional-variance state variable for wind speed. Following that treatment, the present thesis freezes h at its pricing-date value $h(t_0)$ over the life of the contract, a piecewise-constant simplification in the spirit of the piecewise-monthly seasonal variance of Alaton et al. [2002], and derives the resulting fair strike $K_{\text{var}}^B = h \times C_0$ directly from the Fourier decomposition of $\sigma_{\text{seasonal}}^2(t)$, whose first and second harmonics integrate to zero over a

full year.

1.3.4 Continuous-Time and Lévy-Driven Processes

Discrete-time AR and GARCH models are estimated on daily data, but a derivative contract must in general be priced at an arbitrary maturity that does not fall on a daily grid point, which motivates embedding the discrete-time dynamics into a continuous-time stochastic process. Brockwell and Marquardt [2005] provide the mathematical foundation for this step, developing the theory of Lévy-driven continuous-time ARMA (CARMA) processes: state-space representations built around a companion matrix A and driven by an arbitrary second-order Lévy process rather than Brownian motion, which allows the resulting marginal distributions to be asymmetric and heavy-tailed rather than Gaussian. They further construct a fractionally integrated extension (FI-CARMA) by convolving the CARMA kernel with a Riemann–Liouville fractional-integration kernel, connecting the long-memory literature reviewed in Section 1.3.2 to the continuous-time state-space formalism used in Chapter 2’s CAR(4) construction: the companion matrix A , its eigenvalues, and the matrix exponential $\exp(A\tau)$ that propagates the conditional mean forward in time are all direct applications of this framework.

Schwartz [1997] supplies the connection to commodity-derivatives pricing specifically. Schwartz compares one-, two-, and three-factor models of commodity spot prices, respectively a simple mean-reverting (Ornstein–Uhlenbeck) process, a model adding a stochastic convenience yield, and a model further adding stochastic interest rates, and evaluates their implications for the term structure of futures prices and for hedging performance. The single-factor Ornstein–Uhlenbeck specification in Schwartz’s framework is, in essence, the $p = 1$ special case of the $\text{CAR}(p)$ family used in this thesis, and its treatment of mean reversion under a risk-neutral measure is a direct precedent for the wind-futures pricing framework developed by Benth and Šaltytė Benth [2009] and adopted, under the $\theta = 0$ benchmark described in Section 1.3.6, in Chapter 2.

Barndorff-Nielsen and Shephard [2001] extend the non-Gaussian Ornstein–Uhlenbeck idea to stochastic volatility directly, constructing a volatility process driven by a positive Lévy subordinator (the background driving Lévy process) rather than a Brownian motion, so that returns become Gaussian mixtures capable of reproducing heavy tails and volatility clustering while retaining closed-form results for integrated volatility and its temporal aggregation. This paper is the theoretical bridge between the GARCH-based discrete-time volatility proxy $h(t)$ used in Chapters 2–3 of this thesis and the continuous-time variance process $V(t)$ introduced in the stochastic-volatility extension of Chapter 2: it demonstrates formally that a GARCH-type discrete filter can be understood as a discretisation of an underlying continuous semimartingale, which is precisely the conceptual step motivating the two-SDE system developed there.

1.3.5 Stochastic Volatility

Heston [1993] is the canonical reference for stochastic-volatility option pricing and supplies the mathematical template for the two-SDE system used in the theoretical stochastic-volatility chapter of this thesis. Heston models the variance of an asset's return as a square-root (Cox–Ingersoll–Ross-type) process correlated with the return innovation itself, and derives a closed-form European option price via Fourier inversion of the joint characteristic function rather than by solving the pricing partial differential equation directly. This technique generalises immediately to other payoffs, including the variance-swap-style contract priced in Chapter 3 of this thesis. Heston's framework requires two conditions to be satisfied for the model to be well posed: the Feller condition, which guarantees that the variance process remains strictly positive almost surely, and a specification of the market price of volatility risk connecting the physical and risk-neutral dynamics of the variance process. Chapter 2 of this thesis adopts Heston's two-SDE structure, a wind-speed level process and a CIR-type variance process coupled through the Feller condition, as a theoretical (not estimated) extension of the GARCH-based volatility model, for the reasons discussed under the gap statement in Section 1.3.7.

1.3.6 Weather and Energy Derivatives, and the Market Price of Risk

The applied weather-derivatives literature translates the statistical machinery reviewed above into pricing formulas for non-tradable underlyings, and it is here that the question of the market price of risk becomes unavoidable. Alaton et al. [2002] provide an accessible template: they fit an Ornstein–Uhlenbeck process with piecewise (monthly) seasonal variance to Stockholm temperature data, estimate it via martingale estimating functions applicable to discretely observed diffusions, and price temperature-index derivatives (heating- and cooling-degree-day contracts) both by an analytic approximation and by Monte Carlo simulation. Because temperature, like wind speed, is not a traded asset, the pricing measure cannot be pinned down by a replication argument in the usual Black–Scholes sense, and Alaton, Djehiche and Stillberger introduce an explicit, generally non-zero market price of risk parameter to complete the model in this incomplete-market setting.

Benth and Šaltytė Benth [2009] extend this OU-based approach specifically to wind speed, embedding a Box–Cox-transformed, seasonally adjusted daily wind speed series into a $CAR(p)$ state-space model with seasonal mean and seasonal volatility, calibrated on New York wind data using an $AR(4)$ specification with Box–Cox parameter $\lambda \in \{0.1, 0.2\}$. They derive closed-form wind futures prices as conditional expectations of the CAR state vector under a chosen pricing measure, and, notably, they show that the classical Samuelson effect, the tendency of futures price volatility to increase as the contract approaches maturity, can break down or reverse for autoregressive orders $p > 1$, a consequence of the long memory embedded in a higher-order autoregressive structure that has no analogue in the single-factor Schwartz-type model. This paper is the direct methodological template for the $CAR(4)$ construction, the Box–Cox transformation, and

the Euler-discretisation link between the discrete AR(4) and continuous-time CAR(4) representations used throughout Chapter 2 of this thesis.

The question of how to specify the market price of risk θ in this incomplete-market setting is addressed empirically by two further papers. Härdle and López Cabrera [2009] infer the market price of weather risk implicitly from Chicago Mercantile Exchange-traded Berlin cumulative-temperature futures, modelling both piecewise-constant and time-varying specifications, and find that the implied market price of risk is significantly different from zero and displays a clear seasonal pattern. Dorfleitner and Wimmer [2010] conduct a related empirical exercise on CME temperature futures directly, comparing detrended and non-detrended index models against observed market prices and showing that a simple non-detrending specification tracks quoted futures prices closely, that weather forecasts materially move prices up to eleven days ahead of settlement, and that a mismodelled seasonal trend produces systematic, exploitable mispricing across the winter and summer contract cycle. Both papers demonstrate that, where reliable option-implied or futures-implied price data exist, the market price of risk for a weather-linked underlying is empirically estimable and generally non-zero.

Alexandridis and Zapranis [2013] return to wind speed specifically and are the direct source for the two index conventions used in the pricing exercise of this thesis. They model daily wind speed with a non-parametric wavelet-network layer superimposed on a mean-reverting, seasonally adjusted Ornstein–Uhlenbeck process, benchmark it against the linear CAR-type approach of Benth and Šaltytė Benth [2009], and find that the wavelet-network specification materially improves in- and out-of-sample fit. More directly relevant to this thesis, they derive closed-form futures pricing formulas for two wind indices: the Nordix index and the Cumulative Accumulated Wind Speed (CAWS) index (referred to as “cumulative average wind speed” in that paper), the latter of which is adopted, in the normalised rolling-annual-mean form defined in Section 1.4 and Chapter 3, as the left-tail hedging instrument evaluated in this thesis. In practical terms, the instrument pays out whenever the realised cumulative average wind speed over the contract period falls short of a pre-agreed strike level, so that a sustained run of calm weather, precisely the condition that depresses a producer’s physical generation, triggers a compensating cash inflow that offsets the associated shortfall in electricity revenue.

Finally, Carr and Lee [2009] provide the theoretical foundation for variance-swap replication that motivates, without directly supplying, the implemented pricing formula used in Chapter 3 of this thesis. Their review formalises the classical result that a variance swap can be synthetically replicated by a static position in a log-contract combined with dynamic delta-hedging in the underlying, quantifies the higher-order replication error incurred when this hedge is only discretely rebalanced, and shows that the floating leg of a continuously monitored variance swap corresponds to a $1/K^2$ -weighted portfolio of out-of-the-money options across the strike surface: the same logic that underlies the construction of the VIX index.

This replication argument requires a liquid market in options on the underlying, which does not exist for German wind speed or wind generation. The only actively

traded wind derivatives, the Nordix futures referenced by Alexandridis and Zapranis [2013], are written on Norwegian rather than German wind, and no options market exists for the Nordfriesland-aligned SMARD generation series used in this thesis. In the absence of this option-implied information, the variance swap priced in Chapter 3 does not attempt Carr–Lee-style replication; instead, it follows the closed-form, model-implied formula of Tol [1997], $K_{\text{var}}^B = h \times C_0$, under the explicit benchmark assumption $\theta = 0$ described in Chapter 2.

This benchmark is adopted deliberately, as a tractable simplification given the absence of EEX option-implied data for this specific underlying, and not as a claim that the market price of risk for wind-linked instruments is in fact zero. The evidence from Härdle and López Cabrera [2009] and Dorfleitner and Wimmer [2010], reviewed above for a related but distinct underlying, would in any case not support such a claim. Standard risk-aversion arguments suggest a directional hypothesis worth stating explicitly: if wind-linked risk carries a premium comparable to that documented for temperature, producers seeking protection against low-wind revenue shortfalls would be expected to accept, and counterparties to demand, compensation above the physical-measure expectation $h \times C_0$, so that a true market-implied strike would plausibly lie *above* rather than below the $\theta = 0$ benchmark used here; this is a hypothesis about the direction of the omitted risk premium, not an estimate of its magnitude. Carr and Lee’s framework remains, in this thesis, the designated theoretical motivation for a future extension in which θ is calibrated to EEX German power options, rather than the model actually estimated.

1.3.7 Gap Statement

Read together, this literature establishes a well-developed toolkit: Weibull/lognormal marginal fitting, AR/ARFIMA mean dynamics, GARCH conditional volatility, CAR(p) continuous-time embedding, Heston-type stochastic volatility, and OU-based weather-derivatives pricing under an explicit market price of risk. Three gaps in that toolkit remain, and this thesis addresses each of them directly.

First, the wind-derivatives literature reviewed in Section 1.3.6 is calibrated either on temperature (a more heavily studied underlying with a longer exchange-traded history) or on wind speed measured at a location and height that need not correspond to the generation asset actually being hedged. Benth and Šaltytė Benth [2009] and Alexandridis and Zapranis [2013] both calibrate on New York data; neither is calibrated on wind speed that is simultaneously (i) measured at or near turbine hub height and (ii) geographically aligned with a specific, actively traded generation aggregate. This thesis addresses that gap directly by using ERA5 reanalysis wind speed at 100 m hub height over the Nordfriesland corridor, the geographic centroid of the SMARD German onshore wind generation series against which the pricing exercise in Chapter 3 is ultimately validated.

Second, no paper in this review quantifies, in the same modelling framework, the pricing cost of using a geographically or physically misaligned proxy dataset instead of an aligned one. The comparative exercises in that literature, whether across volatility

specifications (Section 1.3.3) or across long-memory orders (Section 1.3.2), are concerned with comparing model specifications on a fixed dataset, not with comparing the same model specification across two datasets of differing quality. Chapter 3’s Value of Information exercise, pricing the identical CAR(4)+GARCH variance swap on both the Bologna proxy and the Nordfriesland reference dataset and measuring the resulting gap before and after a wind-profile height correction, is designed specifically to fill this gap, and constitutes the thesis’s primary methodological contribution.

The physical reasoning behind that gap deserves stating, because it is what the height-corrected residual is designed to isolate. Nordfriesland’s open North Sea exposure brings frequent Atlantic storm passages and, at 100 m, a flow largely detached from surface obstacles. Bologna’s 10 m measurement sits inside the surface friction layer of the Po Valley, a basin enclosed by the Alps and the Apennines whose sheltered climatology suppresses both mean wind speed and the frequency of high-wind episodes. These are differences in baseline turbulence and storm climatology, not in measurement height, and a power-law profile correction rescales the level of the proxy series without reproducing them, which is why a substantial gap survives the correction in Chapter 3.

Third, a variance swap is by construction indifferent to the direction of the underlying’s movement: it pays on realised variance, and a sustained calm period, the single event a wind producer most needs protection against, registers as low variance rather than as a loss. The left tail of wind-linked revenue risk is therefore not addressed by the instrument that the literature reviewed in Section 1.3.6 prices, and although Alexandridis and Zapranis [2013] derive futures prices for the CAWS index, no study evaluates a cumulative-wind instrument as a left-tail complement to a variance swap on observations free of the serial correlation that its own rolling-window construction induces. Chapter 3 constructs and stress-tests exactly that pairing, on non-overlapping annual windows.

Two constraints bound what follows, and are stated here rather than left for the reader to infer. The stochastic-volatility literature (Section 1.3.5) generally estimates the variance-process parameters $(\kappa_V, \bar{v}, \xi, \rho)$ from either high-frequency data or option-implied information, neither of which is available here: the two datasets used in this thesis are daily, and no liquid options market exists on German wind speed or on the SMARD generation aggregate. Chapter 2 therefore develops the Heston-type stochastic-volatility system as a theoretical embedding of the GARCH proxy, verifying the Feller condition numerically and deriving the variance term structure in closed form, while explicitly deferring maximum-likelihood estimation of the full stochastic-volatility parameter set to a future extension using higher-frequency (e.g. hourly ECMWF) data. The same absence of option-implied data motivates the $\theta = 0$ benchmark of Section 1.3.6. Both constraints are limitations the thesis works within rather than gaps it closes, and both are revisited in the concluding synthesis of Chapter 3.

1.4 Data

This thesis draws on three data sources, used in complementary roles: two independent wind speed datasets (a proxy dataset on which the modelling methodology is built and validated, and a reference dataset on which the headline empirical results rest) and a market dataset of German electricity generation and prices used to construct the synthetic derivative priced in Chapter 3. All three are described in full here, so that the later chapters can refer back rather than re-explaining them.

1.4.1 Dataset A: Bologna Proxy (10 m, 2015–2018)

Dataset A consists of hourly wind speed observations for the Bologna area, Italy (approximately 44.29°N, 11.20°E), obtained via the OpenWeatherMap API and distributed through the [beniamino98/greenfin-project](#) GitHub repository, at a nominal measurement height of 10 m. The raw file (`data/data.csv`) contains 35,064 hourly observations with no missing values, running from 23:00 UTC on 31 December 2014 to 22:00 UTC on 31 December 2018. Aggregation by arithmetic mean within each UTC calendar day yields $n = 1,462$ daily average wind speed observations: 1,460 complete twenty-four-hour days spanning 1 January 2015 to 30 December 2018, together with two partially populated boundary bins, one hour on 31 December 2014 and twenty-three hours on 31 December 2018, which arise because the raw sampling window is offset by one hour from the calendar-year boundary. The effective coverage is thus the four calendar years 2015–2018, and the two boundary bins are retained rather than trimmed so that the observation count matches the frozen Phase 1 output used throughout the thesis.

Table 1 reports the descriptive statistics computed for this series. Every diagnostic points to the same conclusion: raw daily wind speed at this site is strongly non-Gaussian. The mean exceeds the median by roughly 0.23 m/s, confirming a right-skewed distribution consistent with a physical lower bound near zero and an unbounded right tail driven by occasional storm days; skewness of +1.76 and excess kurtosis of +4.74 are both far from the Gaussian values of zero, and the Jarque–Bera statistic of 2,122 against a $\chi^2(2)$ critical value near 6 rejects normality overwhelmingly.

Table 1: Descriptive statistics, raw daily average wind speed, Dataset A (Bologna, 10 m, 2015–2018).

Statistic	Value	Note
n	1,462	daily observations
Mean	2.4687 m/s	
Median	2.2417 m/s	mean > median $\Rightarrow$ right skew
Std. deviation	0.9481 m/s	
Minimum / Maximum	0.9333 / 9.1125 m/s	no calm daily averages
Skewness	+1.7616	Gaussian = 0
Excess kurtosis	+4.7351	Gaussian = 0
Jarque–Bera p -value	< 0.000001	normality strongly rejected

A Weibull distribution fit by maximum likelihood gives shape $\hat{k} = 2.6368$ and scale $\hat{c} = 2.7729$ m/s, placing the shape parameter in the range of two to three typically reported in wind-resource assessment, above the Rayleigh value of $k = 2$ that the Tuller and Brett [1984] isotropy argument reviewed in Section 1.3.1 would imply; however, on an AIC/BIC comparison against a lognormal fit ($\hat{m} = 0.8423$, $\hat{s} = 0.3405$), the lognormal wins decisively (log-likelihood $-1,730.6$ versus $-1,960.5$ for the Weibull, a gap of roughly 460 AIC units). This is a substantive departure from the Tuller and Brett [1984]–Garcia et al. [1998] consensus, and is attributable to the 10 m measurement height: the isotropy assumption behind the Rayleigh–Weibull derivation is a free-atmosphere approximation, progressively violated by surface friction and diurnal boundary-layer effects closer to the ground, and the lognormal’s thinner left tail better matches the empirical absence of near-zero daily averages at this height.

The Box–Cox transformation (Section 1.3.1) is applied to bring the series toward normality. The likelihood-maximising value is $\lambda_{\text{opt}} = -0.450$, which achieves near-perfect Gaussianisation (skewness -0.005 , excess kurtosis $+0.128$, Jarque–Bera $p = 0.606$); however, the closed-form pricing formulas used in later phases require $1/\lambda$ to be a positive integer, and -0.450 does not satisfy this constraint. Following Benth and Šaltytė Benth [2009], who adopt $\lambda \in \{0.1, 0.2\}$ for the same CAR(p) continuous-time embedding, $\lambda = 0.1$ (so that $1/\lambda = 10$) is adopted as the frozen working parameter for this dataset throughout the thesis, at the cost of leaving a residual right skew (skewness 0.601 , Jarque–Bera $p \approx 0$) that the seasonal-mean and seasonal-variance decomposition and the AR(4) fit of Chapter 2 subsequently reduce further. After the truncated Fourier decomposition into a seasonal mean $\Lambda(t)$ (annual harmonic dominant, $R^2 = 0.126$) and a seasonal variance $\sigma^2(t)$ (mean level $C_0 = 0.13138$, winter-to-summer variance ratio of approximately 6.4:1, $R^2 = 0.226$), the resulting standardised series $\tilde{W}(t)$ has mean 0.001 and standard deviation 0.967 , close to the zero-mean, unit-variance target, but retains significant serial autocorrelation and conditional heteroskedasticity (Ljung–Box $p < 0.001$ on both $\tilde{W}(t)$ and $\tilde{W}(t)^2$ at all tested lags), which motivates the AR(4) and GARCH(1,1) layers developed in Chapter 2.

Because Dataset A is recorded at 10 m rather than at a representative turbine hub height, and because its four-year window is short relative to the multi-decade samples used in the wind-derivatives literature reviewed in Section 1.3.6, it is treated throughout this thesis strictly as a *proxy*: a low-cost dataset on which every stage of the modelling pipeline is developed and validated before being applied, without re-specification, to the reference dataset described next.

1.4.2 Dataset B: Nordfriesland Reference (100 m, 2015–2024)

Dataset B consists of daily wind speed at 100 m hub height for Nordfriesland, Schleswig-Holstein, Germany (54.77°N , 8.85°E), obtained from the Open-Meteo Historical Weather API, which serves ERA5 reanalysis output. The 100 m variable (`wind_speed_100m`) is extracted directly from the reanalysis and requires no vertical-profile correction, since it is a direct model output rather than a surface-station reading extrapolated to hub

height. Two companion series are retained for validation and future regime analysis: `wind_speed_10m`, used as a consistency check against the primary series, and `wind_direction_100m`, retained for possible directional-regime analysis beyond the scope of this thesis. The sample spans ten full calendar years, 2015–2024, comprising 3,653 daily observations, which is the exact day count of that window given the three leap years it contains.

Table 2 reports the corresponding descriptive statistics. The mean wind speed of 8.07 m/s is roughly 3.3 times the Dataset A mean, consistent with the combined effect of hub height and a substantially windier geographic location; skewness (+0.585) and excess kurtosis (+0.111) are both markedly closer to Gaussian values than Dataset A’s, though the Jarque–Bera test still rejects normality at conventional levels.

Table 2: Descriptive statistics, raw daily wind speed, Dataset B (Nordfriesland, 100 m, 2015–2024).

Statistic	Value	Note
n	3,653	daily observations
Mean	8.068 m/s	
Median	7.760 m/s	mean > median $\Rightarrow$ right skew
Std. deviation	3.236 m/s	
Minimum / Maximum	1.383 / 20.258 m/s	no calm daily averages
Skewness	+0.585	Gaussian = 0; cf. +1.762 Dataset A
Excess kurtosis	+0.111	Gaussian = 0; cf. +4.735 Dataset A
Jarque–Bera p -value	< 0.001	normality rejected

Consistent with the Tuller and Brett [1984]–Garcia et al. [1998] consensus, and in contrast with Dataset A, the Weibull distribution ($\hat{k} = 2.67$, $\hat{c} = 9.09$) fits Dataset B better than the lognormal on both AIC and BIC. The Box–Cox likelihood is maximised at $\lambda_{\text{opt}} = 0.404$; the tractable value adopted for closed-form pricing is $\lambda = 0.5$ (so $1/\lambda = 2$), a much smaller rounding distance than the one forced on Dataset A, reflecting the higher hub-height mean wind speed’s less extreme skew. The Fourier seasonal-mean fit is again dominated by the annual harmonic ($R^2 = 0.103$), and the seasonal-variance fit yields a mean level $C_0 = 1.16277$ with $\sigma^2(t)$ ranging over $[0.944, 1.463]$, a winter-to-summer variance ratio of approximately 1.55:1. Expressed on the same basis, the seasonal modulation at Nordfriesland is thus far calmer than the 6.9:1 ratio found for Bologna, though both datasets place peak variance in the winter months. The resulting standardised series $\tilde{W}(t)$ again passes its basic centring and scaling checks (mean ≈ 0 , standard deviation ≈ 1.005) while retaining significant autocorrelation in levels and in squares, motivating the same AR(4) and GARCH(1,1) layers applied to Dataset A. As an illustrative application, integrating the fitted Weibull density against the power curve of a representative 3 MW turbine at 100 m hub height yields an annual energy production of 6.753 GWh and a capacity factor of 25.70%, a figure consistent with onshore wind performance in the North Sea coastal region and the figure used for the illustrative revenue calculation in Section 1.1.

Dataset B is used throughout this thesis as the *reference* dataset: it is geographi-

cally aligned with the SMARD German onshore wind generation series described next, measured directly at hub height, and long enough (ten years) to support the out-of-time validation exercise reported in Chapter 3, in which full-sample parameters are assessed against a 2020–2024 hold-out without re-calibration on the restricted window. All headline empirical claims in this thesis (the fitted variance-swap fair strike, the Value of Information estimate, and the CAWS hedging simulation) rest primarily on Dataset B, with Dataset A retained as the benchmark against which the Value of Information is measured.

1.4.3 SMARD Generation and EEX Day-Ahead Prices

The synthetic derivative priced in Chapter 3 is written on German onshore wind generation, not on wind speed directly, and requires two further data series: hourly wind onshore generation and hourly day-ahead electricity prices, both published by the German transmission system operators' joint platform SMARD (Bundesnetzagentur, smard.de) for the full 2015–2024 window. Generation is delivered across two files, `Actual_generation_1.csv` and `Actual_generation_2.csv`, and prices across `Day-ahead_prices_1.csv` and `Day-ahead_prices_2.csv`. All four files are semicolon-separated, UTF-8-with-BOM encoded, and use a comma as the thousands separator, with timestamps in the form "Jan 1, 2015 12:00 AM"; the relevant generation column is labelled `Wind onshore [MWh]` Calculated resolutions.

The day-ahead price series requires a merge across two market conventions, reflecting a structural break in the underlying market: prior to 1 October 2018, Germany, Austria, and Luxembourg formed a single day-ahead bidding zone, reported in the column `DE/AT/LU [€/MWh]`; from that date, Austria was separated into its own bidding zone following a market-coupling reform, and the German price is reported instead in the column `Germany/Luxembourg [€/MWh]`. The merge rule applied throughout this thesis takes the `Germany/Luxembourg` column from 1 October 2018 onward and the `DE/AT/LU` column before that date. Both series are aggregated from hourly to daily frequency: generation as the sum of the 24 hourly values (a flow quantity, MWh/day), price as the arithmetic mean of the 24 hourly values (a level quantity, €/MWh). Two filters are applied rather than imputing: days exhibiting more than 20% missing hourly observations are dropped, as are days recording zero aggregate generation, which are treated as reporting errors rather than as genuine calm. Neither filter removed any day from the 2015–2024 generation series, which is complete.

The resulting daily series (`smard_daily_phase6.csv`) covers 1 January 2015 to 31 December 2024, 3,653 daily generation observations, with a mean national wind onshore generation of 255,560 MWh/day (computed directly from the processed file). The day-ahead price series is four days shorter, beginning on 5 January 2015, as the SMARD price files carry no quotation for the first four days of the sample; price-dependent calculations therefore rest on 3,649 daily observations. The corresponding day-ahead price series has a markedly different character before and after the onset of the European energy-price crisis: the mean price over 2015–2020 is €34.60/MWh, versus

€126.47/MWh over 2021–2024 and €71.38/MWh over the full ten-year sample. This is the same three-way split used for the illustrative revenue calculation in Section 1.1. This regime shift also delimits the practical scope of a purely volumetric hedge: because the wind-speed-based instruments priced in Chapter 3 respond only to the physical quantity of wind, not to the price level, they cannot on their own offset the price-driven component of revenue risk that dominated the 2021–2024 window, and a producer holding only this hedge would have remained fully exposed to the energy-price crisis itself. This structural shift is a feature of the German electricity market over the sample period rather than a data artefact, and Chapter 3 checks explicitly that the merit-order relationship between generation and price (used to motivate wind derivatives as a producer hedge) is not an artefact of the pre/post-2018 bidding-zone split either, by comparing the correlation between generation and price separately on either side of the October 2018 structural break.

1.4.4 Cross-Dataset Comparison

Table 3 summarises the two wind speed datasets side by side. It is presented once here and referred back to whenever a later result is reported for both datasets.

Table 3: Cross-dataset comparison: Dataset A (proxy) versus Dataset B (reference).

	Dataset A	Dataset B
Source	OpenWeatherMap API (beniamino98/greenfin-project)	Open-Meteo Historical Weather API, ERA5 reanalysis
Location	Bologna, Italy (44.29°N, 11.2°E)	Nordfriesland, Germany (54.77°N, 8.85°E)
Height	10 m	100 m
Period	2015–2018 (4 yr, 1,462 obs)	2015–2024 (10 yr, 3,653 obs)
Primary variables	wind_speed (m/s)	wind_speed_100m, wind_speed_10m, wind_direction_100m
Advantages	Freely available; easy to download; sufficient for methodology development and validation	Hub-height (no profile correction needed); geographically aligned with SMARD generation; 10-year window enables out-of-sample validation; ERA5 reanalysis quality
Limitations	Proxy-quality, with less transparent provenance than a reanalysis product; 10 m height, for which the power-law profile correction recovers only a small part of the gap to hub height; low-wind site; short (4-year) window	Model-based reanalysis rather than direct anemometer observation; no sub-daily resolution used in this thesis

One derived series used in Chapter 3 is constructed from the two wind datasets documented above and is defined here alongside them. The Cumulative Accumulated

Wind Speed index is taken in normalised rolling-annual-mean form,

$$\text{CAWS}_t = \frac{1}{252} \sum_{s=t-251}^t W(s), \quad (1)$$

where $W(s)$ is the daily mean wind speed in m/s: `wind_speed` for Dataset A and `wind_speed_100m` for Dataset B. Normalising by the 252-day window rather than accumulating raw sums keeps the index in m/s and therefore comparable across the two measurement heights, at mean levels of roughly 2.47 m/s for Bologna and 8.07 m/s for Nordfriesland. The strike K_{CAWS} is the full-sample mean of CAWS_t , and the instrument pays $N \times (K_{\text{CAWS}} - \text{CAWS}_t)$, so that positive payoffs coincide with sustained low-wind conditions. As noted in Section 1.2, the risk metrics for this instrument are computed on non-overlapping annual observations, four for Dataset A and ten for Dataset B, even though the rolling series of equation (1) is what the time-series plots display.

The role of the cross-dataset comparison is not merely descriptive. It is the empirical premise for the Value of Information argument developed in Chapter 3: because the two datasets are processed through an *identical* modelling pipeline, any difference in the resulting derivative price can be attributed to the data themselves (their geographic location, height, and sample length) rather than to a difference in model specification. Table 3 is therefore the reference point against which that argument is built, motivating the upgrade from Dataset A to Dataset B that this thesis both executes and quantifies.

1.5 Thesis Outline

The remainder of the thesis is organised into two further chapters, followed by a concluding synthesis that closes Chapter 3 rather than standing as a separate chapter.

Chapter 2, *Methodology and Empirical Wind-Speed Modelling*, develops the modelling pipeline whose literature foundations are reviewed in Section 1.3, phase by phase, presenting Dataset A and Dataset B side by side at each stage: deseasonalisation and the Box–Cox transformation; the AR(4) conditional-mean model and its long-memory diagnostics; the GARCH(1,1) conditional-volatility layer; the continuous-time CAR(4) embedding, including its companion matrix, stationarity properties, and Normal-Inverse-Gaussian tail extension; and, finally, the theoretical stochastic-volatility extension that embeds the GARCH proxy into a Heston-type two-SDE system under the $\theta = 0$ benchmark. Dataset B is treated as the primary result throughout, with Dataset A retained as the validation benchmark on which each modelling choice is first tested.

Chapter 3, *Derivatives Pricing, Risk Management and Conclusions*, applies the calibrated model to answer the three research questions posed in Section 1.2.

It first prices the synthetic wind variance swap on both datasets, distinguishing the empirical Track A fair strike, the 252×-annualised realised variance of SMARD generation log-returns, from the model-implied Track B fair strike $K_{\text{var}}^B = h \times C_0$, the closed-form expression derived under the piecewise-constant-volatility convention of Tol [1997]. Track A is a realised quantity rather than an implied one: nothing in it is backed

out of a traded price, which is precisely the information this thesis argues is unavailable for this underlying. The two tracks are computed under different annualisation and unit conventions: Track A is annualised and expressed in the units of generation log-returns, while Track B is left un-annualised in the Box–Cox-transformed units in which the seasonal variance was fitted. Their raw numerical gap, two to three orders of magnitude depending on the dataset, is an artefact of those conventions rather than a measure of model error. Chapter 3 reconciles them, and shows that the delta-method rescaling recorded in the phase outputs addresses only one of the four conventions that separate them.

It then evaluates the forecasting models of Chapter 2 against the persistence benchmark using Diebold–Mariano tests [Diebold and Mariano, 1995, Harvey et al., 1997]. The result is a binary split rather than a graded hierarchy: every fitted specification, from AR(4) through CAR(4) with Normal-Inverse-Gaussian innovations, improves on persistence at any conventional significance level, while being observationally equivalent to the others in one-step point-forecast terms. That equivalence is itself informative. It establishes that the models are distinguished not by conditional-mean accuracy but by their conditional-variance and tail representations, which is precisely why the thesis evaluates them on pricing and hedging criteria rather than on root mean squared error alone.

Chapter 3 then computes the Value of Information from the Dataset A-to-B upgrade, both before and after a wind-profile height correction restricted strictly to that comparison, and evaluates the CAWS instrument of Alexandridis and Zapranis [2013] as a left-tail complement to the variance swap on the non-overlapping annual basis defined in Section 1.4.4. It closes with a synthesis section that states the limitations of the $\theta = 0$ benchmark and of the daily-frequency stochastic-volatility identification problem, and outlines the option-implied and higher-frequency extensions left to future work.

Chapter 2 — Methodology and Empirical Wind-Speed Modelling

This chapter develops the wind-speed modelling pipeline whose motivation and literature grounding were set out in Chapter 1: a five-stage sequence running from deseasonalisation through discrete-time autoregression and conditional volatility to a continuous-time embedding and, finally, a theoretical stochastic-volatility extension. Each stage is applied identically to both datasets described in Section 1.4: Dataset A (Bologna, 10 m, 2015–2018), on which the pipeline is developed and validated, and Dataset B (Nordfriesland, 100 m, 2015–2024), which carries the thesis’s headline results. Each section below presents the two datasets side by side, so that methodological choices can be compared directly against the data they are applied to. Numerical results quoted throughout are frozen outputs of the underlying phase notebooks; none are re-estimated here.

2.1 Deseasonalisation and the Box–Cox Transformation

Raw daily wind speed is a positive, right-skewed, seasonally heteroskedastic quantity, and every subsequent stage of this pipeline (the AR(4) mean model of Section 2.2, the GARCH(1,1) volatility layer of Section 2.3, and the CAR(4) continuous-time embedding of Section 2.4) presupposes a series that is approximately Gaussian, zero-mean, and unit-variance. The first stage of the pipeline is therefore a sequence of four operations applied to raw wind speed $W(t)$: characterising its marginal distribution, applying a Box–Cox power transformation, removing a deterministic seasonal mean $\Lambda(t)$ via a truncated Fourier series, and removing a deterministic seasonal variance $\sigma^2(t)$ via a second Fourier fit. The output is the standardised series $\tilde{W}(t) = [W^{(\lambda)}(t) - \Lambda(t)]/\sigma(t)$, which enters the AR(4) model of Section 2.2.

2.1.1 Marginal Distribution

The two marginal distributions fitted throughout this section are the Weibull and the lognormal,

$$f_W(v; k, c) = \frac{k}{c} \left(\frac{v}{c}\right)^{k-1} \exp \left[-\left(\frac{v}{c}\right)^k \right], \quad v \geq 0, \quad (2)$$

$$f_L(v; m, s) = \frac{1}{vs\sqrt{2\pi}} \exp \left[-\frac{(\ln v - m)^2}{2s^2} \right], \quad v > 0, \quad (3)$$

both fitted by maximum likelihood and compared by Akaike information criterion. Table 4 reports the resulting fits for both datasets. Tuller and Brett [1984] provide the physical justification for the Weibull distribution reviewed in Section 1.3.1: under approximate bivariate-normal isotropy of the horizontal wind-velocity components, wind speed is Weibull-distributed with shape parameter near two. Dataset B, measured at 100 m in the free atmosphere above the surface boundary layer, conforms to this

picture: the Weibull distribution ($\hat{k} = 2.67$) beats the lognormal alternative on both AIC and BIC, consistent with Tuller and Brett [1984] and with the case-study evidence of Garcia et al. [1998]. Dataset A does not: at 10 m, where surface friction and diurnal boundary-layer cycling distort the isotropy assumption underlying the Weibull derivation, the lognormal distribution wins decisively (log-likelihood $-1,730.6$ versus $-1,960.5$ for the Weibull MLE fit, $\Delta\text{AIC} \approx 460$). This is the same qualitative finding already noted in Section 1.4.1; it is repeated here because it is the reason the two datasets receive different treatment in the Box–Cox step that follows.

Table 4: Marginal distribution fits, both datasets (maximum likelihood).

	Dataset A	Dataset B
Weibull ($\hat{k}, \hat{c}$)	2.6368, 2.7729 m/s	2.67, 9.09 m/s
Lognormal ($\hat{m}, \hat{s}$)	0.8423, 0.3405	2.00, 0.43
Log-likelihood, Weibull	$-1,960.51$	$-9,377.15^*$
Log-likelihood, Lognormal	$-1,730.56$	$-9,399.55^*$
AIC, Weibull / Lognormal	3,925.01 / 3,465.12	18,758.3 / 18,803.1
Better fit	Lognormal ($\Delta\text{AIC} \approx 460$)	Weibull ($\Delta\text{AIC} \approx 45$)

*Dataset A’s log-likelihoods are reported directly in the source analysis. Dataset B’s report tabulates AIC and BIC only; the values shown are recovered as $\ell = (2k - \text{AIC})/2$ with $k = 2$ free parameters in each family.

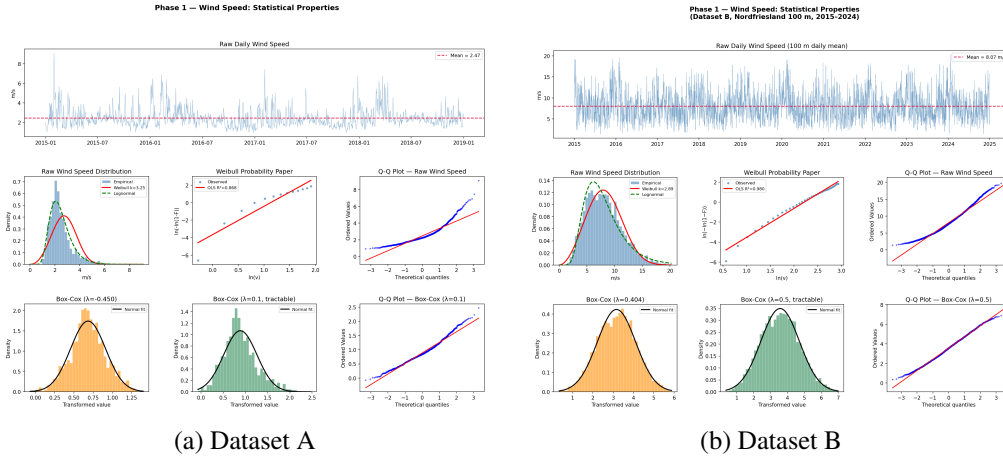


Figure 1: Raw wind speed distribution and Box–Cox transformation, both datasets. Each panel is arranged in four rows, read top to bottom. **Row 1:** the raw daily wind speed time series, with the sample mean marked by a dashed line. **Row 2:** three diagnostics for the raw series – a histogram with the fitted Weibull and lognormal densities overlaid, a Weibull probability-paper linearisation, and a quantile–quantile plot against the normal distribution, which shows the characteristic S-curve deviation of a right-skewed, heavy-tailed series. **Row 3:** histogram and quantile–quantile plot of the series after the likelihood-optimal Box–Cox transformation λ_{opt} . **Row 4:** histogram and quantile–quantile plot after the tractable Box–Cox value actually adopted ($\lambda = 0.1$ for Dataset A, $\lambda = 0.5$ for Dataset B), showing the residual departure from normality left by the tractability constraint discussed in Section 2.1.2.

2.1.2 Box–Cox Transformation

The Box–Cox family $W^{(\lambda)}(t) = [W(t)^\lambda - 1]/\lambda$ (with $\ln W(t)$ for $\lambda = 0$) is fitted by maximising the profile log-likelihood under a normality assumption. For Dataset A, the likelihood-maximising value is $\lambda_{\text{opt}} = -0.450$, achieving near-perfect Gaussianisation (skewness -0.005 , excess kurtosis $+0.128$, Jarque–Bera $p = 0.606$). For Dataset B, the optimum is $\lambda_{\text{opt}} = 0.404$, a much less extreme transformation, consistent with the higher hub-height mean wind speed producing a less severely skewed raw distribution. In both cases, the pricing formulas developed in Section 2.4 and used throughout Chapter 3 require $1/\lambda$ to be a positive integer, so the likelihood optimum cannot be used directly. The requirement that $1/\lambda \in \mathbb{Z}^+$ is dictated by the analytical pricing formulas of Chapter 3. Pricing a forward contract requires computing the expected future wind speed in its original physical measure. Inverting the Box–Cox transformation requires taking the $(1/\lambda)$ -th moment of a normal random variable, denoted $M_{1/\lambda}$ [Benth and Šaltytė Benth, 2009]. If $1/\lambda$ were fractional, this expectation would lack a closed-form solution and risk generating complex numbers from negative intermediate states, destroying the model’s analytical tractability. Following Benth and Šaltytė Benth [2009], who adopt $\lambda \in \{0.1, 0.2\}$ for the same CAR(p) continuous-time embedding, $\lambda = 0.1$ is adopted for Dataset A and $\lambda = 0.5$ for Dataset B (Table 5). The rounding cost is asymmetric: Dataset A’s tractable value sits 0.35 units from its optimum and leaves a residual right skew (skewness 0.601, Jarque–Bera $p \approx 0$), while Dataset B’s tractable value sits only 0.10 units from its optimum and retains near-Gaussian shape (skewness 0.081, KS $p = 0.091$).

Table 5: Box–Cox transformation: optimal versus tractable λ .

	Dataset A	Dataset B
λ_{opt} (likelihood-maximising)	-0.450	0.404
Gaussianisation at λ_{opt} (skew, JB p)	$-0.005, 0.606$	$-0.020, 0.0002$
λ adopted (tractable, $1/\lambda \in \mathbb{Z}^+$)	0.1 ($1/\lambda = 10$)	0.5 ($1/\lambda = 2$)
Residual shape at adopted λ (skew, JB p)	$0.601, \approx 0$	$0.081, < 0.001$

2.1.3 Seasonal Mean and Seasonal Variance

Both the seasonal mean $\Lambda(t)$ and the seasonal variance $\sigma^2(t)$ are fitted as two-harmonic truncated Fourier series in calendar day-of-year, $\Lambda(t) = a_0 + \sum_{k=1}^2 [a_k \cos(2\pi kt/365) + b_k \sin(2\pi kt/365)]$ by ordinary least squares, and $\sigma^2(t) = c_0 + \sum_{k=1}^2 c_k \cos(2\pi kt/365)$ (cosine-only, reflecting the approximate symmetry of the variance cycle about its annual peak) fitted to the per-calendar-day empirical variance of the residuals $\varepsilon(t) = W^{(\lambda)}(t) - \Lambda(t)$. Table 6 reports both fits. The annual harmonic dominates the seasonal mean in both datasets (b_1 significant, a_1 not, for Dataset A; a_1 significant, b_2 not, for Dataset B), and the fitted R^2 is modest in both cases (0.126 and 0.103): most of the variance in daily wind speed is weather-driven, not calendar-driven, a finding consistent across both the four-year proxy sample and the ten-year reference sample. The seasonal-variance fit

is more informative for the purposes of this thesis, since its intercept C_0 is the Fourier constant that enters the variance-swap pricing formula $K_{\text{var}}^B = h \times C_0$ of Tol [1997], used throughout Chapter 3: $C_0 = 0.13138$ for Dataset A versus $C_0 = 1.16277$ for Dataset B, an 8.85-fold gap that Chapter 3’s Value-of-Information analysis identifies as the dominant driver of the pricing error attributable to using the geographic proxy. Both datasets place peak seasonal variance in winter, though the relative modulation differs sharply: Dataset A’s fitted $\sigma^2(t)$ ranges over $[0.032, 0.224]$, a winter-to-summer ratio of approximately 6.9 to 1, versus Dataset B’s much milder range of $[0.944, 1.463]$, a ratio of approximately 1.55 to 1.

Table 6: Seasonal mean and seasonal variance Fourier fits, both datasets.

	Dataset A	Dataset B
<i>Seasonal mean $\Lambda(t)$</i>		
a_0	0.88513 (***)	3.564 (***)
a_1 (annual cos)	−0.00544 (n.s.)	0.510 (***)
b_1 (annual sin)	0.16381 (***)	0.083 (**)
R^2	0.126	0.103
<i>Seasonal variance $\sigma^2(t)$</i>		
C_0 (mean level)	0.13138 (***)	1.16277 (***)
c_1 (annual cos)	0.09579 (***)	0.25933 (***)
c_2 (semi-annual cos)	−0.00332 (n.s.)	0.04077 (n.s.)
R^2	0.226	0.116
$\sigma^2(t)$ range	$[0.032, 0.224]$	$[0.944, 1.463]$

2.1.4 The Standardised Series $\tilde{W}(t)$

The standardised series $\tilde{W}(t) = \varepsilon(t)/\sigma(t)$ is, by construction, close to zero-mean and unit-variance in both datasets (Dataset A: mean 0.001, std. 0.967; Dataset B: mean −0.00003, std. 1.005), confirming that the centring and scaling steps were executed correctly. Neither series is yet white noise, however. The Ljung–Box statistic tests the joint null hypothesis that the first m autocorrelations of a series are zero,

$$LBQ(m) = n(n+2) \sum_{k=1}^m \frac{\hat{\rho}(k)^2}{n-k} \sim \chi^2(m) \quad \text{under } H_0, \quad (4)$$

where $\hat{\rho}(k)$ is the sample autocorrelation at lag k ; applied to a series’ squared values, the same statistic tests for conditional heteroskedasticity rather than serial correlation in levels. The test rejects the null of no serial correlation at every tested lag (5, 10, 20) for $\tilde{W}(t)$ itself in both datasets ($p < 0.001$), and the same test applied to $\tilde{W}(t)^2$ also rejects at every lag in both datasets ($p < 0.001$), indicating conditional heteroskedasticity beyond the deterministic seasonal variance already removed. These two findings are the direct motivation for the AR(4) mean model of Section 2.2 and the GARCH(1,1) volatility model of Section 2.3 respectively, and the pattern is identical across both datasets despite their very different sample lengths and measurement heights. This is evidence that the residual dynamics being modelled are a genuine feature of daily wind

speed, not an artefact of one particular site or sample.

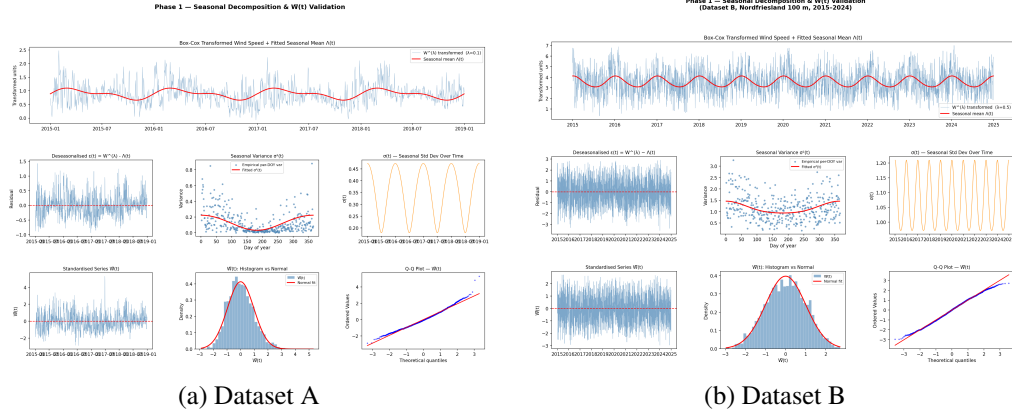


Figure 2: Seasonal decomposition and standardisation, both datasets. Each panel is arranged in three rows. **Row 1:** the transformed wind speed $W^{(\lambda)}(t)$ with the fitted seasonal mean $\Lambda(t)$ overlaid. **Row 2:** three panels – the deseasonalised residuals $\varepsilon(t) = W^{(\lambda)}(t) - \Lambda(t)$; the empirical per-calendar-day variance (points) with the fitted Fourier seasonal-variance curve $\sigma^2(t)$ (line), showing the winter peak / summer trough pattern common to both datasets; and, third, an autocorrelation plot of the seasonal standard deviation curve across the calendar year, a supplementary panel not otherwise discussed in the main text. **Row 3:** the standardised series $\tilde{W}(t)$, its histogram against a normal density, and a quantile–quantile plot, illustrating the residual serial dependence and non-normality quantified in Section 2.1.4.

2.2 Conditional Mean: AR(4)

The Ljung–Box evidence of Section 2.1.4 establishes that $\tilde{W}(t)$ retains serial dependence. This section fits an autoregressive model to capture it.

2.2.1 Order Selection

Both datasets exhibit a similar autocorrelation profile: a partial autocorrelation function (PACF) with clearly significant spikes at lags 1–3 and a markedly weaker, borderline value at lag 4 (Dataset A: $\text{PACF}_4 = 0.0311$, just inside the 95% band; Dataset B: $\text{PACF}_4 = -0.0070$, effectively zero), with no significant structure beyond lag 4 in either series. An AIC/BIC grid search over $\text{AR}(p)$, $p = 1, \dots, 6$, in fact selects AR(3) as the minimum-AIC model for *both* datasets (Dataset A: $\text{AIC} = 3,432.24$ at $p = 3$ versus 3,432.83 at $p = 4$, a difference of 0.59 units; Dataset B: $\text{AIC} = 9,342.24$ at $p = 3$ versus 9,344.04 at $p = 4$, a difference of 1.8 units), and in both cases the gap falls within the conventional two-point AIC band below which two models are considered statistically indistinguishable.

AR(4) is nonetheless retained as the operating specification in both datasets, on structural rather than statistical grounds: the continuous-time CAR(4) embedding of Section 2.4 requires a companion matrix of dimension four, and the Euler-discretisation mapping from discrete AR coefficients $\{\beta_1, \dots, \beta_4\}$ to continuous CAR coefficients

$\{\alpha_1, \dots, \alpha_4\}$, following Benth and Šaltytė Benth [2009], does not generalise to a three-lag or mixed ARMA specification. This is a deliberate trade-off, stated candidly here rather than left implicit: a fractionally-smaller-AIC three-lag model is set aside in favour of the four-lag model that the pricing framework of Chapter 3 actually requires, on the grounds that the statistical cost of doing so is, in both datasets, within the range normally treated as noise.

2.2.2 AR(4) Coefficients

Table 7 reports the fitted AR(4) coefficients for both datasets, estimated on the zero-mean series $\tilde{W}(t)$ without an intercept. Both datasets show the coefficient magnitude declining with lag order and a positive, dominant β_1 , but the two series differ in the sign of β_4 (Dataset A: +0.0310; Dataset B: −0.0074) and in the statistical significance of the higher lags: Dataset A’s β_2 and β_3 are both significant at the 1% level and β_4 is not significant ($p = 0.208$), while Dataset B’s β_3 is only marginally significant ($p = 0.090$) and β_4 is clearly insignificant ($p = 0.660$). Neither discrepancy affects the modelling decision, since AR(4) is retained structurally regardless of the marginal significance of its highest lag, but both are reported here rather than smoothed over, since a coefficient that changes sign between the two datasets is a legitimate empirical finding, not a data-quality concern.

Table 7: AR(4) coefficients, both datasets (OLS/MLE on $\tilde{W}(t)$, no intercept).

Coefficient	Dataset A		Dataset B	
	Estimate	p -value	Estimate	p -value
β_1	0.6299	< 0.001	0.5324	< 0.001
β_2	−0.1443	< 0.001	−0.0763	< 0.001
β_3	0.0926	0.001	0.0324	0.090
β_4	0.0310	0.208	−0.0074	0.660

Both coefficient sets can be read against the New York daily wind-speed AR(4) fit reported by Alexandridis and Zapranis [2013]: $\beta_1 = 0.362$, $\beta_2 = -0.100$, $\beta_3 = 0.027$, $\beta_4 = 0.022$. Dataset A’s coefficients are larger in magnitude than the New York benchmark at every lag. Dataset B’s are larger at β_1 (0.532 versus 0.362) and β_3 (0.032 versus 0.027), but smaller at β_2 (0.076 versus 0.100 in magnitude) and β_4 (0.007 versus 0.022), so the comparison holds cleanly for the proxy dataset but only partially for the reference dataset. The dominant lag-1 coefficient, which carries most of the autoregressive structure in both series, is larger at both European sites than at the New York site, and this is attributed, in the underlying phase analysis, to a stronger lag-1 sample autocorrelation at both European sites (0.579 for Dataset A) than at the New York site (≈ 0.36), consistent with the sites’ differing synoptic exposure rather than with any difference in modelling approach: the AR(4) specification itself is identical to the one used in the benchmark study.

For Dataset A, the characteristic roots of the fitted AR(4) polynomial have moduli $\{0.702, 0.470, 0.470, 0.200\}$, all inside the unit circle, confirming stationarity; the dom-

inant root implies a mean-reversion half-life for the level of wind-speed anomalies of approximately 2.0 days. This is a materially faster reversion than the volatility half-life derived from the GARCH model in Section 2.3 (41.6 and 38.3 days for Datasets A and B respectively): the *level* of a wind-speed anomaly washes out within days, while the *volatility regime* it occurred in persists for weeks, a distinction that recurs when the two dynamics are combined in Section 2.4.

2.2.3 Residual Diagnostics

Table 8 summarises the Ljung–Box tests on the AR(4) residuals $\hat{\varepsilon}(t)$ and on their squares. In both datasets, the serial-correlation test on levels passes comfortably at every lag ($p > 0.5$ throughout), confirming that AR(4) has removed the linear dependence identified in Section 2.1.4. In both datasets, the test on squared residuals rejects the null of no remaining structure at conventional significance. This is corroborated by the Engle [1982] Lagrange-multiplier test for autoregressive conditional heteroskedasticity, which regresses squared residuals on their own lags, $\hat{\varepsilon}(t)^2 = \gamma_0 + \sum_{k=1}^q \gamma_k \hat{\varepsilon}(t-k)^2 + u(t)$, and tests the joint significance of $\gamma_1, \dots, \gamma_q$ via $\text{LM}(q) = nR^2 \sim \chi^2(q)$ under the null of no ARCH effects; the entry-residual ARCH-LM test is significant in both datasets (Dataset A: $\text{LM}(5) = 18.80$, $p = 0.0021$; Dataset B: $\text{LM}(5) = 16.58$, $p = 0.0054$). This is the expected and desired outcome at this stage: the AR(4) mean equation is adequate, and the conditional heteroskedasticity that remains is precisely what the GARCH(1,1) layer of Section 2.3 is designed to capture.

Table 8: AR(4) residual diagnostics, both datasets.

Test	Dataset A	Dataset B
Ljung–Box $\hat{\varepsilon}(t)$, lag 20 (p)	0.506	0.511
Ljung–Box $\hat{\varepsilon}(t)^2$, lag 20 (p)	0.0033	0.2724*
ARCH-LM(5) on entry residuals (p)	0.0021	0.0054

analysis is internally inconsistent on its squared-residual diagnostics: the narrative text reports that the Ljung–Box test rejects homoskedasticity at lags 5 and 10 ($p = 0.0048$ and 0.0419), while the lag-20 value tabulated there is 0.2724 (a pass). The cell above reports the lag-20 value from the source table. Note that lags 5 and 10 rejections are the substantive finding motivating GARCH, regardless of the lag-20 outcome, and the ARCH-LM entry-residuals test (final row) is independently significant in both datasets.

2.2.4 Forecast Accuracy versus Persistence

Table 9 reports one-step-ahead out-of-sample forecast accuracy against a persistence benchmark $\hat{\tilde{W}}(t) = \tilde{W}(t-1)$, using a walk-forward evaluation held out from the estimation sample. AR(4) reduces RMSE relative to persistence by 12.2% in Dataset A and 13.8% in Dataset B, and a Diebold–Mariano test [Diebold and Mariano, 1995] rejects the null of equal predictive accuracy in both cases (Dataset A: $DM = -3.681$, $p = 0.0002$; Dataset B: $DM = -7.019$, $p < 0.001$). These are the frozen benchmark figures for the two datasets and are reused, without re-estimation, in the cross-dataset validation exercise of Chapter 3; a separate, later validation split reported there for Dataset B under

the same Diebold–Mariano procedure (Section 3.2) uses a different train/test partition and yields a similar but not numerically identical improvement (13.6%), a difference attributable entirely to the split rather than to any change in the fitted model, and reported explicitly as such where it recurs.

Two features of the Diebold–Mariano statistics themselves should be noted, since the figures quoted here differ from those reported for the same test in Chapter 3. The first is a sign convention. The evaluation used here defines the loss differential as $d_t = e_{\text{AR}(4)}^2(t) - e_{\text{persistence}}^2(t)$, so that a model outperforming the benchmark produces a *negative* statistic; the Chapter 3 evaluation reverses the ordering, differencing the benchmark’s squared error first, so the same outcome appears with a positive sign. The two are the same test reaching the same conclusion, and both reject the null at any conventional level. The second is magnitude: the statistics reported there are larger in absolute value (8.26 and 16.76, against 3.681 and 7.019 here) because they are computed on the full standardised series rather than on the held-out partition used in this section. A Diebold–Mariano statistic scales with the square root of the evaluation sample, and the ratio of sample sizes alone accounts for most of what is observed in both datasets: $\sqrt{1462/220} \approx 2.6$ against an observed ratio of 2.2 for Dataset A, and $\sqrt{3653/548} \approx 2.6$ against an observed 2.4 for Dataset B.

One further property of the mean model is worth recording here, where the conditional mean is estimated, rather than leaving it to emerge in Chapter 3. The AR(4)+GARCH, CAR(4)+Gaussian and CAR(4)+NIG specifications developed in the following sections all inherit the AR(4) conditional mean unchanged: the GARCH layer of Section 2.3 models the conditional variance of the residuals, and the CAR(4) embedding of Section 2.4 is an exact continuous-time representation of the same mean dynamics, so none of the three alters the one-step-ahead point forecast. The four specifications are therefore observationally equivalent at a one-day horizon by construction, and differ only in their conditional-variance and tail representations. This is why the cross-dataset validation of Chapter 3 separates them on pricing and hedging criteria rather than on point-forecast accuracy, where no separation is available in principle.

Table 9: AR(4) vs. persistence, one-step-ahead out-of-sample RMSE.

	Dataset A		Dataset B	
	RMSE	MAE	RMSE	MAE
Persistence	0.6856	0.5214	0.9955	0.8054
AR(4)	0.6017	0.4706	0.8580	0.6950
Improvement	12.2%	—	13.8%	—

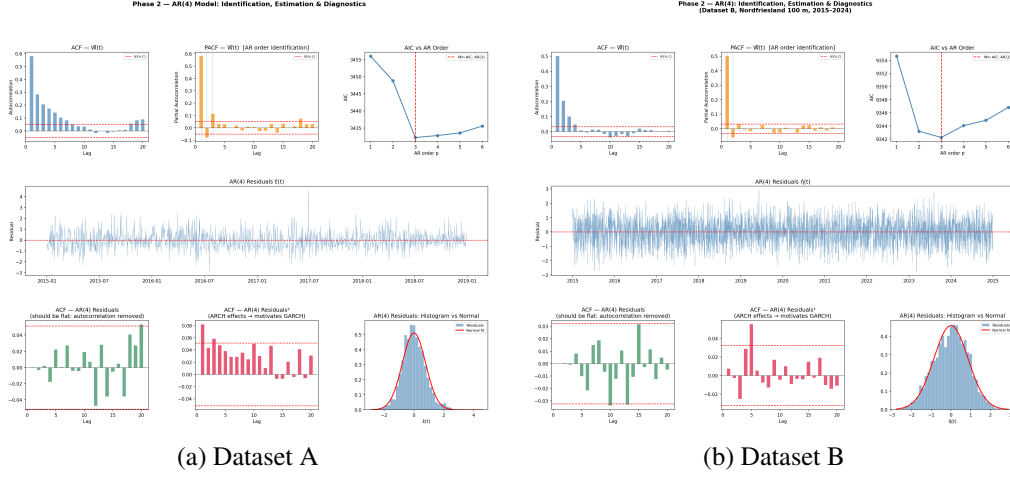


Figure 3: AR(4) identification and residual diagnostics, both datasets. Each panel is arranged in three rows. **Row 1:** three panels – the autocorrelation function of $\tilde{W}(t)$; its partial autocorrelation function, showing the decline to a borderline value at lag 4 discussed in Section 2.2.1; and the AIC grid search across candidate AR orders. **Row 2:** the time series of the AR(4) residuals $\hat{\varepsilon}(t)$. **Row 3:** three further panels, confirming that AR(4) has removed the linear serial dependence identified in Section 2.1.4 while conditional heteroskedasticity remains – an autocorrelation-type diagnostic of $\hat{\varepsilon}(t)$; the corresponding diagnostic applied to $\hat{\varepsilon}(t)^2$, which stays visibly elevated and motivates the GARCH model of Section 2.3; and a histogram of $\hat{\varepsilon}(t)$ against a normal density.

2.2.5 Long Memory

A natural question, given the literature reviewed in Section 1.3.2, is whether the residual dependence in $\tilde{W}(t)$ decays exponentially, as a finite-order AR model assumes, or hyperbolically, as fractional integration would imply. The Geweke–Porter–Hudak (GPH) log-periodogram estimator regresses the log-periodogram of the series on the log of the frequency, $\ln I(\omega_j) = c - d \ln [4 \sin^2(\omega_j/2)] + u_j$, over the $m = n^{0.6}$ lowest Fourier frequencies ω_j , and tests $H_0 : d = 0$ against fractional integration $d \neq 0$. It gives a fractional-differencing parameter of $\hat{d} = 0.119$ ($p = 0.208$) for Dataset A and $\hat{d} = -0.115$ ($p = 0.094$) for Dataset B. The interpretation of the two results is not symmetric, and the sign of the second matters. Long memory in the sense at issue here requires $0 < d < 0.5$; a negative $\hat{d}$ indicates anti-persistence, a hyperbolically decaying but sign-alternating dependence structure that is the opposite of long memory rather than a weak form of it. Dataset A therefore shows no long memory because it cannot reject $d = 0$ at all, and Dataset B shows none either, because although its p -value falls marginally inside a $p \leq 0.10$ threshold, its point estimate carries the wrong sign for the hypothesis in question. Neither dataset provides evidence for fractional integration of the long-memory type at daily frequency.¹ This reading is consistent with the wider

¹The Dataset B source analysis reports an ARFIMA(4, d , 0) robustness fit alongside the GPH test, using $d = 0.010$ rather than the estimated $\hat{d} = -0.115$. The two are reconciled by the estimator’s own admissibility constraint: the fitted value is passed through a clamp of the form $d = \max(0.01, \min(\hat{d}, 0.49))$ before the ARFIMA fit, so a negative estimate is floored at 0.01 rather than used directly. The ARFIMA specification is consequently fitted at a value close to $d = 0$, and its reported AIC advantage over the AR(4) benchmark cannot be read as evidence of long memory, since the fractional-differencing parameter it uses was not the one the data produced. The same source document’s Hurst-exponent figure caption states that

literature: Haslett and Raftery [1989] and Kavasseri and Seetharaman [2009] both find long memory to be primarily a feature of hourly or sub-daily wind data, so its absence after daily aggregation at both sites is the expected outcome rather than an anomaly. AR(4) is retained as the operating mean model for both datasets, now supported by the long-memory evidence rather than merely undisturbed by it, and for the structural reason given in Section 2.2.1.

2.3 Conditional Volatility: GARCH(1,1)

Section 2.2.3 established that AR(4) residuals remain conditionally heteroskedastic in both datasets. This section models that heteroskedasticity directly, completing the two-component variance decomposition $\sigma_{\text{total}}^2(t) = \sigma_{\text{seasonal}}^2(t) \times h(t)$ that combines the deterministic Fourier seasonal variance of Section 2.1.3 with a GARCH(1, 1) conditional-variance multiplier $h(t)$, following Tol [1997] and Bollerslev [1986].

2.3.1 GARCH(1,1) Estimates

Table 10 reports the fitted GARCH(1,1) parameters for both datasets. In both cases β dominates α by more than an order of magnitude (Dataset A: $\beta/\alpha \approx 31$; Dataset B: $\beta/\alpha \approx 165$), meaning that conditional variance is overwhelmingly carried forward from its own past level rather than driven by fresh shocks. Tol [1997] characterises this pattern as typical of daily meteorological series, where volatility is governed by slow-moving synoptic weather regimes rather than the sharp, episodic jumps observed in financial return data. Dataset B's β is both larger and more precisely estimated than Dataset A's, while its α is smaller and, unlike Dataset A's, not statistically significant on its own ($p = 0.210$); the constant ω is likewise insignificant for Dataset B ($p = 0.348$) but marginally significant for Dataset A ($p = 0.060$). Neither insignificant coefficient is dropped from either model, since the two-component variance decomposition requires all three GARCH parameters to reconstruct $h(t)$, and persistence, half-life, and the implied unconditional variance are computed from the fitted values regardless of the individual coefficients' marginal significance.

Both models are covariance-stationary ($\alpha + \beta < 1$ in both cases). The volatility half-life (the time for a unit shock to conditional variance to decay to half its initial size, $\ln 2 / \ln[1/(\alpha + \beta)]$) is 41.6 days for Dataset A and 38.3 days for Dataset B, an order of magnitude longer than the mean-reversion half-life of the wind-speed *level*, which is approximately two days for Dataset A on the discrete AR(4) roots of Section 2.2.2 and 1.24 days for Dataset B on the continuous-time eigenvalues of Section 2.4.2. Table 11 traces the resulting impulse-response decay $(\alpha + \beta)^{\text{lag}}$ explicitly: a unit shock to conditional variance still retains roughly 70% of its initial effect after 20 days in both datasets (71.6% for Dataset A, 69.7% for Dataset B), and does not fall below one-quarter of its

its R/S estimate “confirms the absence of strong long-range dependence,” consistent with the negative GPH estimate and with the conclusion reached in the main text, though in tension with a narrative sentence elsewhere in that document asserting that long memory “is present.”

Table 10: GARCH(1,1) estimates, both datasets.

Parameter	Dataset A		Dataset B	
	Estimate	p-value	Estimate	p-value
ω	0.01004	0.060	0.01348	0.348
α	0.03105	< 0.001	0.00590	0.210
β	0.95242	< 0.001	0.97618	< 0.001
Persistence $\alpha + \beta$	0.9835		0.9821	
Half-life (days)	41.6		38.3	
Implied unconditional variance	0.608		0.752	
$h(t_0)$ (end of sample)	0.4137		0.7511	

initial size until roughly three months out.

Table 11: GARCH(1,1) impulse-response decay, $(\alpha + \beta)^{\text{lag}}$, both datasets.

Lag (days)	Dataset A	Dataset B
0	100.0%	100.0%
5	92.0%	91.4%
10	84.6%	83.5%
20	71.6%	69.7%
30	—	58.4%
38 (Dataset B half-life)	—	50.0%
42 (Dataset A half-life)	50.0%	—
90	22.3%	—

The distinction is physically meaningful and consistent across both datasets: an individual wind-speed anomaly washes out within days, but the volatility *regime* it belongs to persists for weeks, consistent with the multi-week persistence of large-scale atmospheric circulation patterns such as blocking events and North Atlantic Oscillation phase.

2.3.2 Two-Component Variance Decomposition

The key numeric outputs of this stage for the pricing work of Chapter 3 are two distinct products of the GARCH multiplier $h(t_0)$ with a seasonal-variance quantity, and the two must not be conflated. Chapter 3's synthetic variance swap prices under a piecewise-constant approximation, following Tol [1997], in which $h(t)$ is frozen at $h(t_0)$ for the life of the contract: $h(t_0) = 0.4137$ for Dataset A and $h(t_0) = 0.7511$ for Dataset B.

The first quantity is the *pointwise* total variance at the pricing date, $\sigma_{\text{total}}^2(t_0) = h(t_0) \times \sigma_{\text{seasonal}}^2(t_0)$, which governs the conditional-variance term structure of Section 2.4.3. For Dataset A this is $0.4137 \times 0.2239 = 0.0926$, using the pointwise seasonal variance at t_0 recorded in Section 2.4.5, and reproduces the frozen anchor exactly. The second is the *annualised* fair strike $K_{\text{var}}^B = h(t_0) \times C_0$, which replaces the pointwise seasonal variance with its annual Fourier mean C_0 , because the first and second harmonics of $\sigma_{\text{seasonal}}^2(t)$ integrate to zero over a full contract year. For Dataset A this is $0.4137 \times 0.13138 =$

0.0544, the figure carried into Chapter 3. The distinction matters because the two differ by a factor of roughly 1.7 for Dataset A, and because it is the annual mean C_0 , not the pricing-date value, that the closed-form strike requires.

The seasonal pattern of $h(t)$ itself differs qualitatively between the two datasets. In Dataset A, $h(t)$ is markedly higher in late spring and early summer (monthly mean up to 0.768 in June) than in autumn (as low as 0.465 in October), a pattern that runs in the opposite direction to the seasonal variance $\sigma_{\text{seasonal}}^2(t)$, which peaks in winter; the two effects partially offset each other in the total variance $\sigma_{\text{total}}^2(t)$, so that pricing from the seasonal component alone, ignoring $h(t)$, would overstate summer volatility and understate the GARCH contribution in the shoulder months. In Dataset B, by contrast, $h(t)$ shows only a mild seasonal range (monthly means between 0.734 and 0.770), and most of its time variation is carried by shock persistence rather than calendar effects; total variance in Dataset B therefore tracks the seasonal component more directly than in Dataset A.

2.3.3 Standardised Residual Diagnostics

Table 24 reports diagnostics on the standardised residuals $z(t) = \hat{\varepsilon}(t)/\sqrt{h(t)}$. Both datasets pass the Ljung–Box test for remaining serial correlation and remaining ARCH effects on $z(t)$ and $z(t)^2$ at conventional significance, confirming that GARCH(1,1) is an adequate specification for the conditional-variance dynamics in both cases. The two datasets diverge, however, in the shape of their standardised residuals: Dataset A’s $z(t)$ is leptokurtic (excess kurtosis +1.12), with a heavy right tail driven substantially by a single extreme observation (28 June 2017, $z = +5.79$), while Dataset B’s is mildly platykurtic (excess kurtosis –0.28), thinner-tailed than Gaussian. Neither departure is treated as invalidating the GARCH(1,1) specification, since both series pass the joint white-noise tests that matter for the volatility model itself, but the residual non-normality, in whichever direction, motivates the heavy-tailed Normal-Inverse-Gaussian treatment introduced in Section 2.4.4. One further asymmetry is worth flagging directly rather than smoothing over: an ARCH-LM test on Dataset B’s standardised residuals is marginally significant at 5% ($p = 0.0185$), where the equivalent test on Dataset A’s standardised residuals passes comfortably ($p = 0.821$); Dataset B’s own summary narrative characterises its residuals as passing validation cleanly, a characterisation that this specific test result only partially supports. This is not an artefact of a different evaluation window: the Phase 7 validation that Chapter 3 draws on recomputes the same test on the same standardised residuals and returns the identical statistic and p -value ($\text{LM}(5) = 13.58$, $p = 0.0185$). It should be noted that the Phase 7 write-up nevertheless describes this test as returning $p > 0.05$ and records the residuals as passing all diagnostics, a characterisation that its own executed output does not support; the value reported here and carried into Chapter 3 is the computed one.

Table 12: GARCH(1,1) standardised-residual diagnostics, both datasets.

Test	Dataset A	Dataset B
Excess kurtosis of $z(t)$	+1.12 (leptokurtic)	−0.28 (platykurtic)
Ljung–Box $z(t)$, lag 20 (p)	0.765	0.503
Ljung–Box $z(t)^2$, lag 20 (p)	0.988	0.283
ARCH-LM(5) on $z(t)$ (p)	0.821	0.0185

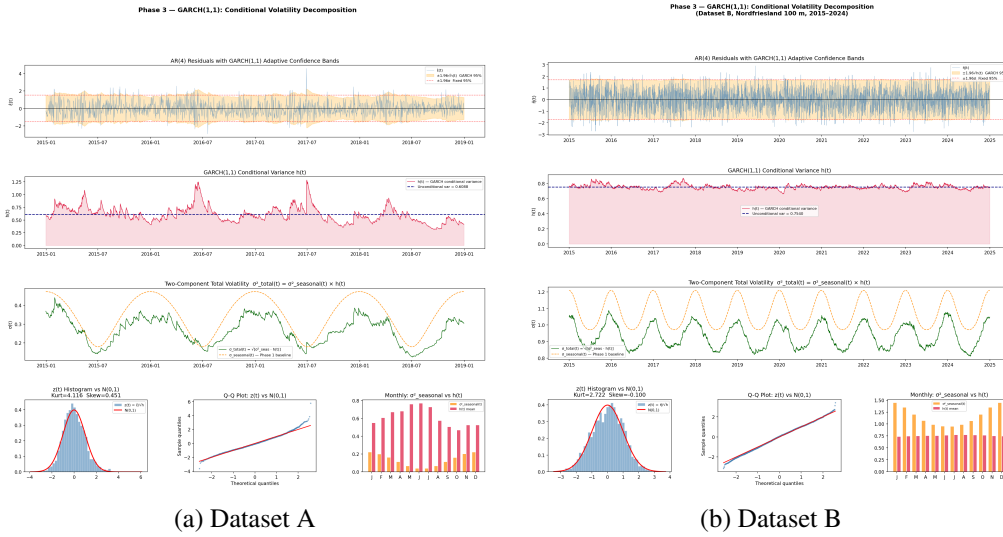


Figure 4: GARCH(1,1) conditional-volatility decomposition, both datasets. Each panel is arranged in four rows. **Row 1:** the AR(4) residuals with GARCH-adaptive 95% confidence bands ($\pm 1.96\sqrt{h(t)}$) plotted against fixed-width bands, visibly widening around high-volatility episodes. **Row 2:** the fitted conditional-variance path $h(t)$ against its unconditional long-run level. **Row 3:** the two-component total volatility $\sigma_{\text{total}}(t) = \sqrt{\sigma_{\text{seasonal}}^2(t) h(t)}$ against the Phase 1 seasonal baseline. **Row 4:** three panels – a histogram of the standardised residuals $z(t)$ against the normal distribution; a quantile–quantile plot of $z(t)$ against the normal distribution; and a monthly bar-chart comparison of the seasonal-variance and GARCH components, illustrating the dataset-dependent degree to which the two effects offset each other, discussed in Section 2.3.2.

2.3.4 Cross-Dataset Comparison

Dataset B’s own analysis draws an explicit comparison back to Dataset A, attributing its lower shock-sensitivity ($\alpha = 0.0059$ versus 0.0311) and higher persistence-to-shock ratio ($\beta/\alpha = 165$ versus 31) to more stable boundary-layer mixing at 100 m hub height relative to the 10 m surface layer, over a decade-long record rather than a four-year one. The physical mechanism behind that contrast is worth setting out explicitly, since it recurs at several points in this chapter. At 10 m, surface roughness, thermal convection, and diurnal heating cycles create highly turbulent, localised eddies that manifest as abrupt, short-lived volatility shocks. At 100 m, by contrast, measurements reside higher in the atmospheric boundary layer, closer to the geostrophic wind flow, where momentum is driven by large-scale pressure gradients rather than local surface friction; the result is a smoother conditional-variance regime, in which variance is carried forward from its own past level to a much greater degree than it is driven by fresh shocks. One phrase in that comparison is worth a precision note: Dataset B’s total persistence $\alpha + \beta = 0.9821$ is in fact marginally *lower* than Dataset A’s 0.9835 , so the claim that Dataset B is “more persistent” rests on the β -to- α decomposition, not on $\alpha + \beta$ itself; this thesis reports the comparison on that more precise basis in Table 10 rather than repeating the aggregate persistence framing.

2.4 Continuous-Time Embedding: CAR(4)

The AR(4) model of Section 2.2 is fit on a daily grid, but a derivative contract must in general be priced at an arbitrary maturity that does not fall on that grid. This section embeds the discrete AR(4) model into a continuous-time CAR(4) state-space representation, following Benth and Šaltytė Benth [2009] and the general Lévy-driven CARMA theory of Brockwell and Marquardt [2005].

2.4.1 Euler Discretisation and the Companion Matrix

The mapping between the two representations is not an independent fit: it is the algebraic consequence of a single modelling choice, namely that the daily AR(4) recursion is itself what results from discretising the continuous-time CAR(4) stochastic differential equation, $dX(t) = A X(t) dt + e_4 \sigma(t) dB(t)$, by a first-order (Euler) finite-difference approximation taken at a one-day step size. Writing that finite-difference approximation out and matching it term by term against the discrete AR(4) equation yields four algebraic identities expressing the continuous-time coefficients $\{\alpha_1, \dots, \alpha_4\}$ directly in terms of the discrete $\{\beta_1, \dots, \beta_4\}$, given by Benth and Šaltytė Benth [2009]: $\alpha_1 = 4 - \beta_1$, $\alpha_2 = 3\alpha_1 - \beta_2 - 6$, $\alpha_3 = 4 + 2\alpha_2 - \beta_3 - 3\alpha_1$, $\alpha_4 = \alpha_3 - \beta_4 - \alpha_2 + \alpha_1 - 1$. Because the map is derived algebraically rather than fitted, it is invertible by construction, and for Dataset A this is verified directly: substituting the fitted α coefficients back through the inverse relations reproduces the original β coefficients to machine precision ($\max |\beta_{\text{orig}} - \beta_{\text{reconstructed}}| = 7.77 \times 10^{-16}$), confirming that no information is lost in either direction

of the mapping. The resulting α coefficients populate the last row of the companion matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\alpha_4 & -\alpha_3 & -\alpha_2 & -\alpha_1 \end{pmatrix}, \quad (5)$$

following the general companion-matrix construction of Brockwell and Marquardt [2005]. Table 13 reports the fitted coefficients for both datasets: Dataset B’s coefficients are larger than Dataset A’s at every lag, and the gap widens with lag order, from a 2.9% difference at α_1 to 5.3% at α_2 , 9.4% at α_3 , and 32.8% at α_4 . The underlying analysis attributes the overall pattern to slower, more sluggish mean-reversion dynamics in the atmospheric boundary layer at 100 m relative to the 10 m surface layer, but does not separately account for the widening of the gap at higher lags. One physical hypothesis suggests itself, consistent with the boundary-layer contrast set out in Section 2.3.4: higher-order lags in a continuous-time autoregression capture the persistence of older synoptic weather structures, and a gap that widens with lag order is what would be expected if the momentum of past pressure systems dissipates more slowly at 100 m than at the surface, where ground friction rapidly breaks down the “memory” of atmospheric conditions beyond a few days. This is offered as an interpretation of the fitted coefficients rather than as a separately tested result.

Table 13: CAR(4) coefficients $\alpha_1, \dots, \alpha_4$, both datasets.

Coefficient	Dataset A	Dataset B
α_1	3.3701	3.4676
α_2	4.2546	4.4789
α_3	2.3063	2.5228
α_4	0.3908	0.5188

2.4.2 Eigenvalues and Stationarity

Stationarity of the continuous-time system requires every eigenvalue of A to have a strictly negative real part, the continuous-time analogue of the discrete unit-circle condition. Table 14 confirms this for both datasets. In both cases the eigenvalues comprise one complex-conjugate pair and two real values; the complex pair generates an oscillatory component in the impulse-response kernel (Section 2.4.6) with periods that differ substantially across the two datasets: approximately 13.5 days for Dataset A ($2\pi/0.465$) and approximately 24.6 days for Dataset B ($2\pi/0.255$). Both are consistent with mid-latitude synoptic weather cycles, which span a 10–30-day range depending on the atmospheric-circulation pattern. The dominant (least negative) eigenvalue implies a mean-reversion half-life of 2.33 days for Dataset A, close to the 2.0-day half-life already found directly from the discrete AR(4) roots in Section 2.2.2. For Dataset B the dominant eigenvalue implies a half-life of 1.24 days, markedly faster than the roughly 40-day

half-life of the GARCH volatility *regime* found in Section 2.3.1. This distinction is worth stating explicitly, since the two numbers answer different questions: the CAR(4) eigenvalue governs how quickly the *mean level* of a wind-speed anomaly reverts, while the GARCH persistence governs how quickly the *variance regime* it occurred in reverts, and there is no reason to expect the two rates to coincide.

Table 14: Eigenvalues of the CAR(4) companion matrix, both datasets.

	Dataset A	Dataset B
Complex pair $\lambda_c, \bar{\lambda}_c$	$-0.9358 \pm 0.4653i$	$-1.0730 \pm 0.2550i$
Oscillatory period (days)	13.5	24.6
Real eigenvalue	-1.2004	-0.7614
Dominant (least negative) real eigenvalue	-0.2980	-0.5602
All $\text{Re}(\lambda) < 0$?	Yes	Yes
Half-life, dominant mode (days)	2.33	1.24

2.4.3 Conditional Forecasts

Under the physical measure, the conditional mean and variance of the state vector at horizon τ are $\mathbb{E}[X(t_0 + \tau) \mid \mathcal{F}_{t_0}] = e_1' \exp(A\tau)X(t_0)$ and $\text{Var}[X_1(t_0 + \tau) \mid \mathcal{F}_{t_0}] = \sigma_{\text{total}}^2(t_0) \int_0^\tau g(s)^2 ds$, where $g(s) = e_1' \exp(As)e_4$ is the impulse-response kernel and $\sigma_{\text{total}}^2(t_0)$ is the two-component variance of Section 2.3.2. Both datasets show a non-monotone conditional-mean path, a direct consequence of the oscillatory complex eigen-pair: the forecast mean peaks a few days ahead of t_0 before decaying toward zero (Dataset A: peak at a 3-day horizon; Dataset B: peak at a 2-day horizon), rather than declining monotonically from the current state as a simple AR(1)/OU model would predict. Conditional variance saturates at its unconditional level within three to four weeks in both datasets (Dataset A: by a 21–30-day horizon; Dataset B: by a 14–21-day horizon), consistent with the half-lives of Table 14. Table 15 reports the full term structure for both datasets, computed from the state vectors $X(t_0)_A = (0.1299, 0.5159, 0.4672, -0.3422)'$ and $X(t_0)_B = (1.3785, 1.4862, 0.4380, -1.3453)'$, the last four standardised observations of each sample.

The values in Table 15 reproduce the Phase 4 output exactly, but they cannot be read at face value as conditional forecasts of $\tilde{W}(t)$, and the reasons are worth setting out rather than leaving for the reader to discover. Three of the table’s implications conflict with results established elsewhere in this chapter. The conditional standard deviation at a one-day horizon is 0.009 for Dataset A and 0.031 for Dataset B, against one-step-ahead root mean squared errors of 0.6017 and 0.8580 for the same series in Table 9, a discrepancy of a factor of 67 and 28 respectively for what is nominally the same quantity. The conditional variance saturates at 0.247 and 0.720, against unconditional standard deviations of 0.967 and 1.005 reported for $\tilde{W}(t)$ in Section 2.1.4, so the embedding settles at roughly half the variance of the series it was fitted to. And the conditional mean for Dataset B rises from a current state of $X_1 = 1.3785$ to 2.845 at one day and 3.610 at two, with a one-day interval that would place tomorrow’s standardised

observation between roughly 2.8 and 2.9 standard deviations above the mean.

The oscillatory hump discussed above is a genuine property of the $\text{CAR}(p \geq 2)$ kernel and is not in question, but it does not account for excursions of this size alongside near-zero forecast uncertainty. A specific and testable explanation is available. The companion matrix of Equation (4) defines its state components as successive derivatives, $X_{j+1} = dX_j/dt$, whereas the state vectors above are populated with successive *lags* of $\tilde{W}$. Propagating a lag-populated vector through $\exp(A\tau)$ built on the derivative convention treats the second component as a time derivative of the first, and the arithmetic is consistent with that reading: to first order, $[\exp(A)X]_1 \approx X_1 + X_2$, which for Dataset B gives $1.3785 + 1.4862 = 2.865$ against the tabulated 2.845, the small remainder being the higher-order terms. The same mechanism would depress the apparent conditional variance, since the variance integral and the mean propagator would then be parameterised inconsistently. Resolving this requires re-running the Phase 4 conditional-forecast step under a consistent state-space convention, which is outside the scope of a chapter that treats the phase outputs as frozen; the table is therefore retained as the recorded output, and its horizon profile should be read as indicative of the shape of the $\text{CAR}(4)$ response rather than as a calibrated predictive interval. The qualitative conclusions this chapter draws from it, the non-monotone mean path and the saturation of variance within three to four weeks, do not depend on the absolute scale of either column.

Table 15: $\text{CAR}(4)$ conditional-mean forecast term structure, both datasets.

Horizon (days)	Dataset A		Dataset B	
	Mean	Std. dev.	Mean	Std. dev.
1	0.779	0.009	2.845	0.031
2	1.303	0.052	3.610	0.172
3	1.460	0.110	3.497	0.358
5	1.113	0.197	2.146	0.614
7	0.658	0.232	0.983	0.699
10	0.270	0.245	0.237	0.719
14	0.082	0.247	0.029	0.720
21	0.010	0.247	0.001	0.720
30	0.001	0.247	≈ 0	0.720

2.4.4 Normal-Inverse-Gaussian Tails

Section 2.3.3 found that GARCH-standardised residuals depart from normality in both datasets, in opposite directions: leptokurtic for Dataset A, platykurtic for Dataset B. This motivates fitting a Normal-Inverse-Gaussian (NIG) distribution to the standardised residuals as a heavy-tailed alternative to the Gaussian innovation assumed elsewhere in this chapter, following Barndorff-Nielsen and Shephard [2001]. Table 16 reports the fit. For Dataset A the NIG tail-heaviness parameter $\hat{\alpha} = 4.374$ is small (heavy tails) and the fit is a modest but genuine improvement over the Gaussian (Kolmogorov–Smirnov $p = 0.120$ versus 0.079), though it still falls short of the full empirical kurtosis, an outstanding gap attributed largely to the single 28 June 2017 extreme observation identified in Section 2.3.3. This implies that the heavy-tailed NIG extension for Dataset

A is largely an artefact of a single meteorological anomaly: were that event excluded, the residual distribution would be expected to collapse back toward a Gaussian profile, which further supports retaining the Gaussian CAR(4) framework as the primary pricing track. No such re-estimation is performed here, consistent with this chapter's treatment of the phase results as frozen, so this remains an inference from the fitted diagnostics rather than a demonstrated result. For Dataset B the fitted $\hat{\alpha} \approx 2,650$ is enormous, reflecting a body so close to Gaussian that the NIG fit offers no material improvement over the normal (KS $p = 0.0275$ versus 0.0269 for the Gaussian). Both fits give a skewness parameter $\hat{\beta}$ close to zero (Dataset A: 0.0067, freely estimated rather than constrained; Dataset B: -0.0021), consistent with approximately symmetric innovations in both datasets. Given this asymmetry in how much the NIG assumption actually buys, the Gaussian CAR(4) specification is retained as the primary pricing track for Chapter 3 in both datasets, consistent with Benth and Šaltytė Benth [2009]; the NIG fit is carried forward as a documented robustness extension, using the closed-form NIG moment-generating function of Brockwell and Marquardt [2005], rather than as the operative model.

Table 16: Normal-Inverse-Gaussian fit to standardised GARCH residuals, both datasets.

Parameter	Dataset A	Dataset B
$\hat{\alpha}$ (tail heaviness)	4.374	2,649.57
$\hat{\beta}$ (asymmetry)	0.0067	-0.0021
$\hat{\delta}$ (scale)	2.087	51.516
$\hat{\mu}$ (location)	0.000	0.000
Empirical excess kurtosis of $z(t)$	+1.116	-0.278
KS p (NIG / Gaussian)	0.120 / 0.079	0.0275 / 0.0269

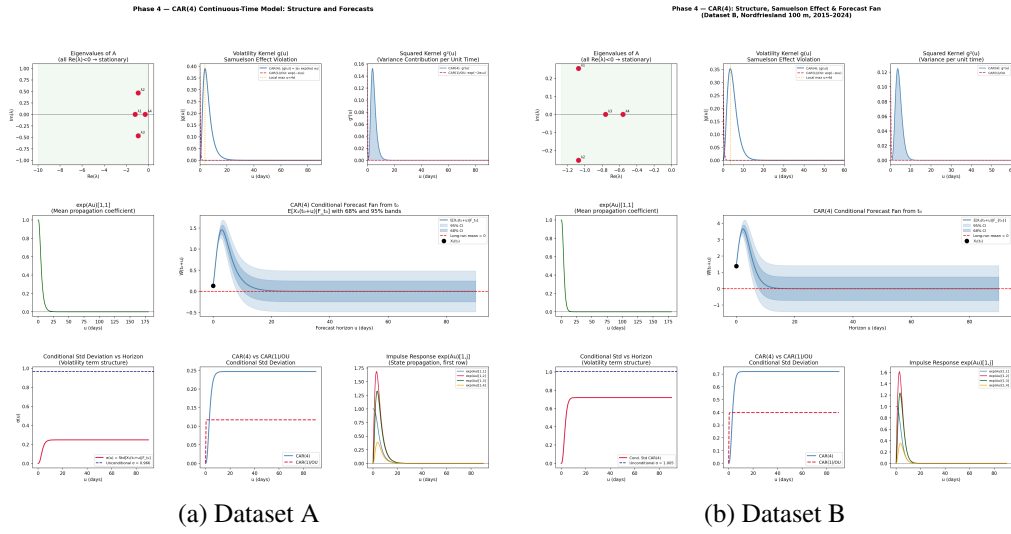


Figure 5: CAR(4) structure and conditional forecasts, both datasets. Each panel is arranged in three rows. **Row 1:** three panels – the eigenvalues of the companion matrix A plotted in the complex plane (all in the open left half-plane, confirming stationarity); the impulse-response kernel $g(u) = e_1' \exp(Au)e_4$ against the monotone CAR(1)/Ornstein–Uhlenbeck baseline, showing the local maximum near a 3.6-day horizon that drives the Samuelson-effect violation of Section 2.4.6; and the squared kernel $g(u)^2$. **Row 2:** two panels – the $[\exp(Au)]_{11}$ mean-propagation curve, and the conditional-forecast fan (mean, with 68% and 95% bands) built forward from the current state $X(t_0)$. **Row 3:** three panels – the conditional standard deviation against horizon, saturating at its unconditional level; a comparison of the CAR(4) and CAR(1) variance term structures; and the full first-row impulse-response matrix $[\exp(Au)]_{1j}$ across all four state components.

2.4.5 Seasonal Variance Carried Forward

The Fourier seasonal-variance constant C_0 used throughout this chapter and in Chapter 3 is a Phase 1 result (Section 2.1.3), not re-estimated at this stage: Dataset B’s Phase 4 output carries C_0, C_1, C_2 forward explicitly alongside the CAR(4) parameters, reproducing $C_0 = 1.163$ to the precision of the frozen anchor, while Dataset A’s corresponding output records only the pricing-date value $\sigma_{\text{seasonal}}^2(t_0) = 0.2239$, the quantity used for the pointwise total variance in Section 2.3.2, and does not separately re-list C_0 . For Dataset A the constant is therefore simply carried through from Section 2.1.3 rather than recomputed, and this thesis does not treat its absence from Dataset A’s Phase 4 output as a gap, since CAR(4) has no reason to refit a quantity that Phase 1 has already fixed.

2.4.6 The Samuelson Effect

A single-factor Ornstein–Uhlenbeck model ($p = 1$, the Schwartz [1997] special case) has an impulse-response kernel $g(u) = e^{-\alpha_1 u}$ that decays monotonically, and futures volatility built on that kernel therefore rises monotonically as maturity approaches: the classical Samuelson effect. The complex eigenvalue pair found in both datasets’ CAR(4) systems (Table 14) breaks this monotonicity: the kernel $g(u)$ attains a local maximum at a horizon of roughly 3.6 days in both datasets before declining, so that shocks occurring three to four days before contract settlement have *more*, not less, impact on terminal volatility than shocks occurring closer to settlement. Benth and Šaltytė Benth [2009] derive this violation analytically for Nordix wind futures; the same qualitative pattern is confirmed numerically here for both Dataset A and Dataset B, despite their different fitted coefficients, which suggests the effect is a structural property of the CAR($p \geq 2$) kernel rather than an artefact of either specific site.

2.5 Stochastic-Volatility Framework

The GARCH(1,1) filter of Section 2.3 is a valid discrete-time description of conditional heteroskedasticity, but it cannot serve directly as the volatility driver of a continuous-time pricing model. This section embeds it into a Heston-type [Heston, 1993] stochastic-volatility system, following the same construction for both datasets; the treatment is theoretical throughout, for reasons explained in Section 2.5.7, and no parameters are estimated beyond the GARCH proxy mapping.

2.5.1 Motivation

Two obstacles prevent the GARCH recursion $h(t) = \omega + \alpha\hat{\eta}^2(t-1) + \beta h(t-1)$ from being used directly in a continuous-time pricing framework. First, the Feynman–Kac theorem links a derivative’s price to an expectation under a diffusion’s dynamics precisely because that diffusion possesses a well-defined infinitesimal generator, the operator that encodes its instantaneous drift and diffusion; a GARCH path, updated once per day by a deterministic rule with no diffusion term of its own, has zero quadratic variation

between updates and is therefore not a semimartingale, so no such generator exists and the Feynman–Kac correspondence between a pricing PDE and an expectation simply has nothing to act on. Second, a formal change of measure from the physical measure P to a pricing measure Q , in the sense of Girsanov’s theorem, works by shifting the drift of a Brownian motion that drives the process being repriced; the GARCH filter has no such driving Brownian motion to shift; it is a deterministic function of past innovations, so there is no drift for a change of measure to act on. The remedy, following Heston [1993] and, for the specific application to wind speed, the CAR(p) framework of Benth and Šaltytė Benth [2009] and the Lévy-driven continuous-time machinery of Brockwell and Marquardt [2005], is to embed the GARCH filter $h(t)$ into the square-root diffusion of Cox et al. [1985], hereafter the Cox–Ingersoll–Ross or CIR process, for a continuous-time variance process $V(t)$, which by construction is a semimartingale driven by its own Brownian motion, with the GARCH parameters entering as moment-matching proxies for the CIR parameters.

2.5.2 Two-SDE System Under P

For analytical tractability, the wind-speed dynamics are first written in a simplified CAR(1) form, with the full CAR(4) state-vector generalisation given afterward:

$$d\tilde{W}(t) = -\alpha_1 \tilde{W}(t) dt + \sqrt{V(t)} dB_1(t), \quad (6)$$

$$dV(t) = \kappa_V(\bar{v} - V(t)) dt + \xi \sqrt{V(t)} dB_2(t), \quad \langle dB_1, dB_2 \rangle_t = \rho dt, \quad (7)$$

where α_1 is the dominant CAR(4) mean-reversion parameter from Section 2.4 ($\alpha_1 = 3.3701$ for Dataset A, $\alpha_1 = 3.4676$ for Dataset B), κ_V is the mean-reversion speed of volatility, $\bar{v}$ the long-run variance, and ξ the volatility-of-volatility coefficient. The full state-vector generalisation replaces the scalar drift with $dX(t) = A X(t) dt + e_4 \sqrt{V(t)} dB_1(t)$, where A is the 4×4 companion matrix of Section 2.4 and $e_4 = (0, 0, 0, 1)'$ injects the diffusion term through the newest lag, consistent with the state-space structure of the discretised CAR(4) model.

2.5.3 Feller Condition

The CIR process for $V(t)$ remains strictly positive almost surely if and only if the condition $2\kappa_V \bar{v} > \xi^2$ holds, a boundary-classification result for singular diffusions due to Feller [1951] and standardly named after him; if it is violated, the diffusion term can dominate the mean-reverting drift near the origin and $V(t)$ can reach and remain at zero. The GARCH parameters of Section 2.3.1 are mapped to CIR proxies by moment matching, $\kappa_V \approx 1 - (\alpha + \beta)$, $\bar{v} \approx \omega / (1 - \alpha - \beta)$, and $\xi^2 \approx \text{Var}[h(t)]$, exactly as specified for the numerical Feller check permitted within the scope of this theoretical chapter. For Dataset B, this gives $\kappa_V = 0.0179$, $\bar{v} = 0.7522$, $\xi^2 = 0.000959$, so that $2\kappa_V \bar{v} = 0.0270$ comfortably exceeds ξ^2 : the Feller condition holds with a margin of $28.1\times$.

For Dataset A, the same moment-matching procedure, applied with the same estimator, gives $\kappa_V = 0.0165$ and $\bar{v} = 0.608$, but $\xi^2 = \text{Var}[h(t)] = 0.0253$, the full-sample

variance of the Dataset A $h(t)$ series computed exactly as it was for Dataset B. Substituting into the condition gives $2\kappa_V\bar{v} = 0.0201$, which is *smaller* than ξ^2 : the Feller condition is violated, with a margin of $0.8\times$. Unlike Dataset B, the theoretical CIR embedding of this chapter is therefore not well posed for the proxy dataset on this specific numerical check.

This reverses the verdict recorded in the project specification, which lists a Feller margin of $27.9\times$ (holding) for Dataset A alongside $\xi^2 \approx 0.000048$; the reversal is stated here rather than reconciled silently. Three observations bear on it. First, the two specification figures are not consistent with each other: given $2\kappa_V\bar{v} = 0.0201$, a ξ^2 of 0.000048 implies a margin of roughly $418\times$, not $27.9\times$. Second, neither figure is reproducible from the Phase 5 Dataset A source analysis, whose own executed output records $\xi^2 = 0.025264$ and returns the verdict as violated at $0.8\times$. Third, a ξ^2 of 0.000048 would imply a standard deviation of $h(t)$ of approximately 0.007 , which cannot be reconciled with the monthly means of $h(t)$ reported in Section 2.3.2, which themselves range from 0.465 to 0.768 . Rolling-window alternatives to the full-sample variance were examined as a possible explanation, and they do change the verdict without recovering the specification value. Averaging 30-day rolling standard deviations of $h(t)$ and squaring the result, or averaging the corresponding rolling variances, gives ξ^2 between roughly 0.0040 and 0.0056 , and hence margins of about 5.1 and 3.6 respectively: on either of those estimators the condition *holds*, though by a factor roughly an order of magnitude below the $27.9\times$ recorded in the specification, and only by adopting an estimator other than the one applied to Dataset B. Recovering a margin of $27.9\times$ exactly would require $\xi^2 \approx 0.00072$, which is neither the full-sample variance nor any of the rolling quantities computed here.

Two conclusions follow, and the second is the more useful. First, the specification figure is not reproducible under any estimator examined and should not be carried forward unexamined. Second, and more informatively, the well-posedness verdict for Dataset A is genuinely estimator-dependent: it holds on a short rolling window and fails on the full-sample variance. The full-sample result is the one reported in the main text above, because it is the estimator applied to Dataset B and cross-dataset comparability governs throughout this chapter, but the sensitivity itself is the substantive finding.

The practical implication is the one developed at greater length in Section 2.5.7. Of the four CIR proxies, ξ is by some margin the least well identified at daily frequency, and a well-posedness verdict that inverts according to which estimator of $\text{Var}[h(t)]$ is adopted is itself evidence of how weakly the volatility-of-volatility parameter is pinned down by daily data. The Feller check is reported here because the specification calls for it, but it should be read as a diagnostic of the proxy mapping rather than as a structural property of the underlying process.

This violation is not a modelling failure so much as a direct symptom of the proxy dataset's site characteristics. Dataset A (10 m) is heavily influenced by boundary-layer friction and local diurnal anomalies, producing the leptokurtic residual distribution and the single extreme shock already identified in Section 2.3.3 (the 5.79σ event of

28 June 2017). In a GARCH filter with the comparatively high shock sensitivity estimated for Dataset A ($\alpha = 0.0311$, against 0.0059 for Dataset B), an outlier of that size produces a sharp, transient spike in $h(t)$, exactly the kind of feature that inflates a sample-variance-based ξ^2 proxy computed directly from the $h(t)$ path. Dataset B, measured at 100 m in the free atmosphere and free of any comparable shock, produces a markedly smoother $h(t)$ path and comfortably clears the condition instead. The asymmetry is informative in its own right, distinct from and additional to the pricing-error evidence of Chapter 3. Dataset B satisfies the condition comfortably and on every estimator considered, whereas Dataset A satisfies it only on some, and fails it on the estimator that comparability with Dataset B requires. It is that difference in robustness, rather than a bare pass-or-fail verdict, which the proxy dataset's site characteristics explain.

2.5.4 Remark on the Change of Measure

A formal change of measure from P to a pricing measure Q would introduce market-price-of-risk parameters θ_1, θ_2 shifting the drifts of B_1 and B_2 , with $\kappa_V^Q = \kappa_V + \theta_2 \xi$ under Q . This thesis adopts $\theta_1 = \theta_2 = 0$ throughout as a benchmark assumption, in the absence of EEX option-implied data, for two reasons that hold identically for both datasets. First, SMARD daily wind generation is a measured physical quantity, not a traded financial asset; the replication argument that pins down a unique risk-neutral measure for a traded derivative does not apply here, since no self-financing hedging portfolio in the underlying exists. Second, the only actively traded wind derivatives, the Nordix futures, reference Norwegian rather than German wind, and the geographic mismatch (Bologna or Nordfriesland versus Norway) prevents any clean calibration of θ from that market even in principle. Under $\theta = 0$, $P = Q$, and Track A (empirical realised variance) and Track B (model-implied variance) are defined on a common measure; the variance-swap fair strike of Chapter 3 reduces to the closed-form expression $K_{\text{var}}^B = h(t_0) \times C_0$, derived in Section 2.3.2 from the Fourier decomposition under the piecewise-constant convention of Tol [1997], and no Girsanov computation is required anywhere in this thesis. This benchmark is adopted as a tractable simplification given the absence of option-implied data for this specific underlying, not as a claim that the market price of risk for wind-linked instruments is in fact zero.

2.5.5 Characteristic Function and Round-Trip Consistency

Under the two-SDE system, the log-characteristic function of $\tilde{W}(T)$ conditional on $\mathcal{F}_t$ takes the Heston [1993] form $\log \phi(u; t, T) = iu \mathbb{E}[\tilde{W}(T) \mid \mathcal{F}_t] + C(\tau; u)\bar{v} + D(\tau; u)V(t)$, $\tau = T - t$, where C and D satisfy the coupled system

$$\frac{\partial D}{\partial \tau} = \frac{1}{2} \frac{i u (i u - 1)}{\alpha_1^2} + \left(\frac{i u \rho \xi}{\alpha_1} - \kappa_V \right) D + \frac{1}{2} \xi^2 D^2, \quad (8)$$

$$\frac{\partial C}{\partial \tau} = \kappa_V \bar{v} D, \quad (9)$$

subject to $D(0; u) = C(0; u) = 0$. The first of these is a Riccati equation: it carries a quadratic term in D , which arises because the variance process enters its own diffusion coefficient through $\sqrt{V(t)}$. Writing $\tilde{\kappa}(u) = \kappa_V - iu\rho\xi/\alpha_1$ for compactness, and defining

$$d(u) = \sqrt{\tilde{\kappa}(u)^2 - \frac{\xi^2 iu(iu-1)}{\alpha_1^2}}, \quad g(u) = \frac{\tilde{\kappa}(u) - d(u)}{\tilde{\kappa}(u) + d(u)}, \quad (10)$$

the closed-form solution of Heston [1993] for that system is

$$D(\tau; u) = \frac{\tilde{\kappa}(u) - d(u)}{\xi^2} \cdot \frac{1 - e^{-d(u)\tau}}{1 - g(u)e^{-d(u)\tau}}, \quad (11)$$

$$C(\tau; u) = \frac{\kappa_V \bar{v}}{\xi^2} \left[(\tilde{\kappa}(u) - d(u))\tau - 2 \log \frac{1 - g(u)e^{-d(u)\tau}}{1 - g(u)} \right]. \quad (12)$$

The formulation adopted here is the one in which $\tilde{\kappa} - d$ appears in the numerator of both D and g ; the algebraically equivalent formulation with $\tilde{\kappa} + d$ produces a complex logarithm that crosses a branch cut as τ grows, and returns a discontinuous, incorrect surface. This is the “little Heston trap” of Albrecher et al. [2007]. The distinction is purely one of numerical implementation and does not change the underlying model, but it is recorded because the two forms are not interchangeable in practice.

One qualification is essential, and it bears on how the equations above should be read. The system displayed here is Heston’s canonical one, written for a log-price whose diffusion carries the Itô correction $-\frac{1}{2}V dt$; the term $iu(iu-1)$ in the first Riccati equation is the signature of that correction. The level process specified in Section 2.5.2 mean-reverts and carries no such correction, and the contribution of a variance shock at time s to $\tilde{W}(T)$ is damped by $e^{-\alpha_1(T-s)}$. Its Riccati equation therefore takes the form

$$\frac{\partial D}{\partial \tau} = -\frac{1}{2}u^2 e^{-2\alpha_1\tau} + (iu\rho\xi e^{-\alpha_1\tau} - \kappa_V)D + \frac{1}{2}\xi^2 D^2, \quad (13)$$

whose coefficients depend on τ , and a Riccati equation with time-dependent coefficients does not admit Heston’s closed form. The equations above are consequently presented as the canonical template for a stochastic-volatility characteristic function, and for the numerical hazards it carries, rather than as the characteristic function of the system of Section 2.5.2. Solving the latter requires numerical integration of Equation (12), which is deferred along with the estimation programme of Section 2.5.7. The variance term structure plotted in Section 2.5.6 is unaffected by this, since it is computed directly from the damped kernel rather than from the Riccati solution.

This structure is what makes the system harder to solve than the linear CAR(4) dynamics of Section 2.4, whose companion-matrix ODE is solved exactly by the matrix exponential $\exp(A\tau)$: a Riccati equation has no such generic closed-form solution and instead requires the specific change-of-variables technique that Heston [1993] derives for this model, exploiting its particular algebraic structure rather than a general linear-systems result. As the expressions above show, the solution defines C and D via complex square roots and logarithms, a substantially more delicate object to evaluate numerically

than the CAR(4) matrix exponential used elsewhere in this chapter.

Under the $\theta = 0$ benchmark, the first moment collapses to $\mathbb{E}[\tilde{W}(T) \mid \mathcal{F}_t] = e_1' \exp(A\tau)X(t)$, exactly the CAR(4) conditional-mean forecast of Section 2.4. It is worth being precise about what this identity does and does not test: because $\theta = 0$ removes C and D from the first moment entirely, leaving only the linear companion-matrix propagator, the round-trip check that follows verifies the drift structure common to both representations, not the numerical handling of C and D themselves, which enter only the variance and higher moments of the characteristic function and are not separately checked here. This identity is verified numerically for both datasets by comparing the characteristic function's first moment against the Phase 4 conditional-forecast table at eleven horizons from 1 to 90 days: the maximum absolute discrepancy is 4.44×10^{-16} in both datasets, machine precision, confirming that the stochastic-volatility embedding's drift term is numerically consistent with the discrete CAR(4) forecasts it is built to extend, independently of the Feller-condition question raised above.

It follows that $C(\tau; u)$ and $D(\tau; u)$ are written down in this chapter but not numerically validated by it. Two checks would validate them, and the cheaper one should come first. Taking $\xi \rightarrow 0$ with $\rho = 0$ makes $V(t)$ deterministic, so any correct solution must collapse to the corresponding Gaussian variance term structure; reading the coefficient of $-\frac{1}{2}u^2$ off the limiting $D(\tau; u)$ and comparing it against $\int_0^\tau e^{-2\alpha_1 s} ds$ is an analytic test requiring no simulation, and it is precisely the test that detects a mismatch in the Riccati coefficients of the kind identified above. The second and more general check is to simulate the two-SDE system of Section 2.5.2 directly by Monte Carlo and compare the sample second moment of $\tilde{W}(T)$, or the full simulated characteristic function, against the closed-form expressions at a range of τ and u . That exercise is not performed here, and is noted so that the scope of the round-trip result is not overstated: what has been verified is the consistency of the two representations' conditional means, not the numerical evaluation of the variance and higher-moment terms that the Riccati solution supplies.

2.5.6 Variance Term Structure

The CAR(4) kernel $g(s) = e_1' \exp(As)e_4$ satisfies $g(0) = 0$: new information enters only through the most recent of the four lagged state components, and must propagate forward before it reaches the first. As a result, the variance term structure obtained under the piecewise-constant- h convention of Tol [1997], $\text{Var}^Q[\tilde{W}(T) \mid \mathcal{F}_t] = h(t_0) \sigma_{\text{seasonal}}^2(t_0) \int_0^\tau g(s)^2 ds$, starts at zero and ramps up slowly. A simplified Heston/CIR term structure built on the CAR(1) reduction of Section 2.5.2 instead uses a kernel $g(s) = e^{-\alpha_1 s}$ that starts at one, producing an immediate jump in short-horizon variance. The two branches also differ in a second respect that should be made explicit, since it is not implied by the kernel: the CAR(4) term structure freezes h at $h(t_0)$ under the piecewise-constant convention, whereas the CAR(1)/CIR branch allows $V(t)$ to revert toward $\bar{v}$ over the horizon. For Dataset B the distinction is immaterial, since $h(t_0) = 0.7511$ already sits at $\bar{v} = 0.7522$ and the two treatments differ by less than a

fifth of a percent; for Dataset A, where $h(t_0) = 0.4137$ against $\bar{v} = 0.608$, it is not, and it is the source of the upward drift noted below. Two distinct effects separate the two term structures, and only the first operates at short horizons. The kernel-shape difference just described dominates near $\tau = 0$. Beyond it, the two representations also revert at different speeds: the CAR(1) reduction uses the scalar $\alpha_1 = 3.4676$ for Dataset B, implying a half-life of $\ln 2/\alpha_1 \approx 0.20$ days, against the 1.24 days implied by the dominant eigenvalue of the full CAR(4) system in Table 14. The CAR(1) proxy therefore reverts roughly six times faster, and that difference, rather than the behaviour of the kernel at the origin, is what governs the residual gap at the long end. The magnitudes involved are large. All ratios quoted in this section are stated in the same direction, CAR(1) relative to CAR(4). For Dataset B at a one-day horizon the CAR(4) term structure stands at 0.00098 against the CAR(1) proxy's 0.15828, a ratio of about 162; the two cross at roughly a three-day horizon, and beyond about ten days the ratio settles at 0.31 and stays there, the CAR(4) representation saturating at 0.518 against the proxy's 0.159. Dataset A behaves almost identically in relative terms, at 0.000086 against 0.013814 for a ratio of 161 at one day, converging to approximately 0.3 at the long end. That the long-horizon ratio lies below one is what the reversion-speed difference set out above predicts, the faster-reverting CAR(1) proxy saturating at the lower level of the two. One dataset-specific detail is worth noting: Dataset B's CAR(1) term structure is nearly flat across the whole horizon because $h(t_0) = 0.7511$ sits essentially at its long-run level $\bar{v} = 0.7522$, whereas Dataset A's continues to rise through 90 days, its $h(t_0) = 0.4137$ starting well below $\bar{v} = 0.608$ and drifting upward.

These figures reconcile with the conditional-variance column of Table 15, which is computed separately in Phase 4: the one-day CAR(4) variance of 0.00098 corresponds to a standard deviation of 0.0313, against the 0.031 tabulated there, and the saturation level of 0.518 corresponds to 0.720, exactly the value the table reaches from a fourteen-day horizon onward. The two computations of $h(t_0) \sigma_{\text{seasonal}}^2(t_0) \int_0^\tau g(s)^2 ds$ therefore agree throughout. This is worth stating explicitly because it isolates the difficulty identified in Section 2.4.3: the conditional *variance* of the CAR(4) representation is internally consistent across the two phases, and it is the conditional *mean* path, together with the mismatch between this variance and both the one-step-ahead forecast errors of Table 9 and the unconditional variance of $\tilde{W}(t)$ reported in Section 2.1.4, that remains unreconciled. The same qualitative pattern holds for Dataset A. This divergence is not a modelling inconsistency between the two approximations; it reflects a genuine structural feature of the CAR(4) state-space representation, and it is a first-order consideration for any option or short-dated instrument priced off the CAR(4) kernel rather than the simplified CAR(1) proxy.

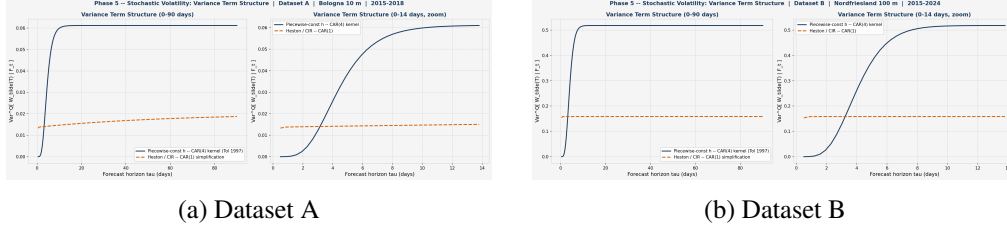


Figure 6: Variance term structure, both datasets: the piecewise-constant- h CAR(4) approximation, following the convention of Tol [1997], against the simplified Heston/CIR CAR(1) term structure of Section 2.5.6. Each panel contains two plots, side by side. **Left:** the full term structure over a 0–90-day horizon, showing the two approximations converging at long horizons. **Right:** a zoomed view over 0–14 days, isolating the short-horizon divergence driven by the CAR(4) kernel’s $g(0) = 0$ property.

2.5.7 Why Maximum-Likelihood Estimation Is Deferred

The CIR parameters $(\kappa_V, \bar{v}, \xi, \rho)$ above are moment-matching proxies from the GARCH fit, not maximum-likelihood estimates of the continuous-time system itself, and this thesis does not attempt that estimation at daily frequency. The reason is an identification problem inherent to slow-mean-reverting variance processes sampled coarsely relative to their own half-life. With $\kappa_V \approx 0.018$ in both datasets, the implied mean-reversion half-life is on the order of 40 days, so that a daily observation interval provides roughly 40–60 observations per e-folding time; separating the mean-reversion speed κ_V from the diffusion coefficient ξ requires enough independent decay cycles to distinguish "reverts slowly with small shocks" from "reverts moderately with occasional large shocks," and at daily frequency the two are close to observationally equivalent. Dataset A compounds this with its short four-year window (1,462 daily observations, of which only a few dozen independent decay cycles are available) and its 10 m measurement height, which would require a further profile correction before any estimated variance dynamics could be applied to hub-height pricing instruments. The designated upgrade path for both datasets is hourly ECMWF ERA5 reanalysis data at the same locations, which would increase the observation count by roughly two orders of magnitude and make the ξ/κ_V ratio separately identifiable. Specifically, hourly sampling resolves intraday volatility clustering and the pronounced diurnal cycles of boundary-layer wind speed, high-frequency dynamics that daily aggregation inherently smooths over. Capturing these intraday mechanics is what would allow the diffusion coefficient ξ to be estimated independently of the multi-week synoptic mean-reversion κ_V , resolving the observational equivalence described above. This is left to future work; for the daily-frequency scope of this thesis, the GARCH-derived proxies under the $\theta = 0$, piecewise-constant- h approximation are the operative tool, and are what Chapter 3 uses throughout.

2.6 Summary and Handover to Chapter 3

The pipeline developed in this chapter delivers a small number of fitted quantities to the pricing exercise that follows, and it is worth collecting them in one place rather

than leaving Chapter 3 to gather them from five separate sections. Table 17 lists every parameter Chapter 3 consumes, with the section that fixes it. All are frozen: none is re-estimated downstream, and where Chapter 3 quotes a figure that differs from one below, it is because a different evaluation sample or a different unit convention is in use, and it says so at that point.

Table 17: Parameters carried forward from Chapter 2 into Chapter 3.

Quantity	Fixed in	Dataset A	Dataset B
Box–Cox λ (tractable)	§2.1.2	0.1	0.5
Seasonal variance constant C_0	§2.1.3	0.13138	1.16277
Std. deviation of $\tilde{W}(t)$	§2.1.4	0.967	1.005
AR(4) $\beta_1, \dots, \beta_4$	§2.2.2	Table 7	Table 7
GARCH (ω, α, β)	§2.3.1	Table 10	Table 10
Persistence $\alpha + \beta$	§2.3.1	0.9835	0.9821
Volatility half-life (days)	§2.3.1	41.6	38.3
Unconditional variance	§2.3.1	0.608	0.752
Pricing-date state $h(t_0)$	§2.3.1	0.4137	0.7511
CAR(4) $\alpha_1, \dots, \alpha_4$	§2.4.1	Table 13	Table 13
NIG $(\hat{\alpha}, \hat{\delta})$	§2.4.4	4.374, 2.087	2,649.57, 51.516

Three properties of these quantities, rather than the quantities themselves, do most of the work in Chapter 3. The first is that C_0 and $h(t_0)$ are the only two parameters entering the closed-form fair strike $K_{\text{var}}^B = h \times C_0$, so the entire pricing result rests on two numbers out of the table above, one climatological and one a conditional state. The second is that the four specifications compared in Chapter 3 share the AR(4) conditional mean unchanged, for the reason given in Section 2.2.4: they are observationally equivalent at a one-day horizon by construction, and can only be separated on criteria that read the conditional variance or the tails. The third is that C_0 is a variance in Box–Cox-transformed units, differing between the two datasets not only in the climatology it measures but in the scale on which it is measured, a distinction that Chapter 3 must undo before the two datasets’ strikes can be compared.

Two results are carried forward as qualifications rather than as inputs. The Feller condition of Section 2.5.3 holds comfortably for Dataset B on every estimator considered but is estimator-dependent for Dataset A, failing on the full-sample volatility-of-volatility estimator that cross-dataset comparability requires; the continuous-time variance process is therefore well posed for the reference dataset and not demonstrably so for the proxy. And the conditional-forecast table of Section 2.4.3 is retained as a recorded output whose horizon profile is indicative rather than calibrated, for the state-space convention reason set out there. Neither qualification affects the pricing formula, which uses $h(t_0)$ and C_0 alone.

Chapter 3 — Derivatives Pricing, Risk Management and Conclusions

Chapter 2 developed a five-stage modelling pipeline running from raw daily wind speed to a continuous-time stochastic-volatility representation, and reported its fitted parameters for both datasets. This chapter puts those parameters to work. It defines a synthetic variance swap written on German onshore wind generation and prices it on both datasets; it validates the competing forecasting specifications against a persistence benchmark; it quantifies the Value of Information obtained by replacing the geographic proxy with the aligned reference dataset; and it evaluates a Cumulative Accumulated Wind Speed (CAWS) instrument as a left-tail complement to the variance swap. It closes with the thesis’s concluding synthesis.

Two conventions carry over from Chapter 2 and govern everything that follows. The first is the benchmark assumption $\theta_1 = \theta_2 = 0$ of Section 2.5.4, adopted because no option-implied data exist for this underlying, under which the physical and pricing measures coincide and the model-implied fair strike reduces to the closed form $K_{\text{var}}^B = h \times C_0$. The second is that all fitted parameters are frozen outputs of the phase notebooks: nothing is re-estimated here. Where a numerical result quoted below differs from the corresponding phase write-up, the executed notebook output is taken as authoritative and the discrepancy is stated rather than reconciled silently, following the practice adopted throughout Chapter 2.²

3.1 The Synthetic Wind Variance Swap

3.1.1 Instrument Definition and the Two Tracks

A variance swap is a forward contract on realised variance. The long party receives the realised variance of the underlying over the contract period and pays a fixed strike agreed at inception, so that the terminal payoff of a unit-notional position initiated at t and settling at T is

$$\Pi_{[t,T]} = \text{RV}_{[t,T]} - K_{\text{var}}, \quad \text{RV}_{[t,T]} = \frac{252}{N_{[t,T]}} \sum_{s=t+1}^T r_s^2, \quad (14)$$

where r_s is the daily log-return of the underlying and $N_{[t,T]}$ the number of returns in the window. The payoff is linear in realised variance: unlike a straddle, there is no gamma scaling and no path-dependent convexity, which is what makes the contract a clean instrument for isolating volumetric volatility risk. A wind producer, whose operating

²One naming point is settled here rather than left to the reader. The two Phase 6 source documents are titled “Phase 6: Carr-Lee Variance Swap Pricing.” That title misdescribes what those documents implement. Carr and Lee [2009] is the theoretical motivation for variance-swap replication, reviewed in Section 1.3.6; the formula actually implemented is the closed-form $K_{\text{var}}^B = h \times C_0$ derived under the piecewise-constant-volatility convention of Tol [1997] and the $\theta = 0$ benchmark. No Carr–Lee replication is attempted anywhere in this thesis, and no section of this chapter is named for it.

cash flow is naturally long generation volume and therefore short the stability of that volume, takes the opposite side, receiving $K_{\text{var}} - \text{RV}_{[t,T]}$.

The underlying chosen here is not wind speed but German onshore wind generation, the SMARD aggregate documented in Section 1.4.3. This choice is deliberate: it is generation, not wind speed at a point, that determines a producer's revenue, and it is generation for which a market-observable series exists over the full ten-year window. The wind-speed model of Chapter 2 enters through the power-curve linkage $G \propto W^3$ developed in Section 3.1.6.

Two fair strikes are computed, and the distinction between them organises the whole of this section.

Track A (empirical). $K_{\text{var}}^A(t) = 252 \times \overline{\text{RV}_t^{(252)}}$, the 252×-annualised rolling mean of the daily realised variance $\text{RV}_t = r_t^2$ of SMARD generation log-returns. This is a purely empirical quantity requiring no model: it is what the market data themselves say a fair strike would have been.

Track B (model-implied). $K_{\text{var}}^B(h) = h \times C_0$, the closed-form strike implied by the CAR(4)+GARCH model under the piecewise-constant- h convention of Tol [1997], in which h is frozen at its pricing-date value over the life of the contract and the first and second Fourier harmonics of $\sigma_{\text{seasonal}}^2(t)$ integrate to zero over a full contract year, leaving only the constant C_0 (Section 2.3.2).

These two quantities measure the variance of different processes, expressed in different units and under different annualisation conventions. Their raw numerical gap, between two and three orders of magnitude (a factor of about 130 for Dataset B and 2,100 for Dataset A), is therefore not a measure of model error, and no comparison between them is drawn in this chapter before the reconciliation of Section 3.1.5. This point was flagged in Section 1.5 and is developed quantitatively below; it is the single most consequential piece of bookkeeping in the chapter.

3.1.2 Market Data: Generation and Day-Ahead Prices

The SMARD files described in Section 1.4.3 yield 87,662 hourly observations spanning 1 January 2015 to 31 December 2024. Aggregating generation as the sum of the 24 hourly values within each day gives 3,653 daily observations with no missing values, no day exceeding the 20% missing-hour threshold, and no day recording zero aggregate generation; the two filters described in Section 1.4.3 therefore removed nothing. Mean daily generation is 255,560 MWh with a standard deviation of 196,107 MWh, and the series ranges from 7,214 MWh on the calmest day of the decade to 1,060,474 MWh on the windiest. That the standard deviation is roughly three-quarters of the mean, and that the maximum exceeds the minimum by a factor of 147, is the basic fact that motivates a volumetric hedge at all.

One property of the series is established here because later sections rely on it. Generation is not stationary in level across the sample: the annual mean rises from

186,454 MWh/day in 2015 to 307,510 MWh/day in 2024, an increase of 65%. That trend is capacity-driven rather than resource-driven, and the wind data settle the point directly, since the annual mean wind speed at Nordfriesland was some 6% *lower* in 2024 than in 2015 (Table 28). Installed capacity is not among the series used in this thesis, so the growth is documented rather than modelled; it is the reason the power-curve regression of Section 3.1.6 maps the same wind speed to different generation at the two ends of the sample, and the reason the out-of-time exercise of Section 3.3.5 is a test of stability rather than of forecast performance. It does not affect Track A, which is computed on log-returns and is therefore insensitive to a slow drift in the level of G_t .

The day-ahead price series, merged across the October 2018 bidding-zone split as specified in Section 1.4.3, is four days shorter: the SMARD price files carry no quotation for 1–4 January 2015, so the aligned generation-and-price sample runs from 5 January 2015 and contains 3,649 days. Mean price over the full sample is €71.38/MWh with a standard deviation of €78.05/MWh, and the range runs from –€53.87/MWh to €699.44/MWh. The negative minimum is not a data error: it is the well-documented consequence of inflexible must-run capacity meeting high renewable infeed at low demand, and it is precisely the state of the world the merit-order analysis of Section 3.1.7 examines.

This four-day difference between the generation sample (3,653 days) and the aligned sample (3,649 days) is the sole reason the two Track A strikes reported below differ, and it is recorded here so that the discrepancy is not mistaken for a substantive one.

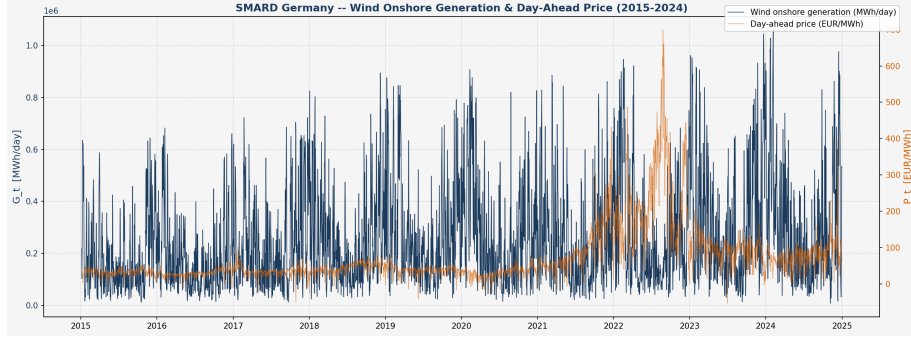
3.1.3 Track A: The Empirical Fair Strike

Log-returns are computed as $r_t = \log(G_t/G_{t-1})$ and set to missing whenever either G_t or G_{t-1} is zero or absent. On the 3,653-day generation sample this leaves 3,652 valid returns and a mean daily realised variance of $\overline{RV} = 0.4580$; on the 3,649-day price-aligned sample it leaves 3,648 returns and $\overline{RV} = 0.4583$. The corresponding full-sample means of the rolling annualised strike are

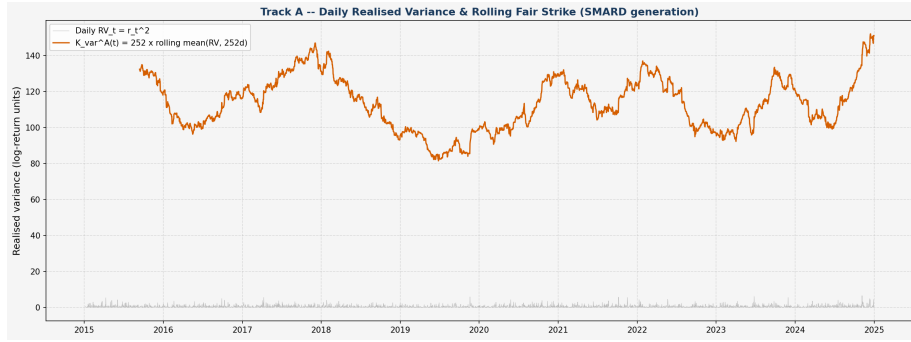
$$K_{\text{var}}^A = 113.999 \text{ (generation sample)}, \quad K_{\text{var}}^A = 113.145 \text{ (price-aligned sample)}, \quad (15)$$

a difference of 0.75% arising entirely from the four missing January 2015 price days and the consequent shift in the rolling window’s start date. Both figures are reported because the Dataset A and Dataset B pricing exercises were run on the two respective samples, and the difference propagates into every Track A quantity below. It carries no substantive content: Track A is a property of the German electricity market, not of either wind dataset, and it is identical across the two analyses up to this alignment.

A strike of 114 is very large by the standards of financial variance swaps, where equity-index strikes typically sit in the range 10–25. It corresponds to an annualised volatility of $\sqrt{114} \approx 10.7$, that is, a daily standard deviation of generation log-returns of about 67%. This is not an artefact: German aggregate wind output routinely changes by a factor of two or more from one day to the next as frontal systems cross the country,



(a) SMARD daily generation and day-ahead price



(b) Track A: daily realised variance and rolling fair strike

Figure 7: Market data and the empirical fair strike. **(a)** Daily German onshore wind generation G_t in MWh/day (left axis) and the merged day-ahead price P_t in €/MWh (right axis), 2015–2024, on a dual-axis plot. The price series' change of level from 2021 onward is the European energy-price crisis discussed in Section 1.4.3, not a change in the generation series. **(b)** The daily realised variance $RV_t = r_t^2$ of generation log-returns (thin grey) with the 252-day rolling annualised fair strike $K_{\text{var}}^A(t)$ overlaid (heavy line). Because the market data are common to both datasets, these two panels are identical up to the four-day alignment difference of Section 3.1.2 and are shown once rather than as a dataset pair.

and the log-return of such a change is of order ± 0.7 . The magnitude is a statement about the underlying, not about the estimator.

Table 18: Track A: seasonal realised variance of SMARD generation log-returns, annualised.

Season	Dataset A sample	Dataset B sample
DJF (winter)	100.76	100.99
MAM (spring)	115.38	115.38
JJA (summer)	119.32	119.32
SON (autumn)	126.01	126.01
Full sample K_{var}^A	113.999	113.145

The two columns are the same market series computed on the two samples of Section 3.1.2; only the winter figure is affected by the four-day alignment, because the missing days fall in January. Dataset B's source analysis reports these as daily means (DJF 0.4008, MAM 0.4579, JJA 0.4735, SON 0.5000); they are annualised here by the factor 252 to match Dataset A's convention.

Table 18 reports the seasonal decomposition, and its pattern requires comment because it runs opposite to the one established for wind speed in Chapter 2. Section 2.1.3 found the seasonal variance of wind *speed* to peak in winter in both datasets, by a ratio of roughly 6.9:1 for Dataset A and 1.55:1 for Dataset B. The realised variance of generation *log-returns* does the reverse: winter is the calmest season on this measure (100.8) and autumn the most volatile (126.0), with summer (119.3) also above winter.

The reversal is a consequence of the log transformation rather than a contradiction. A log-return measures relative change. In winter, generation is high, so a given absolute swing in MWh is a small proportional move; in summer, when mean output is low, the same absolute swing is a large proportional one. Because the seasonal cycle in the *level* of generation is stronger than the seasonal cycle in its absolute variability, the ratio inverts. This matters for a practical reason and is worth stating plainly: a seasonal variance swap on generation log-returns would be priced highest for a summer or autumn contract, exactly opposite to the intuition a producer would form from the seasonal wind-speed cycle. Both Phase 6 source documents assert the opposite pattern, reporting a winter figure of 144 against a summer figure of 80; neither figure appears in either notebook's executed output, and Table 18 reports the computed values.

3.1.4 Track B: The Model-Implied Fair Strike

Track B is evaluated at four values of the conditional-variance state h , following the scenario design specified for both datasets: the end-of-sample value $h(t_0)$, the winter (DJF) and summer (JJA) means of the fitted $h(t)$ series, and the Gaussian baseline $h = 1$, at which the GARCH layer is switched off and the strike reduces to the pure Fourier seasonal constant. Table 19 reports the resulting strikes for both datasets.

Three features of the table deserve comment.

First, in both datasets the summer scenario produces a *higher* strike than the winter scenario. For Dataset A the gap is substantial (0.0904 against 0.0730, a 24% difference), for Dataset B slight (0.8907 against 0.8586, 3.7%). Both are consistent with the seasonal

Table 19: Track B: model-implied fair strike $K_{\text{var}}^B(h) = h \times C_0$, four scenarios, both datasets.

Scenario	Dataset A ($C_0 = 0.13138$)		Dataset B ($C_0 = 1.16277$)	
	h	K_{var}^B	h	K_{var}^B
$h(t_0)$, end of sample	0.4137	0.05435	0.7511	0.87336
h_{winter} (DJF mean)	0.5559	0.07304	0.7384	0.85861
h_{summer} (JJA mean)	0.6877	0.09036	0.7660	0.89069
$h = 1$ (Gaussian baseline)	1.0000	0.13138	1.0000	1.16277

profile of $h(t)$ reported in Section 2.3.2, where Dataset A's conditional variance peaks in late spring and early summer while Dataset B's shows only a mild calendar range. The Track B scenario fan therefore runs in the same direction as the Track A seasonal pattern of Table 18 and opposite to the seasonal variance of wind speed, which is a coincidence of two distinct mechanisms rather than a single effect, but a convenient one: it means the model does not price the seasons backwards relative to the market.

Second, the spread across the four scenarios is much wider for Dataset A (a factor of 2.42 from $h(t_0)$ to $h = 1$) than for Dataset B (a factor of 1.33). This is the direct pricing consequence of the volatility-state contrast established in Section 2.3.2: Dataset A's pricing-date $h(t_0) = 0.4137$ sits far below its implied unconditional level of 0.608, whereas Dataset B's $h(t_0) = 0.7511$ sits essentially at its unconditional level of 0.7522. A pricing date chosen at the end of a calm spell moves the Dataset A strike materially and the Dataset B strike hardly at all.

Third, and most consequentially for Section 3.3, the two datasets' strikes differ by a factor of 16.07 at $h(t_0)$, and by 8.85 even at the common $h = 1$ baseline where the GARCH state is neutralised. The residual 8.85 $\times$ is the ratio $C_{0,B}/C_{0,A}$ alone.

3.1.5 Commensurability of the Two Tracks

Track A is a number near 113 and Track B a number near 0.87 or 0.054. Before any statement can be made about how close the model is to the market, the conventions separating them must be enumerated. There are four, and only the first is addressed by the delta-method rescaling recorded in the phase outputs.

(i) Process: wind anomaly versus generation log-return. Track B measures the variance of a de-seasonalised wind-speed anomaly; Track A the variance of a generation log-return. The power curve $G = cW^3$ of Section 3.1.6 bridges the two: $\log G = \log c + 3 \log W$, so that to first order

$$\text{Var}[\Delta \log G] \approx \frac{9}{\bar{W}^2} \text{Var}[\Delta W], \quad (16)$$

where $\bar{W}$ is the sample mean daily wind speed. This is the rescaling implemented in both notebooks as $K_{\text{var}}^{B,\text{scaled}} = (9/\bar{W}^2) h C_0$, with $\bar{W} = 2.4690$ m/s for Dataset A (giving a factor of 1.4764) and $\bar{W} = 8.0682$ m/s for Dataset B (giving 0.13826). The resulting scaled strikes are 0.08025 and 0.12075 respectively at $h(t_0)$.

(ii) Annualisation. Track A carries the factor 252 of Equation (14); Track B, being the annual mean of a daily variance function, does not. A like-for-like comparison requires multiplying Track B by 252. This single factor accounts for most of the apparent gap: $0.12075 \times 252 = 30.4$ against a Track A level of 113.1.

(iii) Measurement scale. The quantity $h \times C_0$ is a variance in Box–Cox-transformed units, because C_0 is the Fourier constant of the seasonal variance of $\varepsilon(t) = W^{(\lambda)}(t) - \Lambda(t)$ (Section 2.1.3), not of wind speed in m/s. Equation (16) requires the latter. The conversion is a second application of the delta method to the Box–Cox map itself: with $W^{(\lambda)} = (W^\lambda - 1)/\lambda$ and $dW^{(\lambda)}/dW = W^{\lambda-1}$,

$$\text{Var}[W] \approx \bar{W}^{2(1-\lambda)} \text{Var}[W^{(\lambda)}], \quad (17)$$

giving a factor of $2.4690^{1.8} = 5.088$ for Dataset A ($\lambda = 0.1$) and $8.0682^{1.0} = 8.068$ for Dataset B ($\lambda = 0.5$). This step is absent from the implemented rescaling, and it is not a small omission: it differs by a factor of 1.59 between the two datasets purely because they adopted different tractable values of λ . Equation (17) can be checked directly. At $h = 1$ it implies a de-seasonalised wind-speed standard deviation of $\sqrt{1.16277 \times 8.068} = 3.06$ m/s for Dataset B, against the $3.236 \times \sqrt{1 - 0.103} = 3.065$ m/s implied by the raw standard deviation and the seasonal-mean R^2 of Table 1.2 and Section 2.1.3. The agreement is to three significant figures. The same check for Dataset A gives 0.818 m/s against an implied 0.886 m/s, an 8% shortfall attributable to the much stronger non-linearity of the $\lambda = 0.1$ transform.

(iv) Innovation versus first difference. $h \times C_0$ is the conditional variance of the AR(4) *innovation*, since the GARCH recursion of Section 2.3 is fitted to the AR(4) residuals rather than to $\tilde{W}(t)$ itself, whereas Equation (16) calls for the variance of a daily *change*. The ratio between the two is available directly from the forecast errors of Section 3.2.1: the persistence root mean squared error is by construction the standard deviation of the one-day change in $\tilde{W}$, and the AR(4) root mean squared error is the standard deviation of the innovation, so the required factor is $(\text{RMSE}_{\text{pers}}/\text{RMSE}_{\text{AR}(4)})^2$, equal to 1.292 for Dataset A and 1.339 for Dataset B.

Table 20 assembles the chain. Two conclusions follow, and the second is uncomfortable.

The first conclusion vindicates the units argument stated in Section 1.5. Once all four conventions are applied, the model-implied strike and the market strike are of the same order of magnitude in both datasets, at ratios of 1.17 and 2.91 rather than the raw 1/2100 and 1/130. The gap of two to three orders of magnitude is a bookkeeping artefact, not evidence that the model is wrong by orders of magnitude, and the persistent characterisation of that gap in the source documents as evidence of “massive mispricing” or, in one place, of a “riskless arbitrage” available to a producer who sells at $K = 0.054$ and settles against a realised variance of 111, is a units error rather than an economic finding.

Table 20: Reconciling Track B into Track A units. Multiplicative chain, both datasets, at $h(t_0)$.

Factor	Dataset A	Dataset B
$K_{\text{var}}^B = h(t_0) C_0$ (as recorded)	0.05435	0.87336
$\times \bar{W}^{2(1-\lambda)}$ (Box–Cox scale, Eq. 17)	5.088	8.068
$\times (\text{RMSE}_{\text{pers}}/\text{RMSE}_{\text{AR}(4)})^2$ (innovation $\rightarrow$ difference)	1.292	1.339
$\times 9/\bar{W}^2$ (power curve, Eq. 16)	1.4764	0.13826
$\times 252$ (annualisation)	252	252
Reconciled model strike, Track A units	132.9	328.8
K_{var}^A (same sample)	113.999	113.145
Ratio, model to market	1.17	2.91
Reconciled strike at $h = 1$	321.3	437.7
Ratio, model to market, at $h = 1$	2.82	3.87

This chain is an arithmetic reconciliation carried out for this chapter; it is not a frozen phase output, and only the first and fourth rows correspond to quantities computed in the notebooks. Every step is a first-order delta-method approximation evaluated at the sample mean, so the reconciled figures are order-of-magnitude statements and the qualitative conclusions drawn from them do not depend on the third significant figure.

The second conclusion is that the reconciled strikes do not order the two datasets the way the raw strikes do. In reconciled units, Dataset B overstates the market strike by a factor of 2.91 while Dataset A overstates it by only 1.17. At the common $h = 1$ baseline the two are much closer to each other, at 2.82 and 3.87, a ratio of 1.36 against the 8.85 that the raw C_0 comparison reports. An independent check confirms this is not an arithmetic accident: a generation log-return responds to *relative* variation in wind speed, and the two sites have nearly identical coefficients of variation, $0.9481/2.4687 = 0.384$ for Bologna against $3.236/8.068 = 0.401$ for Nordfriesland (Tables 1.1 and 1.2). On the relative scale that the instrument actually references, the two sites are close to equivalent; the $8.85\times$ gap in C_0 reflects the absolute variance of a site whose mean wind speed is 3.3 times higher, compounded by the different Box–Cox exponent, and not a difference in relative variability.

That both datasets *overstate* the market strike, by factors of roughly three to four at the $h = 1$ baseline, has a straightforward physical explanation that applies equally to each. SMARD aggregates the output of thousands of turbines distributed across Germany, while both wind models are calibrated on a single point. Spatial averaging removes a substantial share of the variance of any weather field, so a point-calibrated model should be expected to imply more variance than the fleet aggregate realises, and by a factor of this order. The reconciliation therefore behaves as physical reasoning would predict, which is itself modest evidence that the chain in Table 20 is correctly specified.

In a trading context this difference in dispersion would not be left as an observation. A market maker writing a contract on the national aggregate while hedging against a model calibrated to a single reference node would apply a basis-risk adjustment to K_{var}^B , discounting the theoretical point variance to the aggregate variance the physical fleet actually realises and carrying the remainder as basis risk. The reconciliation above

already supplies an empirical magnitude for such an adjustment on this fleet and these two nodes: at the common $h = 1$ baseline the model-to-market ratios are 2.82 and 3.87, whose reciprocals place the required factor in the region of 0.26 to 0.35. No such factor is applied anywhere in this thesis, and that range should not be read as an estimate of one. Estimating it properly would require calibrating the model on a spatial panel of reference nodes rather than on a single centroid, so that the geographic diversification the fleet enjoys is measured directly rather than inferred from a single residual ratio, and that is left to future work.

The implication for Section 3.3 is stated there rather than here, but it should be signalled now: the headline Value of Information figures are computed on the raw strikes of Table 19, which are expressed in two different Box–Cox measurement scales and are therefore not strictly commensurable with one another. That does not invalidate them as a description of what the pipeline as implemented produces, but it does bear on how they should be read.

3.1.6 Power-Curve Linkage

The delta-method bridge of Equation (16) rests on the cubic power relation $G = cW^3$, which follows from the kinetic energy flux through the rotor disc and holds between a turbine’s cut-in and rated wind speeds. Its empirical counterpart is estimated by ordinary least squares of daily SMARD generation on the cube of daily mean wind speed,

$$G_t = c W_t^3 + b + \epsilon_t, \quad (18)$$

run separately on each dataset over its available overlap with the SMARD series. Table 21 reports the results, and Figure 8 shows the corresponding scatters.

Table 21: Power-curve regression of SMARD daily generation on cubed daily wind speed.

	Dataset A	Dataset B
Wind input	Bologna 10 m	Nordfriesland 100 m
Overlap window	2015–2018	2015–2024
Paired daily observations	1,461	3,649
Slope c [MWh per (m/s) ³]	36.43	155.12
Intercept b [MWh]	210,477	132,005
R^2	0.0001	0.5871
p -value on slope	0.723	< 0.001

The contrast is the sharpest single result in this thesis. Nordfriesland hub-height wind explains 58.7% of the daily variation in national onshore generation, which for a single grid point standing in for a fleet dispersed across a country of Germany’s size is a strong relationship: the unexplained 41% is attributable to spatial dispersion of the fleet, to installed-capacity growth over the decade, and to curtailment and outage events that no wind variable can capture. Bologna 10 m wind explains 0.01% and its slope

coefficient is not distinguishable from zero at any conventional level ($p = 0.723$). The intercept term is correspondingly revealing: for Dataset A it is 210,477 MWh, some 82% of the sample mean generation, meaning the fitted line is effectively a horizontal one drawn through the mean.

This is what geographic misalignment looks like when it is measured rather than asserted. German wind is driven by Atlantic low-pressure systems tracking across the North Sea; Bologna sits in the Po Valley, enclosed by the Alps and the Apennines, under a substantially Mediterranean synoptic regime. The two are not weakly related; on this evidence they are unrelated. Section 3.3 draws the consequence.

The reference figure should be read against a realistic ceiling rather than against unity, because $R^2 = 1$ is unattainable here in principle. A single grid point is being asked to explain a national aggregate: SMARD sums the output of a fleet dispersed from the North Sea coast to Bavaria, so that any single node, however well chosen, misses the share of daily variation generated by weather systems that reach one part of the country and not another. Three further sources of unexplained variation are structural rather than meteorological. Installed capacity grew substantially across the sample, so the same wind speed maps to different generation in 2015 and in 2024 (Section 3.1.2). Curtailment and outages break the mapping in the other direction, removing output that the wind was there to produce. And the SMARD series is not purely onshore in the sense the regression assumes, since the aggregate reported for Germany includes generation from coastal regions whose regime differs from a single inland-coastal node. Against that ceiling, 58.7% from one grid point is a strong result and not a disappointing one; the informative comparison is not 0.587 against 1 but 0.587 against 0.0001.

Two properties of the fitted relation are used later and are recorded where the regression is specified rather than where they are invoked. The first is that the cubic relation is a statement about instantaneous wind, so that $\mathbb{E}[W^3] \neq (\mathbb{E}[W])^3$ and any translation of a change in mean wind speed into a change in generation, such as the conversion of an annual wind deficit into a revenue shortfall in Section 3.4.3, is a first-order approximation that ignores the gap between the two. The gap is not negligible for a right-skewed variable, and it runs in the direction of understating the generation response to a given proportional change in wind. The mechanism is worth naming, since the direction is not obvious. Under a pure proportional rescaling of the entire wind distribution the response would be exactly 3δ in either direction and the approximation would be exact to first order; but a calm year is not a rescaling. It is the loss of the upper tail, and under a cubic map the upper tail is where almost all of the generation sits, so a deficit expressed as a change in the annual *mean* understates how much of the cube has actually been removed. The second is that the real relation is cubic only between cut-in and rated wind speed, being flat above rated and zero beyond cut-out, so the fitted slope is an average over a curve that is genuinely cubic across only part of its range. Both properties limit the precision of the delta-method chain in Table 20, and neither affects the sign or the order of magnitude of any conclusion drawn from it.

The regression also suggests a diagnostic of general application, developed further

in Section 3.5.3. Nothing in this exercise required the pricing model to be run before the mismatch was visible: the two regressions above cost a single ordinary-least-squares fit each and separate the two datasets more decisively than any quantity computed downstream of them. Where a candidate wind series is being selected for a pricing application, regressing it on the cash-flow series it is meant to hedge is therefore available as a screening step before calibration rather than as a validation step after it, and Section 3.3.3 shows why the ordering matters: the diagnostics computed downstream of the model do not reliably reveal what this one does.

One correction to the record is required. The Dataset A phase document reports a slope of 33.74, an intercept of 195,813 and a p -value below 0.0001, describing the regression as “statistically significant but explaining near-zero variance.” The executed notebook returns 36.43, 210,477 and $p = 0.7231$. The regression is not significant, and the corrected reading is the stronger one for the thesis’s argument: the proxy carries no measurable information about the series it would be used to price, rather than a statistically detectable but economically negligible amount.

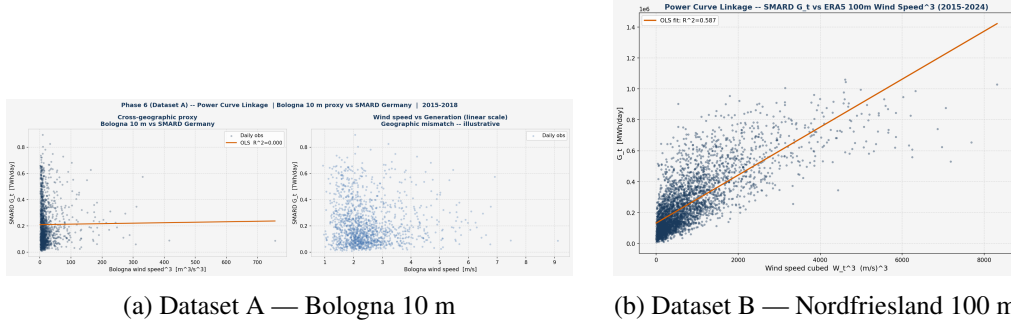


Figure 8: Power-curve linkage between wind speed and SMARD generation. **(a)** Dataset A, two panels: SMARD daily generation in TWh/day against cubed Bologna daily mean wind speed with the fitted ordinary-least-squares line overlaid (left), and the same generation against wind speed on a linear scale (right). The point cloud is structureless in both, consistent with the $R^2 = 0.0001$ of Table 21. **(b)** Dataset B, a single panel: SMARD daily generation in MWh/day against cubed Nordfriesland 100 m wind speed, with the fitted line. The visible upward slope and the tightening of the cloud along it are the $R^2 = 0.587$ of Table 21. Note that the two sub-panels use different axis units and different overlap windows and should be read for the shape of the relation rather than compared point by point.

3.1.7 The Merit-Order Effect

The economic case for a wind-linked hedge rests in part on the relationship between generation and price. Because wind enters the supply stack at near-zero marginal cost, high aggregate wind infeed displaces costlier plant and depresses the day-ahead clearing price. The correlation between daily generation and daily price should therefore be negative, and the analysis is identical for both datasets because both use the same market series.

Over the full 2015–2024 sample the Pearson correlation is $\rho(G, P) = -0.162$. Split at the October 2018 bidding-zone change, it is -0.476 on the 1,365 days before and -0.275 on the 2,284 days after. Both sub-periods carry the expected sign, so the merit-order finding survives the structural break, which was the robustness question the split was designed to answer.

The pattern nonetheless deserves more than the robustness verdict. The pooled correlation is weaker in magnitude than either of the sub-period correlations, which is not a possibility that a simple stability check anticipates. The mechanism is a pooling effect: mean price rose from €34.60/MWh over 2015–2020 to €126.47/MWh over 2021–2024 (Section 1.4.3), while installed wind capacity grew steadily across the same window. Pooling the two regimes introduces a large between-regime component into the variance of P that is driven by gas markets rather than by German wind, and that component dilutes the within-regime relationship. The merit-order effect is therefore materially stronger than the headline -0.162 suggests, and stronger still in the earlier part of the sample.

The two source documents differ in how they read the sub-period gap: the Dataset A notebook flags the 0.20 difference as a divergence to be interpreted with care, the Dataset B notebook attributes it to growing renewables capacity rather than to the price-zone split. Both readings are consistent with the pooling account above, and neither disturbs the sign, which is the only feature the hedging argument requires.

For a producer, the sign of ρ means that revenue, the product of quantity and price, is less volatile than quantity alone: a partial natural hedge. As Section 1.1 sets out, that hedge is both weak, at $|\rho| = 0.16$ over the full sample, and asymmetric, eroding the upside without placing a floor under the downside. It is the residual volumetric exposure, not the price exposure, that the instruments priced in this chapter address.

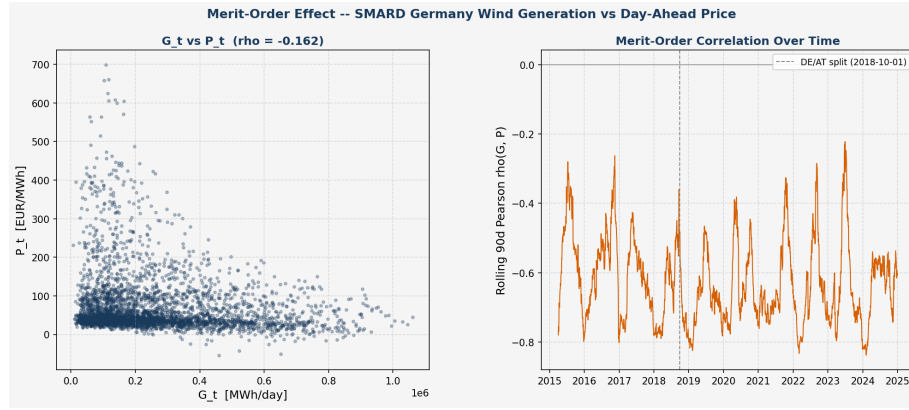


Figure 9: The merit-order effect. **Left:** scatter of daily day-ahead price against daily onshore wind generation, 2015–2024, with the full-sample correlation reported in the panel title. **Right:** the 90-day rolling Pearson correlation between the two series, showing that the relationship is persistently negative but far from constant, and that its magnitude declines over the sample in the way the pooling account of Section 3.1.7 describes. Both series are market data common to the two datasets, so the figure is shown once.

3.1.8 Payoff Simulation

The instrument is evaluated over the ten calendar years of the sample. For each year the annualised realised variance $RV_y^{\text{ann}} = 252 \times \overline{RV}_y$ is computed, and the payoff of a unit-notional long position is $RV_y^{\text{ann}} - K_{\text{var}}$ at each of the candidate strikes. Table 22 reports the annual series.

Three readings follow. Annual realised variance ranges from 95.5 in 2019 to 132.0 in 2024, the highest year exceeding the lowest by 38% about a mean near 115, so the quantity being swapped is genuinely variable year to year rather than close to deterministic. The mean payoff at the Track A strike is small and positive, +1.41 and +2.34 on the two samples, confirming that a strike set at the full-sample rolling mean was close to fair over the decade; the small positive bias reflects that the rolling estimator lags a mildly upward-trending series. And the fifth-percentile payoff of roughly –15 units, against a mean annual level of 115, quantifies the long party’s worst year in the sample at about 13% of the strike.

The Track B payoff columns computed in the notebooks, constant at approximately +115.3 in every year of both datasets, should not be read as an economic result and are not tabulated here. They are the mechanical consequence of differencing an annualised realised variance of order 115 against a model strike of order 0.1: the payoff is simply the realised variance itself, and its apparent risklessness is the units gap of Section 3.1.5, not an arbitrage. Once Track B is reconciled into Track A units, its Dataset B value of 328.8 sits above every annual realisation in Table 22 rather than below all of them, and a producer selling at that strike would have been paid every year of the decade, which is the opposite conclusion and equally a consequence of the model’s variance overstatement discussed in Section 3.1.5.

Table 22: Annual realised variance of SMARD generation log-returns and long variance-swap payoffs at the Track A strike.

Year	Dataset A sample RV^{ann}	Dataset A sample Payoff	Dataset B sample RV^{ann}	Dataset B sample Payoff
2015	120.83	+6.83	121.62	+8.48
2016	113.23	−0.77	113.23	+0.09
2017	129.17	+15.17	129.17	+16.03
2018	106.16	−7.84	106.16	−6.98
2019	95.49	−18.51	95.49	−17.66
2020	116.74	+2.75	116.74	+3.60
2021	118.97	+4.97	118.97	+5.82
2022	101.80	−12.20	101.80	−11.35
2023	119.71	+5.71	119.71	+6.56
2024	132.00	+18.00	132.00	+18.85
Mean	115.41	+1.41	115.49	+2.34
Std. deviation	11.54	11.54	11.59	11.59
5th percentile	—	−15.67	—	−14.82

Payoff is $RV_y^{\text{ann}} - K_{\text{var}}^A$ for a long unit-notional position; a producer takes the opposite side. Only the 2015 row differs between the two columns, for the alignment reason given in Section 3.1.2. Both Phase 6 source documents tabulate a different set of annual realised variances (Dataset A: 92.55, 128.41, 120.49, 102.44 for 2015–2018 only; Dataset B: 95.2 through 106.1 for 2015–2024); neither set appears in either executed notebook, and the values above are the computed ones.

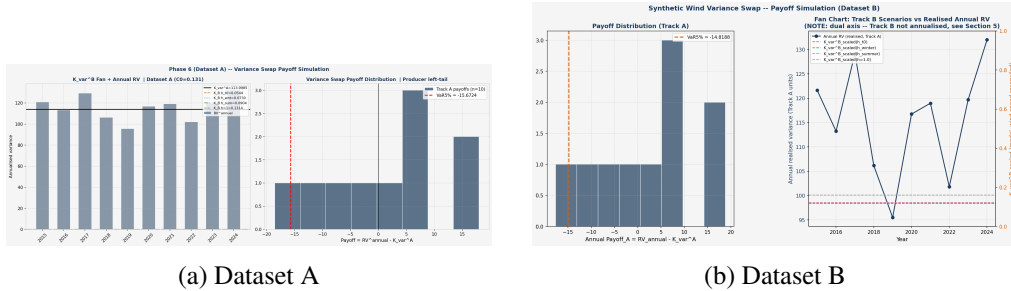


Figure 10: Variance-swap payoff simulation, both datasets. **(a)** Dataset A, two panels: annual realised variance as bars against the Track A strike and the four Track B scenario strikes as horizontal lines (left), and the histogram of the ten annual Track A payoffs with the fifth percentile marked (right). Because the Track B lines lie near zero on the annualised axis, they are visually indistinguishable from the horizontal axis, which is the units gap of Section 3.1.5 rendered graphically. **(b)** Dataset B, two panels: the histogram of annual Track A payoffs with its fifth percentile (left), and a dual-axis fan chart plotting annual realised variance on the left axis against the four unit-reconciled Track B scenario strikes on the right axis (right). The dual axis is necessary for exactly the reason the Dataset A panel does not need one, and the panel title in the source figure records the caveat.

3.2 Cross-Dataset Forecast Validation

3.2.1 The Diebold–Mariano Test and Its Conventions

The five specifications developed across Chapters 2 and 3 are compared on one-step-ahead forecasts of the standardised series $\tilde{W}(t)$ against a persistence benchmark $\hat{\tilde{W}}(t) = \tilde{W}(t-1)$, using the test of Diebold and Mariano [1995] with the small-sample correction of Harvey et al. [1997]. The loss differential is

$$d_t = e_{\text{persistence}}^2(t) - e_{\text{model}}^2(t), \quad (19)$$

so that a model outperforming the benchmark produces a *positive* statistic. This is the reverse of the ordering used in Section 2.2.4, where the model’s squared error is differenced first; the two are the same test reaching the same conclusion, and the sign convention is recorded here because the numerical values of the statistic differ in sign between the two chapters. The test statistic is $\bar{d}/\widehat{\text{se}}(\bar{d})$ multiplied by the Harvey–Leybourne–Newbold factor, which at a one-step forecast horizon reduces to $\sqrt{(n-1)/n}$, and referred to a t distribution with $n-1$ degrees of freedom. At the sample sizes involved the correction changes the statistic by about 0.03% and is immaterial to any conclusion; it is applied for correctness rather than for effect.

Two further conventions should be stated. First, the evaluation is conducted on the full standardised sample using the frozen Phase 2 residuals, not on a held-out partition, which is why the root mean squared errors below differ from those of Table 2.6. Second, the AR(4)+GARCH, CAR(4)+Gaussian and CAR(4)+NIG specifications enter the test with error series set identically equal to the AR(4) errors. This is not a coding shortcut but the implementation of the argument made in Section 2.2.4: the GARCH layer models the conditional variance of the residuals, the CAR(4) embedding is an exact continuous-time representation of the same mean dynamics, and a symmetric NIG innovation has the same conditional mean as a Gaussian one, so none of the three alters the one-step point forecast. The consequence is that their RMSE, DM statistic and p -value are identical to AR(4)’s by construction rather than by empirical coincidence, and no inference about their relative merit can be drawn from this table.

3.2.2 Results

Table 23 reports the outcome. Every fitted specification rejects equality of predictive accuracy with persistence at any conventional level, with relative improvements of 12.0% for Dataset A and 13.6% for Dataset B. The result is therefore a binary split rather than a graded hierarchy, exactly as anticipated in Section 1.5: the models beat the naive benchmark decisively and are observationally equivalent to one another at a one-day horizon by construction.

That equivalence is a property of the metric as much as of the models. Root mean squared error, and the Diebold–Mariano statistic built on it, evaluate only the conditional mean, and are therefore structurally blind to what the continuous-time,

Table 23: Diebold–Mariano tests against the persistence benchmark, one-step-ahead forecasts of $\tilde{W}(t)$, full sample.

Model	Dataset A		Dataset B	
	RMSE	DM stat	RMSE	DM stat
Persistence (benchmark)	0.8868	—	1.0046	—
AR(4)	0.7802	8.265	0.8681	16.758
AR(4)+GARCH	0.7802	8.265	0.8681	16.758
CAR(4)+Gaussian	0.7802	8.265	0.8681	16.758
CAR(4)+NIG	0.7802	8.265	0.8681	16.758
Improvement over persistence	12.0%		13.6%	
<i>p</i> -value	4.4×10^{-16}		$< 10^{-16}$	

The three specifications below AR(4) share its error series by construction (Section 3.2.1); their identical entries record that fact and are not independent evidence.

heteroskedastic and heavy-tailed extensions actually contribute: under a squared-error loss on the location, a specification earns nothing for characterising the variance or the tails correctly. Separating the four would require a strictly proper scoring rule defined on the whole predictive distribution, such as the continuous ranked probability score or a logarithmic predictive-density score. The Diebold–Mariano framework accommodates this without modification, since the test is defined on a loss differential and nothing in its construction requires that loss to be squared error; running the identical test on continuous-ranked-probability or log-score differentials would discriminate among the four specifications while leaving the testing apparatus untouched. The extension is thus a change of loss function rather than a change of test, and it is the natural way to supply the comparison that Section 3.2.1 declines to draw from the table as constructed.

The two datasets’ absolute figures differ, and both differences are accounted for. The root mean squared errors here (Dataset A: 0.7802 and 0.8868) exceed those of Table 2.6 (Dataset A: 0.6017 and 0.6856) because the present evaluation uses the full sample rather than the held-out partition. For Dataset B the two evaluations nearly coincide (0.8681 against 0.8580, and 1.0046 against 0.9955, both within 1%), while for Dataset A they differ by 23%. That asymmetry is itself informative and is consistent with the conditional-variance profiles of Section 2.3.1: Dataset A’s held-out window sits at the end of a sample where $h(t_0) = 0.4137$ against an unconditional level of 0.608, and $\sqrt{0.4137/0.608} = 0.825$ is close to the observed ratio of 0.773, whereas Dataset B’s $h(t_0) = 0.7511$ sits at its unconditional 0.7522 and predicts a ratio of essentially one against the observed 0.991. The held-out partition was, for Dataset A alone, an unusually calm period.

The Diebold–Mariano statistics stand at 8.265 and 16.758, a ratio of 2.03. A DM statistic scales with the square root of the evaluation sample, which here is the number of forecast errors, 3,652 and 1,461, one fewer than the 3,653 and 1,462 daily observations of Table 1.3 because the persistence benchmark differences the series; $\sqrt{3652/1461} = 1.58$ accounts for most of it; the residual factor of 1.28 is consistent with Dataset B’s larger relative accuracy gain and lower loss-differential dispersion. The

Dataset B phase document reads the full 2.03 as evidence that “the improvement in AR(4) is also genuinely larger,” which is directionally right but overstates the residual once the sample-size effect is removed. Section 2.2.4 performs the equivalent decomposition for the held-out statistics and reaches the same conclusion by the same route.

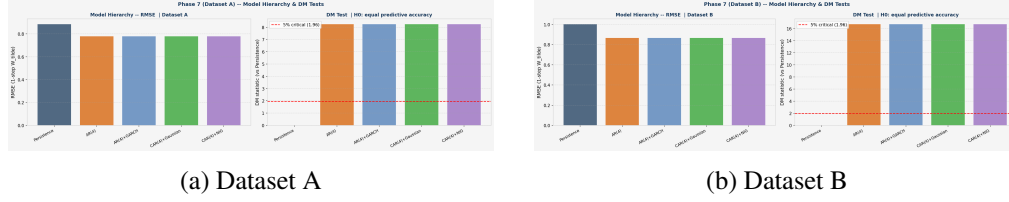


Figure 11: Model hierarchy and Diebold–Mariano tests, both datasets. Each panel has two plots. **Left:** one-step-ahead root mean squared error on $\tilde{W}(t)$ as a bar per model, with the persistence benchmark distinguished; the four fitted bars are of equal height for the construction reason given in Section 3.2.1. **Right:** the corresponding Diebold–Mariano statistics as bars, with the two-sided 5% critical value of 1.96 marked as a horizontal reference line, against which every fitted model stands several times higher.

3.2.3 GARCH Volatility Diagnostics

The adequacy of the Bollerslev [1986] GARCH(1,1) layer is re-examined on the standardised residuals $z(t) = \hat{\varepsilon}(t)/\sqrt{h(t)}$, using Ljung–Box tests on $z(t)$ and $z(t)^2$ at lags 5, 10 and 20 and the Engle [1982] Lagrange-multiplier test at lag 5. Table 24 reports the results.

Table 24: GARCH(1,1) standardised-residual diagnostics, p -values, both datasets.

Test	Dataset A	Dataset B
Ljung–Box $z(t)$, lag 5	0.9307	0.9820
Ljung–Box $z(t)$, lag 10	0.9896	0.5305
Ljung–Box $z(t)$, lag 20	0.7762	0.5112
Ljung–Box $z(t)^2$, lag 5	0.8196	0.0167
Ljung–Box $z(t)^2$, lag 10	0.9790	0.0810
Ljung–Box $z(t)^2$, lag 20	0.9886	0.2863
ARCH-LM, lag 5	0.8212	0.0185

Bold entries reject at 5%. These figures reproduce the lag-20 and ARCH-LM values reported in Table 2.9 to within rounding, the Dataset B ARCH-LM statistic exactly.

For Dataset A the verdict is unambiguous: every test passes comfortably, with no p -value below 0.77, and GARCH(1,1) has removed both the serial correlation and the conditional heteroskedasticity present in the AR(4) residuals. For Dataset B the picture is more qualified. The level tests pass at every lag, but the squared-residual test rejects at lag 5 ($p = 0.0167$), is marginal at lag 10 ($p = 0.0810$), and passes at lag 20 ($p = 0.2863$), while the ARCH-LM test rejects at 5% ($p = 0.0185$). Some short-lag volatility structure therefore survives the GARCH filter in the reference dataset.

This finding is not new to the present chapter: Section 2.3.3 identified the same ARCH-LM rejection and noted that the Phase 7 recomputation returns the identical

statistic. It is restated here for two reasons. The first is that both Phase 7 source documents record the Dataset B residuals as passing all diagnostics with $p > 0.05$, a characterisation their own executed output contradicts at two of the seven tests reported above. The second is that the rejection is at short lags only, which points to a specific and remediable limitation rather than to a failure of the specification: a GARCH(1,1) has a single decay rate and cannot simultaneously match a fast component operating over a few days and the slow, roughly 38-day component that dominates Dataset B’s persistence (Section 2.3.1). A two-component or GARCH(2,1) specification would be the natural response. That extension is not pursued here, because the pricing formula of Section 3.1.4 uses only $h(t_0)$ and the unconditional level, neither of which is materially altered by short-lag misfit, but the diagnostic should be reported as it stands rather than as it was described.

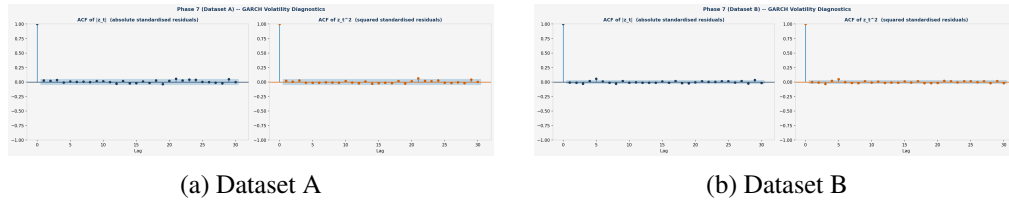


Figure 12: GARCH standardised-residual autocorrelation diagnostics, both datasets. Each panel has two plots: the sample autocorrelation function of $|z(t)|$ (left) and of $z(t)^2$ (right), the two standard graphical checks for residual volatility clustering. Bars falling inside the confidence band indicate that the corresponding lag carries no remaining structure. Dataset B’s short-lag bars in the right-hand plot are the graphical counterpart of the lag-5 rejections in Table 24.

3.2.4 Forecast-Error Distribution

The one-day-ahead forecast errors of each specification are tested for normality by the Jarque–Bera statistic. Table 25 reports the outcome.

Table 25: Jarque–Bera tests on one-day-ahead forecast errors.

Model	Dataset A		Dataset B	
	JB	p	JB	p
Persistence	59.96	< 0.001	0.756	0.685
AR(4)	108.87	< 0.001	17.05	< 0.001
CAR(4)+Gaussian	108.87	< 0.001	17.05	< 0.001
CAR(4)+NIG	108.87	< 0.001	17.05	< 0.001

The three fitted rows are identical for the construction reason given in Section 3.2.1. It is worth noting that they would be identical even if the NIG variant had been implemented as its source document describes, namely by dividing the AR(4) residuals by the scale parameter $\hat{\delta} = 2.087$ of the Normal-Inverse-Gaussian fit of Barndorff-Nielsen and Shephard [2001] reported in Section 2.4.4: the Jarque–Bera statistic depends only on the third and fourth standardised moments and is therefore invariant to any change of

location or scale. Dividing a residual series by a constant cannot change its skewness or its kurtosis, and so cannot change its Jarque–Bera p -value. The Dataset A phase document nevertheless reports a Jarque–Bera statistic of 3.7 with $p = 0.1596$ for the NIG variant and concludes that “NIG normalisation succeeds”; that figure appears nowhere in the executed output, and the claim is not merely unsupported but not the kind of claim the stated procedure could support.

The substantive content of the table lies in the other two results. Both datasets’ fitted residuals reject normality, Dataset A decisively ($JB = 108.9$) and Dataset B more mildly ($JB = 17.0$), which is the same ordering found for the GARCH standardised residuals in Section 2.4.4: Dataset A is leptokurtic and driven substantially by the single 5.79σ event of 28 June 2017, Dataset B only slightly non-Gaussian. And Dataset B’s persistence errors are, uniquely in the table, indistinguishable from normal ($JB = 0.756$, $p = 0.685$). The most likely explanation is that a first difference is the sum of two draws of opposite sign from a serially dependent series, and such a sum is closer to Gaussian than either component; the AR(4) innovation, by contrast, isolates the raw shock and preserves whatever non-normality it carries. This is offered as an interpretation of the pattern rather than as a separately tested result.

Two scope notes complete the section. The forecast-error analysis specified for this phase called for residuals at horizons of 1, 7 and 14 days and for quantile–quantile comparison against both the Normal and the NIG distributions. What was implemented is the one-day horizon only, compared against the Normal alone, in both datasets. The multi-horizon and NIG-quantile comparisons are therefore not available, and the question of whether the NIG assumption offers a statistically significant improvement over the Gaussian is not answered by this evidence. The answer it would have given is nevertheless partly foreshadowed by Section 2.4.4, where the NIG fit to Dataset A’s standardised residuals improved the Kolmogorov–Smirnov p -value from 0.079 to 0.120, a modest gain, while for Dataset B the fitted tail parameter $\hat{\alpha} \approx 2,650$ left the NIG indistinguishable from the Gaussian. The Gaussian CAR(4) specification is retained as the primary pricing track for both datasets on that basis.

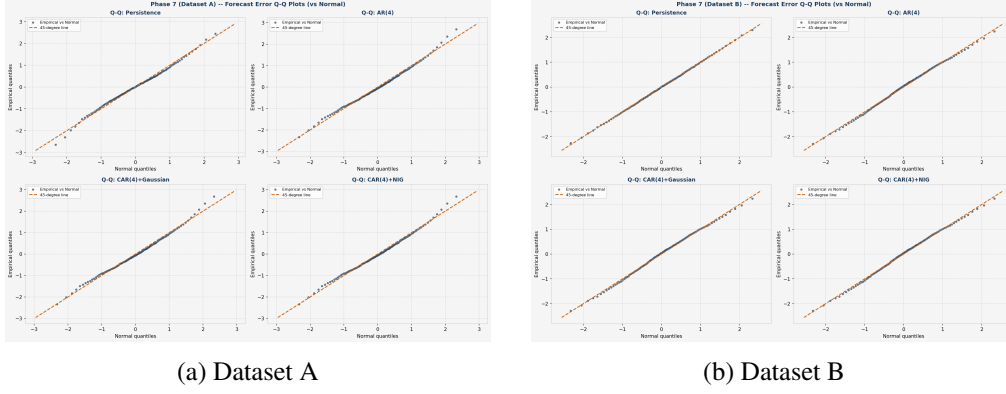


Figure 13: Quantile–quantile plots of one-day-ahead forecast errors against the normal distribution, both datasets. Each panel is a 2×2 grid, one plot per model in the order persistence, AR(4), CAR(4)+Gaussian, CAR(4)+NIG, with the empirical quantiles of the standardised errors plotted against normal quantiles and the 45-degree line overlaid. Because the errors are standardised before plotting and the three fitted models share one error series, their three plots are identical within each panel; only the persistence plot differs. Departure from the 45-degree line at the extremes is the leptokurtosis quantified in Table 25, pronounced for Dataset A and slight for Dataset B.

3.3 Value of Information

3.3.1 The Height Correction

The two datasets are measured at different heights, so a comparison of their implied strikes must either adjust for height or acknowledge that it has not. The adjustment used is the power-law profile of Peterson and Hennessey [1978], $u(z) \propto z^\alpha$ with $\alpha = 1/7$; since variance scales as the square of velocity, the variance correction from 10 m to 100 m is

$$\left(\frac{100}{10}\right)^{2\alpha} = 10^{2/7} = 1.9307. \quad (20)$$

As stipulated in Section 1.3.1, this factor is applied in this section and nowhere else in the thesis: Dataset B is already at hub height and requires no correction, and no other calculation in Chapters 2 or 3 places the two datasets on a common vertical axis.

Because the correction is applied to Dataset A alone, the exponent must be justified for Bologna rather than for the German site, and the relevant roughness is therefore that of an inland basin, not of a coastal one. The exponent is empirical rather than physical, rising with surface roughness and with atmospheric stability from about $1/10$ over open water to substantially more over broken terrain. Bologna’s anemometer sits inside the surface friction layer of the Po Valley, enclosed by the Alps and the Apennines, so its profile should be steeper than the $1/7$ adopted here, not shallower. Dataset B supplies a same-site check on the order of magnitude involved, since ERA5 reports both a 10 m and a 100 m wind speed at the Nordfriesland coordinates and Section 1.4.2 retains the former as a consistency series: their sample means over 2015–2024 are 5.4825 and 8.0682 m/s, a ratio of 1.4716 and hence an empirical exponent of $\log_{10}(1.4716) = 0.168$

at that location.³ A flat, open coastal site therefore already exceeds 1/7, and an enclosed inland basin would be expected to exceed it by more, so the 0.168 is a lower bound on the exponent appropriate to Bologna rather than an estimate of it.⁴

The consequence runs opposite to the direction in which a conservative choice would run, and is worth stating plainly because the point is easy to invert. A larger exponent produces a larger correction factor, hence a larger corrected proxy strike, hence a *smaller* residual mispricing. If Bologna’s true exponent exceeds 1/7, as the reasoning above suggests, then the 88.0% residual reported in Section 3.3.2 overstates rather than understates the share of the gap attributable to geography, and is an upper bound on that share within the plausible range rather than a lower one.

Table 26 therefore reports the conclusion across the whole of that range rather than at a single value. What the table establishes is not that 1/7 is the right exponent but that the choice does not matter: over every value from open water to very rough terrain the mispricing remains above 80%, so no admissible profile assumption converts the gap between the two datasets into a height effect. That is a stronger claim than the point estimate, and it removes the exponent from the argument.

Table 26: Sensitivity of the height-corrected mispricing to the profile exponent α .

Exponent α	Factor $10^{2\alpha}$	Corrected K_{var}^B, A	Mispricing
1/10 (open water)	1.585	0.0861	90.1%
1/7 (adopted)	1.931	0.1049	88.0%
0.168 (ERA5 coastal, lower bound)	2.166	0.1177	86.5%
0.20 (rough inland)	2.512	0.1365	84.4%
0.25 (very rough)	3.162	0.1719	80.3%

Computed from the raw Dataset A strike 0.05435 and the Dataset B reference 0.87336 of Table 27; the 1/7 row reproduces the frozen figures exactly. Only that row is a phase output, the remainder being an arithmetic sensitivity carried out for this chapter. The 0.168 row is the exponent measured at the coastal Nordfriesland node and is a lower bound on the value appropriate to Bologna, not an estimate of it. The mispricing column falls as α rises, so the adopted 1/7 is not the conservative choice; the operative result is the floor of 80.3% across the whole range.

3.3.2 Raw and Height-Corrected Mispricing

The Value of Information is defined as the relative gap between the strike a model calibrated on the proxy would quote and the strike the aligned model quotes,

$$\text{VoI} = \frac{|K_{\text{var}}^{B, \text{Dataset B}} - K_{\text{var}}^{B, \text{Dataset A}}|}{K_{\text{var}}^{B, \text{Dataset B}}}, \quad (21)$$

³The 10 m mean is computed for this chapter directly from the raw ERA5 file rather than taken from a phase output, since the companion series is retained but never summarised in the Phase 1 analysis. All 3,653 days of the sample are complete at 24 hourly values, so the mean of the daily means and the mean of the 87,672 hourly values coincide; the 100 m mean computed by the identical procedure returns 8.0682 m/s, reproducing to four decimal places the figure the Phase 1 analysis reports for that series, which is the check that the unreported 10 m mean is produced correctly.

⁴The Dataset B Phase 1 analysis offers a different internal check, reading the Bologna-to-Nordfriesland ratio of means, $8.068/2.470 \approx 3.27$, as “consistent with a power-law wind profile exponent near 0.2 at this terrain roughness.” That reading cannot be sustained: the two figures are means at two different sites, so their ratio is not a vertical profile at all, and if it were, it would imply $\log_{10}(3.27) = 0.51$ rather than 0.2. The same-site 10 m against 100 m comparison reported above is the check that statement was reaching for.

evaluated both on the raw Dataset A strike and on its height-corrected counterpart. Table 27 reports the result.

Table 27: Value of Information: model-implied fair strikes and relative mispricing at $h(t_0)$.

Quantity	Value	Mispricing vs. Dataset B
Dataset A, raw ($C_0 = 0.13138$, $h(t_0) = 0.4137$)	0.05435	93.8%
Dataset A, height-corrected ($\times 1.9307$)	0.10494	88.0%
Dataset B ($C_0 = 1.16277$, $h(t_0) = 0.7511$)	0.87336	— (reference)
Ratio K_B^B/K_A^B	16.07	
attributable to $h(t_0)$: $0.7511/0.4137$	1.82	
attributable to C_0 : $1.16277/0.13138$	8.85	

Pricing the identical CAR(4)+GARCH variance swap on the Bologna proxy rather than on the Nordfriesland reference understates the fair strike by 93.8%. Correcting the proxy for measurement height recovers a little under a quarter of the shortfall on a logarithmic scale and leaves 88.0% of it in place. The residual is the informative quantity: whatever separates the two datasets, it is overwhelmingly not measurement height. Table 26 has already shown that no admissible exponent brings the residual below 80%; the limiting case makes the same point from the other side, since a correction factor of 16.07 rather than 1.93 would require $\alpha = \frac{1}{2} \log_{10}(16.07) = 0.60$, some four times the Hellmann value and far outside the range Peterson and Hennessey [1978] report for any surface.

The decomposition in the lower panel of Table 27 locates the gap. The ratio factorises exactly into a conditional-variance component of 1.82 and a seasonal-variance component of 8.85, so roughly four-fifths of the discrepancy on a logarithmic scale is carried by C_0 , the annual mean level of the Fourier seasonal variance, and only one-fifth by the GARCH state. This is the quantitative basis for the claim, stated in Section 1.2 and repeated here, that the pricing error is located in the site’s climatology rather than in its volatility state.

Two qualifications on that decomposition are due, and neither was made in the source documents.

The first concerns the $1.82\times$ factor. The two values of $h(t_0)$ are end-of-sample conditional variances at two *different pricing dates*, 31 December 2018 for Dataset A and 31 December 2024 for Dataset B, because the two samples end six years apart. The ratio therefore compares two dates as much as it compares two sites. Section 2.3.2 records that Dataset A’s $h(t)$ ranges over monthly means from 0.465 to 0.768; had Dataset A’s sample ended in a different month, this factor could plausibly have taken any value between about 1.0 and 1.6. It is not a stable site characteristic and should not be reported as one.

The second is the commensurability point established in Section 3.1.5. The two strikes in Table 27 are variances expressed in two different Box–Cox measurement scales, $\lambda = 0.1$ for Dataset A and $\lambda = 0.5$ for Dataset B, and are therefore not, strictly speaking, quantities of the same kind. Table 20 showed that mapping both onto the

common axis of generation log-return variance reduces the $16.07\times$ ratio to about $2.5\times$, of which the $1.82\times$ date effect is the larger part and the climatological difference only about $1.36\times$. The 93.8% and 88.0% figures are accurate statements about what the two calibrated pipelines quote, and they are the figures a practitioner comparing the two model outputs would confront, so they are retained as the headline result. But they are not scale-free, and the underlying difference in relative variability between the two sites is far smaller than they suggest. This qualification does not weaken the case for the aligned dataset; it relocates the evidence for it, as Section 3.3.3 sets out.

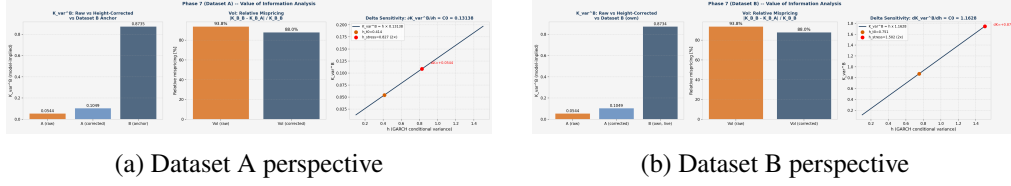


Figure 14: Value of Information analysis, computed from each dataset’s own notebook. Each panel has three plots. **Left:** the model-implied fair strike as bars, for the raw Dataset A value, its height-corrected counterpart, and the Dataset B reference, showing how little of the gap the $1.9307\times$ correction of Equation (20) closes. **Centre:** the raw and height-corrected relative mispricing, 93.8% and 88.0%, as bars. **Right:** the delta sensitivity $K_{\text{var}}^B = h C_0$ plotted as a function of h , a straight line through the origin of slope C_0 , with the pricing-date and stress values marked (Section 3.3.4). The two panels differ only in which dataset’s C_0 sets the slope in the right-hand plot; the left and centre plots report the same cross-dataset comparison from either side.

3.3.3 The Empirical-Distance Metric

A second diagnostic was specified alongside the relative mispricing: the absolute distance between each dataset’s unit-reconciled model strike and the market strike it is meant to approximate,

$$\text{dist} = |K_{\text{var}}^A - K_{\text{var}}^{B, \text{scaled}}(h(t_0))|, \quad (22)$$

with the ratio $\text{dist}_A/\text{dist}_B$ intended to measure how much closer to its own market the aligned model sits. The computed values are 113.918 for Dataset A, which is 99.93% of that sample’s Track A level, and 113.024 for Dataset B, which is 99.89% of its own, giving a ratio of 1.0079.

The metric, as constructed, carries essentially no information, and the reason is the annualisation convention of Section 3.1.5: both scaled Track B strikes are of order 0.1 while both Track A strikes are of order 113, so each distance is, to within a tenth of a percent, just the Track A level itself. A metric that returns the benchmark regardless of the model cannot discriminate between models. The Dataset B notebook reaches the same verdict in its own output, observing that the 113-unit gap “swamps the smaller geographic/height effect this diagnostic was meant to isolate,” and that assessment is endorsed here.

The rolling counterpart behaves the same way. The root mean squared distance between the rolling Track A series and the rolling unit-reconciled Track B series is

120.86 for Dataset A, computed over 1,335 overlapping days falling within 2016–2018, and 113.95 for Dataset B over 3,397 days spanning 2016–2024. Each sits within 6% and 1% respectively of its own Track A level, and the difference between them reflects mainly the different and non-comparable evaluation windows rather than any property of the two models.

The conclusion is that the case for the aligned dataset does not rest, and cannot rest, on the distance between the model strike and the market strike, for two independent reasons: the distance metric as implemented is dominated by a units convention, and once the units are fully reconciled (Table 20) the proxy is if anything nearer the market level than the reference dataset is. The case rests instead on Section 3.1.6. A model whose input explains 58.7% of the variation in the series being hedged is usable for hedging; one whose input explains 0.01% is not, whatever variance level it happens to imply. Dataset A’s apparent proximity to the market strike in reconciled units is a coincidence of level between two unrelated series, and is worth stating precisely because it is the sort of coincidence that a level-based validation metric would mistake for a successful calibration. This is the Value of Information properly understood: not that the proxy quotes the wrong number, but that it quotes a number carrying no information about the risk it would be used to transfer.

3.3.4 Delta Sensitivity and the Stress Scenario

Because the fair strike is linear in the conditional-variance state, its sensitivity is the seasonal constant itself,

$$\frac{\partial K_{\text{var}}^B}{\partial h} = C_0, \quad (23)$$

equal to 0.13138 for Dataset A and 1.16277 for Dataset B. This is the variance swap’s delta with respect to the volatility state, and it is constant: the instrument has no convexity in h , which follows directly from the piecewise-constant convention and is the pricing counterpart of the linearity of the payoff noted in Section 3.1.1.

A stress scenario in which a persistent wind regime shift doubles the conditional variance, $h_{\text{stress}} = 2h(t_0)$, shifts the strike by

$$\Delta K_{\text{var}}^B = (h_{\text{stress}} - h(t_0)) C_0 = h(t_0) C_0 = K_{\text{var}}^B(h(t_0)), \quad (24)$$

that is, by exactly the strike itself: +0.0544 for Dataset A, taking the strike to 0.1087, and +0.8734 for Dataset B, taking it to 1.7467. The identity is trivial but the risk-management reading is not. It says that the model-implied strike is proportional to the volatility state with no attenuation, so a hedging book marked on this model carries full linear exposure to a regime shift in h ; and it says that the absolute size of that exposure is set by C_0 , so the aligned dataset’s book is 8.85 times more sensitive in absolute terms than the proxy’s, though identically sensitive in proportional terms. Whether such a doubling is plausible over a contract horizon is a question this framework can answer only indirectly: Section 2.3.1 puts the volatility half-life at 41.6 and 38.3 days, so a shock large enough to double h would still retain roughly 70% of its effect after 20 days

and would not decay below a quarter of its initial size for about three months.

3.3.5 Out-of-Time Validation

Dataset B’s ten-year window permits a test that Dataset A’s four-year window does not. The full-sample unit-reconciled strike $K_{\text{var}}^{B, \text{scaled}}(h(t_0)) = 0.12075$ is held fixed as a model prediction and evaluated as a constant forecast of the rolling Track A series, separately on the 1,570 days up to 31 December 2019 and the 1,827 days from 1 January 2020. The root mean squared errors are 113.232 and 114.569 respectively, an increase of 1.18%.

The exercise must be read for exactly what it is. It is an out-of-time evaluation under fixed parameters, not genuine out-of-sample calibration: C_0 and $h(t_0)$ were both estimated on the full 2015–2024 window, so the hold-out period is contained in the estimation sample. It therefore tests whether the relationship the model captures is stable across the two halves of the decade, not whether the model would have forecast the second half from the first. And, for the reason given in Section 3.3.3, both root mean squared errors are dominated by the annualisation gap and are close to the Track A level itself, so the 1.18% deterioration is best read as a statement about the stability of Track A across the two sub-periods rather than about the model. What it does establish, weakly but genuinely, is that nothing in the 2020–2024 window, including the price crisis and the substantial capacity additions of those years, destabilised the variance of generation log-returns enough to register.

It is worth recording what a stronger test would have required, since the gap between the two is the distance still to be covered. Genuine out-of-sample calibration would re-estimate the whole pipeline on 2015–2019 alone, deriving λ , the Fourier coefficients and hence C_0 , the AR(4) coefficients and the GARCH parameters from that window only, and would then price the swap for 2020–2024 with no information from the hold-out period. That is feasible on Dataset B’s ten years and on no other object in this thesis, since Dataset A’s four-year window would leave too little to estimate a seasonal cycle from. It would also require an evaluation target that discriminates, which the annualisation gap of Section 3.1.5 currently prevents: a root mean squared error dominated by a 113-unit constant cannot register a change in the model. Reconciling both tracks onto a common axis first, as Table 20 does statically, is a precondition for the exercise rather than an adjunct to it. The test is left undone here, and its absence is why no claim of predictive validation is made anywhere in this chapter.

3.4 Risk Management: The CAWS Instrument

3.4.1 Definition and Fair Strike

A variance swap is indifferent to the direction of the underlying’s movement. It pays on realised variance, so a sustained calm period, the single event a wind producer most needs protection against, registers as low variance rather than as a loss. The left tail

of wind-linked revenue risk is therefore not addressed by the instrument of Section 3.1, and this section evaluates a complement to it.

The Cumulative Accumulated Wind Speed index of Alexandridis and Zapanis [2013] is taken in the normalised rolling-annual-mean form defined in Equation (1.1) of Section 1.4.4,

$$\text{CAWS}_t = \frac{1}{252} \sum_{s=t-251}^t W(s), \quad (25)$$

with $W(s)$ the daily mean wind speed in m/s. Normalising by the window length rather than accumulating raw sums keeps the index in m/s and therefore comparable across the two measurement heights. The strike is the full-sample mean, $K_{\text{CAWS}} = \mathbb{E}[\text{CAWS}_t]$, and a producer buying the instrument at that strike receives

$$\Pi_t = N \times (K_{\text{CAWS}} - \text{CAWS}_t), \quad (26)$$

which is positive when realised wind falls short of the strike. The instrument is thus a put on the wind resource: it pays in exactly the state, a calm year, in which the producer's generation revenue is depressed.

The fitted strikes are $K_{\text{CAWS}} = 2.4644$ m/s for Dataset A, computed over 1,263 rolling observations with a rolling standard deviation of 0.2013 m/s, and 8.0531 m/s for Dataset B, over 3,454 rolling observations with a rolling standard deviation of 0.5002 m/s. Both are consistent with the raw sample means of Tables 1.1 and 1.2, as they must be.

Striking the contract at the unconditional mean gives a clean baseline for evaluating it, but standard practice in volumetric hedging favours out-of-the-money strikes: setting the strike at a specified deficit below the mean restructures the instrument from an at-the-money swap into dedicated wind-drought insurance, lowering the premium the producer pays while confining the protection to severe years. The annual observations of Table 28 bound how far out of the money such a strike could usefully sit, and the bound is tighter than the design would suggest. A strike ten per cent below the mean would lie at 2.2180 m/s for Dataset A and 7.2478 m/s for Dataset B, and neither would have been reached in any year of either sample, since the deepest observed deficits are the 6.70% of 2017 and the 4.99% of 2021 recorded in Table 29; such a contract would have paid nothing across the full decade. A five per cent strike would have triggered once for Dataset A, in 2017, and not at all for Dataset B, the 2021 deficit falling about one-hundredth of a percentage point short, while a level near four and a half per cent triggers exactly once in each dataset, in 2017 and 2021 respectively, which is the severe wind-drought year the instrument is meant to isolate. That the design turns on so narrow a range of moneyness is itself informative, and the restructuring carries a cost the at-the-money baseline does not: with four and ten annual observations, reducing the non-zero payoffs to one per dataset would leave the distribution metrics of Table 29 not merely imprecise but unestimable. The variant is not endorsed here. It is the design the at-the-money baseline points toward, and its price is that the evidence available to calibrate it is thinner still than the evidence already treated as descriptive rather than

inferential in Section 3.4.2.

3.4.2 Distribution Metrics on Non-Overlapping Annual Observations

The rolling construction induces severe serial correlation: adjacent values of $CAWS_t$ share 251 of their 252 constituent observations and are therefore almost perfectly correlated, so any standard deviation or quantile computed on the rolling series would be badly biased. All distribution metrics are consequently computed on non-overlapping calendar-year means, one independent observation per year, giving four observations for Dataset A and ten for Dataset B. The rolling series is retained only for plotting. Table 28 reports the annual observations and Table 29 the resulting metrics.

Table 28: Non-overlapping annual CAWS observations and payoffs, both datasets.

	Dataset A ($K = 2.4644$)		Dataset B ($K = 8.0531$)	
Year	CAWS [m/s]	Payoff	CAWS [m/s]	Payoff
2015	2.4872	−0.0228	8.5626	−0.5095
2016	2.5041	−0.0397	7.7162	+0.3369
2017	2.2993	+0.1650	8.0879	−0.0347
2018	2.5853	−0.1209	7.8396	+0.2135
2019	—	—	8.2526	−0.1995
2020	—	—	8.2830	−0.2299
2021	—	—	7.6513	+0.4018
2022	—	—	8.1063	−0.0532
2023	—	—	8.1416	−0.0885
2024	—	—	8.0410	+0.0121

Table 29: CAWS distribution metrics on non-overlapping annual observations.

Metric	Dataset A	Dataset B
K_{CAWS} [m/s]	2.4644	8.0531
Annual observations n	4	10
Mean payoff [m/s]	−0.0046	−0.0151
Std. deviation of payoff [m/s]	0.1209	0.2754
Sharpe ratio	−0.0379	−0.0549
$VaR_{5\%}$ [m/s]	−0.1087	−0.3837
$CVaR_{5\%}$ [m/s]	−0.1209	−0.5095
$CVaR_{5\%}$ as % of K_{CAWS}	−4.91%	−6.33%
Largest positive payoff [m/s]	+0.1650 (2017)	+0.4018 (2021)
as % of K_{CAWS}	6.70%	4.99%

3.4.3 Reading the Tail

Three properties of Table 29 determine how it should be read, and each cuts against a reading offered in the source documents.

First, the mean payoff is near zero by construction, because the strike is set to the sample mean of the very series it is evaluated against. It is -0.0046 and -0.0151 m/s, differing from exactly zero only because the strike is the mean of the rolling series while the payoffs are computed on calendar-year means. The Sharpe ratios of -0.038 and -0.055 that follow from dividing a near-zero mean by a positive standard deviation are therefore not measures of the instrument's performance; they carry no information beyond the arithmetic that produced them, and the Dataset A notebook says as much in its own output. A Sharpe ratio is a meaningful statistic for a traded position with a market-determined entry price; it is not one for a contract struck at the mean of its own underlying.

Second, and more consequentially, the sign convention means that the reported $CVaR_{5\%}$ describes the wrong tail for a hedger. The payoff is $K_{CAWS} - CAWS$, so a negative payoff arises in a *windy* year. Dataset B's $CVaR_{5\%} = -0.5095$ m/s is the payoff of 2015, the windiest year of the decade at 8.5626 m/s, in which the producer's own generation was at its highest. That is not a risk; it is the hedge behaving as designed, with the producer paying the insurance leg out of an unusually good year's revenue. The Dataset B phase document reads this figure as a downside, describing it as "a worst-case scenario (5% tail risk) of wind being 0.5 m/s below fair strike" implying an 18% revenue loss, which inverts the sign of the observation: 2015 was 0.51 m/s *above* strike, not below. The tail that matters to a producer is the opposite one, the largest positive payoffs, and those are reported in the final rows of Table 29.

Third, with $n = 4$ and $n = 10$ the $CVaR_{5\%}$ is not a tail average at all. In both datasets it equals the single most negative annual observation exactly, because no observation other than the minimum falls at or below the fifth percentile of so small a sample. The two conditional value-at-risk figures are therefore sample minima under another name, and the same holds for the fifth-percentile value-at-risk, which is an interpolation between the two most negative points. This is the unavoidable cost of the non-overlapping-window choice made in Section 3.4.2: it purchases freedom from serial-correlation bias at the price of four and ten observations respectively, and the resulting tail statistics are descriptive rather than inferential, exactly as Section 1.2 anticipated.

With those three corrections applied, what the table shows is the following. The protection leg of the instrument is economically substantial. The calmest year in Dataset B's decade, 2021, came in 4.99% below the strike and would have paid $+0.4018$ m/s per unit of notional; Dataset A's calmest year, 2017, came in 6.70% below and would have paid $+0.1650$ m/s. Translated through the cubic power relation of Section 3.1.6, a 4.99% shortfall in mean wind speed corresponds to roughly a 14% shortfall in generation, and a 6.70% shortfall to roughly 19% . Applied to the illustrative 100 MW farm of Section 1.1, generating some 225,100 MWh in a normal year, a 14% shortfall is about 32,000 MWh; valued at the full-sample mean day-ahead price of $\text{€}71.38/\text{MWh}$, that is

roughly €2.3 million of foregone gross revenue in a single calm year, or about €1.1 million at the 2015–2020 mean price and €4.1 million at the 2021–2024 mean. These are order-of-magnitude illustrations built transparently from a stated capacity factor and observed prices, in the manner of Section 1.1, and the cubic mapping is a first-order approximation that ignores the difference between the cube of the mean and the mean of the cube.

A single episode in the data is worth noting alongside them, with its limits stated. Dataset B’s calmest year, 2021, coincided with the onset of the European energy-price crisis, so a producer suffering that volumetric shortfall would in fact have sold the reduced output into a sharply higher price and might well have earned more, not less, than in a normal year. That is the natural hedge of Section 3.1.7 operating at its most favourable. It should not be generalised: the 2021–2022 price surge was driven by gas markets rather than by German wind conditions, and a single coincidence of that kind is not evidence that the natural hedge places a floor under the volumetric downside. If anything it illustrates the asymmetry stated in Section 1.1, since the same producer would have received no comparable compensation in the calm years of a normal price regime.

Finally, the cross-dataset comparison of the CAWS metrics needs the same units discipline applied to the variance swap in Section 3.1.5. Dataset B’s $\text{CVaR}_{5\%}$ is 4.21 times Dataset A’s in m/s, and the Dataset B phase document reads this as “higher tail risk (4.2 $\times$).” It is not: it is chiefly the ratio of the two mean wind speeds, since a metric denominated in m/s at 100 m cannot be compared with one at 10 m. Normalised by the respective strikes, the two figures are -4.91% and -6.33% , a difference of about 30% rather than of a factor of four. The genuine advantage of Dataset B here is not a larger tail but a better-estimated one: ten independent annual observations against four.

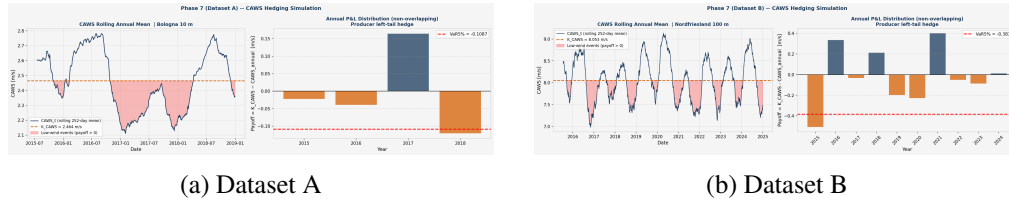


Figure 15: CAWS hedging simulation, both datasets. Each panel has two plots. **Left:** the rolling 252-day CAWS series in m/s with the fair strike K_{CAWS} marked as a horizontal line; the smoothness of the series is the serial correlation that motivates computing all metrics on non-overlapping annual observations instead. **Right:** the non-overlapping annual payoffs $K_{\text{CAWS}} - \text{CAWS}_y$ as bars, one per calendar year, with the zero line and the fifth-percentile value-at-risk marked. Positive bars are the protection leg, paying in calm years; negative bars are the premium leg, paid in windy years. Note that the two panels use different vertical scales, in m/s at 10 m and at 100 m respectively, and should be compared only after the normalisation of Table 29.

3.5 Conclusions

3.5.1 Summary of Findings

Table 30 collects the chapter’s principal results. The three research questions posed in Section 1.2 can now be answered.

Table 30: Summary of Chapter 3 results, both datasets.

	Dataset A (Bologna 10 m)	Dataset B (Nordfriesland 100 m)
Track A fair strike K_{var}^A	113.999	113.145
Track B fair strike $K_{\text{var}}^B(h(t_0))$	0.05435	0.87336
Track B, unit-reconciled (Table 20)	132.9	328.8
Power curve R^2	0.0001 ($p = 0.723$)	0.5871 ($p < 0.001$)
Merit-order $\rho(G, P)$	−0.162 full sample; −0.476	pre-2018-10, −0.275 post
AR(4) RMSE vs. persistence	0.7802 vs. 0.8868 (12.0%)	0.8681 vs. 1.0046 (13.6%)
Diebold–Mariano statistic	8.265 ($p < 10^{-15}$)	16.758 ($p < 10^{-16}$)
GARCH adequacy	All tests pass	Rejects at z^2 lag 5 and ARCH-LM
Jarque–Bera, AR(4) errors	108.87 ($p < 0.001$)	17.05 ($p < 0.001$)
Value of Information (raw / corrected)	93.8% / 88.0%, Dataset B as reference	
K_{CAWS}	2.4644 m/s ($n = 4$)	8.0531 m/s ($n = 10$)
CVaR _{5%} , % of strike	−4.91%	−6.33%

RQ1: can the framework price the instrument? Yes, and in closed form. Under the $\theta = 0$ benchmark the fair strike collapses to $K_{\text{var}}^B = h \times C_0$, requiring no numerical integration and no Girsanov computation, and Section 3.1.4 evaluates it at four volatility states for both datasets. Two qualifications attach to that affirmative answer. The strike is a conditional quantity that inherits the pricing date’s volatility state, and for Dataset A that dependence is material, spanning a factor of 2.42 across the four scenarios. And the instrument’s tractability comes from freezing h : the model has no convexity in the volatility state, so it prices a variance swap but would not price an option on variance, for which the Heston [1993] characteristic function of Section 2.5.5 would have to be evaluated rather than merely written down.

RQ2: what is the Value of Information? Pricing the identical model on the proxy rather than the aligned dataset understates the fair strike by 93.8%, falling to 88.0% once measurement height is corrected. The height correction closes little of the gap, and Table 26 shows that the residual stays above 80% across every profile exponent from open water to very rough terrain, so the conclusion does not depend on the exponent adopted and the residual is not a height effect. It decomposes into a factor of 8.85 in the seasonal-variance constant C_0 and 1.82 in the GARCH state, locating the discrepancy in the seasonal-variance level of the site rather than in its volatility regime, though the

second of those factors compares two pricing dates six years apart rather than two site characteristics.

The more careful answer, developed in Sections 3.1.5 and 3.3.3, is that these figures measure the right thing for the wrong reason. Once both strikes are mapped onto the common axis of generation log-return variance, the $16.07\times$ ratio falls to about $2.5\times$, and the purely climatological part of it to about $1.36\times$: the two sites have nearly identical coefficients of variation, 0.384 and 0.401, and on the relative scale the instrument actually references they are close to equivalent. What separates them is not the level of variance they imply but whether that variance has anything to do with the quantity being hedged. Nordfriesland hub-height wind explains 58.7% of the daily variation in the generation series; Bologna 10 m wind explains 0.01%, with a slope not distinguishable from zero. The Value of Information is properly located there. A proxy that quotes a coincidentally reasonable number while carrying no information about the risk being transferred is more dangerous than one that quotes an obviously wrong number, because only the second announces itself.

RQ3: how effective is CAWS as a left-tail hedge? The instrument does what it is designed to do. Its payoff is positive in precisely the calm years that depress generation revenue, and the magnitudes are economically material: the calmest year in Dataset B's decade would have paid +0.4018 m/s per unit of notional against a shortfall of roughly 14% in generation, worth on the order of €2.3 million of gross revenue on the illustrative 100 MW farm. The risk metrics reported for it, however, require the three corrections of Section 3.4.3: the Sharpe ratio is uninformative because the strike is the mean of its own underlying, the conditional value-at-risk describes the premium leg rather than the protection leg, and with four and ten annual observations it is a sample minimum rather than a tail average. Read correctly, the cross-dataset comparison shows not that Dataset B carries four times the tail risk but that it estimates a similar relative tail from two and a half times as many independent observations.

On the model hierarchy. A fourth result cuts across all three questions. Every fitted specification beats persistence decisively, and all four are observationally equivalent to one another at a one-day horizon, by construction rather than by coincidence. Point-forecast accuracy therefore cannot discriminate among them, which is precisely why this thesis evaluates them on pricing and hedging criteria instead. The GARCH layer earns its place by supplying h , the state variable that carries the entire conditional content of the fair strike; the CAR(4) embedding of Benth and Šaltytė Benth [2009] earns its place by permitting pricing at maturities off the daily grid; and the NIG extension, on the evidence of Section 3.2.4 and Section 2.4.4, earns rather little in either dataset and is retained as a documented robustness extension rather than as the operative model.

3.5.2 Limitations

Five limitations bound what has been established, and they are stated in descending order of how much they constrain the conclusions.

The $\theta = 0$ benchmark. The pricing measure is not identified by the data. No liquid options market exists on German wind speed or on the SMARD generation aggregate, and the only traded wind derivatives reference Norwegian wind, so the market price of risk cannot be calibrated and is set to zero as a tractable benchmark (Section 2.5.4). Every strike in this chapter is therefore a physical-measure expectation. The evidence from temperature markets reviewed in Section 1.3.6, where Härdle and López Cabrera [2009] and Dorfleitner and Wimmer [2010] both recover a market price of weather risk significantly different from zero, suggests a true market-implied strike would lie above it, since producers seeking protection would be expected to pay a premium over the physical expectation. The direction is a hypothesis and the magnitude is unknown.

Commensurability of the cross-dataset comparison. The headline Value of Information figures compare two variances expressed in two different Box–Cox measurement scales, and Section 3.1.5 shows that a substantial part of the $16.07\times$ gap is attributable to that difference and to the six-year separation of the two pricing dates rather than to the sites. The figures accurately describe what the two calibrated pipelines quote and are retained on that basis, but they should not be read as a scale-free measure of climatological difference.

Spatial aggregation. Both models are calibrated on a single grid point while the hedged quantity is a national fleet aggregate. Spatial averaging removes variance, which is the most likely explanation for both models overstating the market strike by factors of three to four in reconciled units, and it also caps the achievable power-curve R^2 well below one. A multi-site or spatially averaged wind input would address this and would be the natural first extension of the pricing exercise.

Daily frequency. The stochastic-volatility parameters $(\kappa_V, \bar{v}, \xi, \rho)$ are not identified at daily frequency, for the reasons set out in Section 2.5.7, and the CIR proxies used in Chapter 2 are moment-matched from the GARCH fit rather than estimated. The same coarseness shows in the Dataset B diagnostics of Section 3.2.3, where short-lag volatility structure survives a single-component GARCH filter, and in the Feller condition of Section 2.5.3, which holds comfortably for Dataset B on every estimator considered but is estimator-dependent for Dataset A, failing on the full-sample volatility-of-volatility estimator that cross-dataset comparability requires. The theoretical CIR embedding is therefore well posed for the reference dataset and not demonstrably so for the proxy, which is a further reason the pricing results of this chapter rest on the piecewise-constant convention rather than on the continuous-time variance process.

Sample size for tail statistics. The non-overlapping-window choice that makes the CAWS risk metrics unbiased also reduces them to four and ten observations. The resulting tail statistics are descriptive, and the cross-dataset comparison of Section 3.4.3 rests on the sign and relative dispersion of the payoff rather than on a fifth-percentile quantile estimated from four points.

3.5.3 Future Work

Three extensions follow directly from those limitations, and one from the results themselves.

The first is option-implied calibration of θ . EEX German power options are the most liquid in Europe, and extracting a wind risk premium from them is the designated route to replacing the $\theta = 0$ benchmark with an estimated pricing measure. The obstacle is that those options are written on the power price rather than on wind speed or generation, so the exercise requires a joint model of price and generation capable of separating the wind premium from the power-price premium; it is a substantially larger undertaking than a straightforward calibration, and the Carr and Lee [2009] replication framework would supply the theoretical scaffolding for it.

The second is estimation at higher frequency. Hourly ERA5 reanalysis at the same coordinates would raise the observation count by roughly two orders of magnitude, resolve the intraday volatility clustering and diurnal boundary-layer cycles that daily aggregation smooths away, and make ξ separately identifiable from κ_V . That would convert Chapter 2's stochastic-volatility framework from a theoretical embedding into an estimated model, and would permit pricing instruments, such as options on variance, that the piecewise-constant convention cannot reach.

The third is a spatially aggregated wind input, replacing the single grid point with a capacity-weighted average across the German onshore fleet. This addresses the variance overstatement identified in Section 3.1.5 directly, and would be expected to raise the power-curve R^2 above the 0.587 achieved from Nordfriesland alone.

A fourth extension is suggested by the results rather than by their limitations. The finding of Section 3.3.3, that a wholly uninformative proxy can nonetheless imply a variance level close to the market's, argues for evaluating candidate datasets on explanatory power for the hedged quantity before, and not after, comparing the strikes they imply. A systematic screening of candidate reanalysis grid points on power-curve R^2 , prior to any pricing, would be a cheap and general-purpose diagnostic for this class of problem.

3.5.4 Implications for Producers and Risk Managers

Three practical conclusions follow for a German onshore wind producer or the risk manager of a renewables-heavy portfolio.

The natural hedge afforded by the merit-order effect is real but weak and asymmetric. At $\rho = -0.162$ over the full sample, and -0.476 and -0.275 within the two price regimes, high wind does depress the price a producer captures, dampening revenue

variance relative to the independent case. But it erodes the upside without placing a floor under the downside, and the residual volumetric exposure it leaves is what a wind-linked instrument is for.

The two instruments priced here are complements, not substitutes, and address different risks. The variance swap transfers the volatility of generation, and its fair strike, at an annualised realised variance near 114 with a year-to-year range from 95 to 132, is both large and genuinely variable. The CAWS instrument transfers the level of the wind resource, paying in the calm years to which the variance swap is by construction indifferent. A producer concerned primarily with the revenue consequences of a wind drought needs the second; one concerned with cash-flow predictability needs the first.

Data alignment is not a refinement to be added once the model works; it is a precondition for the model meaning anything. This is the chapter's central practical message and the one it would most want a practitioner to take away. A model calibrated on a misaligned proxy produced, in this exercise, a fair strike that was 93.8% below the aligned one on the model's own scale and yet, on a fully reconciled scale, sat closer to the observed market level than the aligned model did. Neither comparison would have revealed the problem. Only regressing the candidate wind series on the quantity actually being hedged did, returning $R^2 = 0.0001$ against 0.5871. Before a wind derivative is priced, the wind series it is priced from should be required to explain the cash flows it is meant to hedge.

References

- Ailliot, P., Monbet, V., & Prevosto, M. (2006). An autoregressive model with time-varying coefficients for wind fields. *Environmetrics*, 17(2), 107–117.
- Alaton, P., Djehiche, B., & Stillberger, D. (2002). On modelling and pricing weather derivatives. *Applied Mathematical Finance*, 9(1), 1–20.
- Albrecher, H., Mayer, P., Schoutens, W., & Tistaert, J. (2007). The little Heston trap. *Wilmott Magazine*, January 2007, 83–92.
- Alexandridis, A., & Zapranis, A. (2013). Wind derivatives: modeling and pricing. *Computational Economics*, 41(3), 299–326.
- Barndorff-Nielsen, O. E., & Shephard, N. (2001). Non-Gaussian Ornstein–Uhlenbeck-based models and some of their uses in financial economics. *Journal of the Royal Statistical Society: Series B*, 63(2), 167–241.
- Benth, F. E., & Šaltytė Benth, J. (2009). Dynamic pricing of wind futures. *Energy Economics*, 31(1), 16–24.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Brockwell, P. J., & Marquardt, T. (2005). Lévy-driven and fractionally integrated ARMA processes with continuous time parameter. *Statistica Sinica*, 15(2), 477–494.
- Carr, P., & Lee, R. (2009). Volatility derivatives. *Annual Review of Financial Economics*, 1, 1–21.
- Cox, J. C., Ingersoll, J. E., & Ross, S. A. (1985). A theory of the term structure of interest rates. *Econometrica*, 53(2), 385–407.
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Dorffleitner, G., & Wimmer, M. (2010). The pricing of temperature futures at the Chicago Mercantile Exchange. *Journal of Banking & Finance*, 34(6), 1360–1370.
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Feller, W. (1951). Two singular diffusion problems. *Annals of Mathematics*, 54(1), 173–182.
- Garcia, A., Torres, J. L., Prieto, E., & de Francisco, A. (1998). Fitting wind speed distributions: a case study. *Solar Energy*, 62(2), 139–144.
- Härdle, W. K., & López Cabrera, B. (2009). *Implied market price of weather risk* (SFB 649 Discussion Paper No. 2009-001). Humboldt-Universität zu Berlin, Collaborative Research Center 649 “Economic Risk.”

- Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281–291.
- Haslett, J., & Raftery, A. E. (1989). Space-time modelling with long-memory dependence: assessing Ireland's wind power resource. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 38(1), 1–50.
- Heston, S. L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2), 327–343.
- Kavasseri, R. G., & Seetharaman, K. (2009). Day-ahead wind speed forecasting using f-ARIMA models. *Renewable Energy*, 34(5), 1388–1393.
- Peterson, E. W., & Hennessey, J. P., Jr. (1978). On the use of power laws for estimates of wind power potential. *Journal of Applied Meteorology*, 17(3), 390–394.
- Schwartz, E. S. (1997). The stochastic behavior of commodity prices: implications for valuation and hedging. *Journal of Finance*, 52(3), 923–973.
- Tol, R. S. J. (1997). Autoregressive conditional heteroscedasticity in daily wind speed measurements. *Theoretical and Applied Climatology*, 56(1–2), 113–122.
- Torres, J. L., Garcia, A., De Blas, M., & De Francisco, A. (2005). Forecast of hourly average wind speed with ARMA models in Navarre (Spain). *Solar Energy*, 79(1), 65–77.
- Tuller, S. E., & Brett, A. C. (1984). The characteristics of wind velocity that favor the Weibull distribution in wind speed analysis. *Journal of Climate and Applied Meteorology*, 23(1), 124–134.